

Bio 250 Why We Get Sick

Numbered Transcript

BIO 250 • Dr. Benjamin Bikman

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Lecture 1: Metabolic Origins of Disease

At a glance — Bikman on the metabolic origins of disease.

- 1.1 I am delighted to be teaching you guys for the next little while. I am a metabolic scientist, and that forms the way I view the world to a degree, but certainly the way I look at health and disease. Why would a guy like me, why would I devote my life to understanding metabolic disorders? It's because of how prevalent they are and how relevant metabolic disorders have become. In the time we have together, we're going to explore the metabolic origins of chronic disease, what I like to call the plagues of prosperity, and we'll go through them.
- 1.2 But let's just paint the picture, just so we know how bad things really are when it comes to metabolism and health. This was a paper published just a few years ago, and let's use this actually as an opportunity to get oriented with the way I'll be presenting. Whenever you see this bold, big title, that's actually the title of a manuscript that I'm citing. And then I may share a quote or just share the data from it. And then lastly, I'll share the citation. I'll make sure that you have that citation that you guys, anyone at any moment, can look this up.
- 1.3 So back to the manuscript itself. This paper found that 88% of US adults, well, adults within the US, were considered metabolically unhealthy. So that's a pretty sobering statistic, right? In other words, only 12% of us were considered metabolically fit. Now, it's very popular nowadays within the United States, especially to criticize the United States and think we're the worst at everything. But metabolically speaking, we're far from the worst, as much as we may feel otherwise.
- 1.4 If we look around the world, we see that metabolic disorders are much more common in South Asia, in particular. In fact, that's something I'll come to later when we talk about the differences in fat cells across different ethnicities. But also it's much worse in the Middle East, where this instance of metabolic disorders are much higher than we have in the US. And in the Pacific Islands, it's significantly worse. In Southeast Asia, and even in North America, we are not the most metabolically challenged. Our neighbor to the south has us beat here.
- 1.5 So, as bad as it is here, where, again, we documented that 88% of adults were considered metabolically unfit, it's not as bad as it is elsewhere. So in other words, this is a global problem, and these issues continue to get worse. And so we need new ways of viewing it and correcting it. And that's some of what we'll be talking about over the time we have together. So back to this statistic again, 88% of adults being metabolically unfit.
- 1.6 Now, what is metabolic health? How do we define it? I want to present to you two general definitions, and then we'll dive into each of them in a little more detail. The first is the metabolic syndrome itself, which you've probably heard of. That's a more

common term, although it's a newer term for an old problem. And then this idea of metabolic flexibility. And as I define these, I'm going to introduce a few new characters. And this is all just to set the stage of having a common understanding of what metabolic health is.

- 1.7 First, the metabolic syndrome. The metabolic syndrome is a constellation of disorders, and they're all common in and of themselves. But when they come together, they reflect a deeper problem. And again, the study that I showed earlier actually measured these metrics, and they found that it was 88% of adults who had at least one of these problems. Again, in other words, only 12% had none.
- 1.8 Now, what are the problems? I have them mentioned here, but let's just briefly review them. The first problem is having an excessively large waist circumference. That's pretty self-explanatory, but men and women carry fat differently, and thus it's no surprise that men and women will have different levels of acceptable fatness around that midsection. And we'll talk about the relevance of that midsection, that central adiposity, in just a little while. But again, suffice to say, if there's too much fat here, that reflects a metabolic problem.
- 1.9 Blood pressure is another easy one. Interestingly, not only will we go into more detail with fat control very soon, but we'll also talk about hypertension and high blood pressure, how that is itself reflective of a metabolic problem. And then high blood sugar will become very obvious, how that connects to metabolic issues. And then lastly, what we can just generally call dyslipidemia, when blood lipids aren't behaving. This would be reflected as a high triglyceride level or a low HDL cholesterol.
- 1.10 Now, it's interesting to note, and this foreshadows a topic we'll get to later, even within this well-established, well-supported metabolic syndrome definition, you see that, while we focus on some blood lipids, the most famous is absent, which is one called LDL cholesterol. The conventional clinical care is obsessed with measuring LDL cholesterol, and yet it doesn't even make the team here when we're looking at these problems. We'll revisit that later.
- 1.11 Now, metabolic syndrome, I'd said, I alluded to earlier that this is a problem that has a newer name. It wasn't always called the metabolic syndrome. When this disorder, this constellation of disorders, was first documented, it was referred to as the insulin resistance syndrome. This was documented by a scientist named Gerald Reaven, one of my academic heroes. And again, he called it the insulin resistance syndrome. I prefer that term. It's not as sexy as metabolic syndrome is. Anything with the word metabolism just oozes appeal.
- 1.12 With insulin resistance syndrome, however, I prefer it because it's more descriptive. It lets the patient or lets the individual understand what is the nature of the problem. What is the common soil from which all of these weeds are growing? Calling it the metabolic syndrome is pretty vague. It doesn't really help you understand what you can

address. Insulin resistance syndrome used to be the name I preferred, and I'm not trying to bring it back. I'm not trying to make it happen. If you will, metabolic syndrome is the term. I just want you to know that it used to have another name, and that other name is a little more accurate.

- 1.13** Now, let's move on to the next one, a very interesting topic referred to as metabolic flexibility. Now, to talk about metabolic flexibility, I have to introduce a very important player or character within the story that I'm telling you, and that is the humble hormone insulin. We're going to be talking about insulin a great deal in the time we have together. And it is a humble little hormone produced at just picomole concentrations. It has such a small presence within the blood, and it is literally a teeny little peptide of a hormone. By way of hormones, it's very small. It would be fighting in the welterweight division, but it packs a serious punch.
- 1.14** So I'm just introducing it for now. Insulin is a hormone that we make from the beta cells of our pancreas. And the pancreas is a long, slender little gland tucked right beneath our stomach that connects to our small intestine. So it connects into the intestinal tract right after the stomach. And again, one of the cells within the pancreas, which is a very busy cell, a busy tissue, there's a lot of real estate within the pancreas occupied by lots of different cells. But the beta cells are the manufacturers of the hormone insulin.
- 1.15** Insulin's primary stimulus is high glucose levels or blood sugar levels. When blood sugar levels start to climb, that would become lethal if it maintained that higher level for any real period of time. So insulin swoops in and saves the day. Insulin being elicited, being stimulated with the high glucose will then come gushing out of the beta cells into the blood, and essentially, and we'll revisit this, escorting the glucose from the blood into the various cells of the body, thereby lowering blood glucose. And insulin, having done its most famous job, controlling blood sugar, then retreats into the background and goes away.
- 1.16** Now, I was very careful with what I just said. I said that lowering glucose is insulin's most famous job, but it is far from its most important. Insulin is, in fact, despite its teeny weight and very low levels in the blood, one of the most important hormones in the body. It regulates numerous processes. Basically, any aspect of inflammation, almost any aspect of immunity, is going to be impacted by insulin. So nutrient metabolism, something we'll talk a lot about more in just a moment, namely, what does the body do with fats and proteins and glucose? Protein synthesis, stimulating protein synthesis within muscles and other tissues, regulating the growth of the tissue, controlling the movement of electrolytes, sodium, potassium, into and out of cells, and much, much more. We'll get into more of that in a bit.
- 1.17** Now, we've introduced insulin. Let's come back to this idea of metabolic flexibility. Human metabolism is like a hybrid. We have two fuels that are primarily providing the body energy at any moment. In truth, there are a little more, but they are much, much smaller contributors to what we consider fuel use or metabolism within the body. So at

any moment, the body is burning a mix of sugar, blood sugar, or fat. We're fat-burning or we are sugar-burning. And this is entirely dictated by the hormone insulin, completely dictated. If insulin is elevated, the body has no choice but to shift from, say, a balanced burning to relying almost completely on sugar-burning.

1.18 And this is what is reflected in what we call commonly the fed state. We go eat a meal, we go eat a sandwich, we're out to lunch. Insulin will be increased because blood sugar goes up. Well, one of the ways insulin helps reduce blood sugar is by telling the body to burn sugar, to burn the glucose. However, while it does that, that means fat-burning is relatively lower. Now, however, let's say we ate, and now it's been several hours. We have gone from the fed state, high insulin, to the fasted state, a state where insulin has come down to fasting levels.

1.19 When insulin has come down, the metabolic engine shifts. The hybrid shifts its fuel over from relying on glucose as the fuel to fat as the fuel. So this concept of being able to shift between the fuel sources is what we call metabolic flexibility. You eat, you burn sugar. You stop eating, you start to burn fat. Again, all dictated by insulin. However, there is a metabolic phenomenon referred to as metabolic inflexibility. And in this case, the person is living a life of almost constantly or chronically elevated insulin. In this situation, even when the person is fasted, as we would say, maybe they haven't eaten in six hours, that would be a fasted state, they're still primarily burning glucose as their fuel. Their metabolic engine is stuck.

1.20 The fuel source is stuck on sugar-burning, all because the elevated insulin won't allow the body to shift over. The metabolic flexibility is gone. All right. Now, these are the two definitions of metabolic health that we can rely on. When we look at the scope of the problem, when we appreciate that 88% of adults in the United States, and even worse abroad in many countries, are considered metabolically unhealthy, this is how we can look at it. Now, these are distinct issues. Metabolic syndrome is not the same as metabolic flexibility, but they have one thing in common, which is that they are both derivative of insulin resistance, which is something I've alluded to so far.

1.21 And before I define it, I just want to further impress upon you the relevance of this issue. Virtually every chronic disorder is derivative in some way, is somehow influenced by insulin resistance. Like, for example, the big ones that we're all afraid of, all of which we'll get into in more detail. Heart disease, type 2 diabetes, certain cancers, fatty liver disease, the most common liver problem, and even obesity, this dysregulated fat tissue growth. All of these are influenced by insulin resistance. Where insulin resistance is either the direct cause, like, as for example, type 2 diabetes, or if it's not the cause, it's exacerbating it, as is the case with cancers, which, again, we'll get to eventually.

1.22 And then beyond these chronic lethal disorders are the ones that are more inconvenient but heartbreaking and tragic, which is infertility—in men and women, something we'll discuss later, just because these are becoming so common. But even these at their core are initiated by insulin resistance. And then a host of other

problems, from the musculoskeletal, like arthritis, to neurological. Many, many neurological disorders, including dementia, Alzheimer's disease, is very influenced, if not directly caused, by insulin resistance.

- 1.23** So to this point, I've set the stage, defining metabolic health and introducing the relevance of this problem and the prevalence of it as well, globally. With this as preamble, if you will, these are the three distinct topics that we'll be discussing in the time we have together. Firstly, we will explore what insulin resistance is, with some other subtopics within that. And then next, we will discuss, why does it matter? What are the diseases that come from it, and how is it connecting? How is insulin causing these disorders? And then lastly, the good news, which is the "what can you do about it" part.
- 1.24** So that is a happy ending. So, while, in the time we have together, it will seem like I'm telling a bit of a horror story, and insulin resistance is the villain of the story, there is a happy ending. There is resolution, and it's very real, and it's very immediate. All right. Now, as I just mentioned, I'm telling you a horror story. Well, it is all the rage these days to talk about, or rave these days, to talk about how to give a backstory to a villain, right? So insulin resistance is the villain, and let's define it now. Let's explore. Let's really dive into what it is and where it is coming from.
- 1.25** And to understand insulin resistance, let's look at the level of a cell. So here I've portrayed a cell. This could be any cell within the body, because literally every cell of the body has an insulin receptor—a little dock, a little door, if you will, that only insulin will come and knock on. Now, I said literally every cell, and I don't use the term literally too liberally, like the kids do these days. I do mean literally every cell in the body, from brain cells to bone cells, from lung cells to liver cells, and everything in between, all of them have a door that insulin can come and knock on.
- 1.26** And when insulin knocks on that door, it elicits a cellular action. And I'm just going to call it an action right now because, not surprisingly, what insulin does at a brain cell is not the same as what it might do at the bone cell. So we'll just keep that kind of generic for now and get into some specifics later. But do you recall what was insulin's most famous effect? Not its most important. The most famous effect was lowering blood glucose. So I'm going to just superimpose that right here, where we have glucose outside the cell, in the blood, for example, and one of insulin's actions at some cells of the body, like, for example, muscle cells. It will do this.
- 1.27** Well, insulin, after knocking on the door, will open up another door for the glucose to come into the cell, allowing the blood sugar levels to come down and allowing the cell to use that for energy, burning sugar, getting into glucose-burning mode. However, over time, due to some insults, the action of insulin can become compromised. In other words, insulin isn't working well at that cell anymore, whatever cell we would be talking about. In this example, one consequence of this would be that blood glucose levels start to rise. If insulin can't knock on the door very well and can't open the

glucose doors very well, it's not surprising that the glucose levels will steadily start to climb.

- 1.28** At the same time, though, the body is so clever at making adjustments. It is so determined to balance things out and to return to what we call homeostasis, to keep everything normal, that it knows, it senses insulin isn't working as well as it used to. What used to be a big, robust action is now a rather modest action. And one consequence of this, one effort in order to try to overcome it, is to increase the production of insulin. And what was once one polite molecule of insulin knocking on the door becomes an angry mob, pounding on the doors of the cell, all in an effort to take what is now a diminished action and attempt to return it to normal.
- 1.29** All right. Now, in this wordy explanation of defining the villain in the story here, there are two fundamental parts. There are two critical aspects. This is a coin that has two sides. There cannot be one side without the other. If you're holding the coin, you have both sides. There are two sides to this story or to the character. First, some cells aren't responding to insulin. That is the obvious part of what gives the name insulin resistance its name. This is where that part would come from. Some cells aren't responding.
- 1.30** But notice the use of my word "some". Some cells are as responsive to insulin as ever. And that becomes critically important in light of, flipping the coin over, the next part of insulin resistance. When we look at the whole body, another feature of this, whereas we have been looking at the cell level at the moment, but always these come together. If insulin isn't working well, it's some cells, and within the whole body, insulin levels are higher than they were before, a condition that we call hyperinsulinemia. "Emia" just means in the blood, "hyper" means a lot. Hyperinsulinemia.
- 1.31** Now, this is important. This is commonly overlooked. There can be no such thing as insulin resistance within the body without accompanying hyperinsulinemia. They always come together like two sides of a coin. Now, I'm going to slightly adjust this paradigm because it's one we're going to revisit. But the way I just explained this, I explained to you how the compromised insulin action causes an increase in insulin production. That's how I explained it a moment ago. We're going to complicate that more in a bit, but I want to keep this paradigm. I want to point it out so that it's keeping in mind.
- 1.32** Now, let's shift the paradigm even more and look at the lifetime of an individual who's moving through time over the years. They're coming in for an annual visit with their physician, and every year they get their blood sugar levels or their glucose levels measured. That is, after all, one of the primary clinical outcomes. We always measure glucose. I will make the case that we measure it too much or focus on it too much. But over time, the person is getting higher and higher blood sugar levels.
- 1.33** If we superimpose insulin, which is commonly not measured—I'm trying to make the case that we should, by the end of our time together—insulin would look a little

something like this over the years of the person. That long before the glucose levels, indeed, even decades before the glucose levels ever start to climb, insulin has been fighting this cold war within the body, a silent war behind the scenes, only behind the scenes because we don't look at it.

- 1.34** Now, regardless, with this perspective in mind, I want to split it in two. This first period of time, while glucose levels are normal, but insulin levels are elevated, is insulin resistance. And again, it's important to note that it is so often missed, which is a topic we'll dive into more in detail later, because the glucose levels stay normal. But it's the insulin that is significantly elevated. Now, however, eventually, due to several insults to the body, as well as the potential for insulin production to come down a bit, then we have the glucose levels start to climb. Then it becomes clinically relevant because we focus on glucose and we say, "You now have diabetes."
- 1.35** But again, this was happening years, even decades, after the insulin was already elevated. But it's now in this state of type 2 diabetes, where both insulin is elevated and glucose. It's important to note that first point, because commonly in type 2 diabetes, clinicians will say that insulin has become insufficient. Well, that's a relative term. Insufficient to what? Insufficient, perhaps to where it needs to be to control the glucose. But the body is, it's not working anyway. The high insulin isn't helping anyway.
- 1.36** And so what happens in some is that insulin levels come down a bit, which exacerbates the climbing glycemia or the glucose levels. But importantly, and I was very careful in how I created this, if it's type 2 diabetes, the insulin does not go to zero. It doesn't go even to as low as it used to be when they were insulin sensitive. It comes down from its peak, but it still may be multiples higher than it used to be. But our failure to look at insulin means we failed to detect the problem.
- 1.37** Now, let's add to this view. I've been talking about what insulin resistance is and differentiating it now compared to type 2 diabetes, but also impressing upon you the fact that there is no type 2 diabetes without insulin resistance, which is a disease we will revisit later much more explicitly. But let's now look at different tissues of the body and how they influence this. Insulin resistance starts in the fat cells. You could imagine a row of dominoes. When I'm playing with my kids, we stack up the dominoes and then hope none of the kids knock it over. But the dominoes are there waiting to fall, but something has to tip the domino over, which we'll get to.
- 1.38** But there is a first domino that then tumbles into the rest. That first domino, when it comes to insulin resistance, is the adipose tissue, or the fat cells. When this tissue falls, if you will, it then starts to spread the insulin resistance, through ways we'll define later, to other tissues of the body. And when these tissues, muscle, liver, and the alpha cells of the pancreas, something we will revisit later—not the beta cells, but the alpha cells—when they become insulin resistant, I deliberately placed them above this hash line because when they fall, now glucose levels start to go crazy. But until they become

insulin resistant, it's just the insulin is struggling, but it is able to keep the glucose in check.

- 1.39** Now, let's shift this one last time. I promise we'll leave this paradigm in just a moment. To look at this in another way, when insulin resistance is only affecting the fat cells, we have this state of high insulin but normal glucose. And then when the insulin resistance has moved to these other tissues, now the high insulin is accompanied by elevated glucose as well, which we would define as type 2 diabetes. Wouldn't it be nice if we could have detected the problem here, rather than waiting for it to get to that point?
- 1.40** All right. So we know what insulin resistance is. We've identified the villain. Now it's time for the backstory. We'll explore how did insulin break bad, if you will. How did it go from working for us to working against us? I want to present to you three primary causes. Now, when I use the word primary, I'm being very careful as I've devoted my life to understanding this obscure topic that no one really cares about, but I'm hoping more people will. Thank you for all pretending to care for now and indeed caring later. But there are three primary causes.
- 1.41** I define primary based on how validated are the causes that I'm about to introduce you to. In other words, if it's primary, I say that it has been validated in isolated cell cultures, cells growing in little petri dishes. It has been further validated in animal models, within the laboratory setting, rodents, and then lastly, validated in the pinnacle of all creation, the humans. So if all three of these conditions are met, or if all three of these models have shown a result, I consider it primary.
- 1.42** And in fact, there are conveniently three causes—stress, inflammation, and hyperinsulinemia, chronically elevated insulin itself. Each of these is capable of causing insulin resistance on their own throughout the body. Now, as luck would have it, we tend to live in a culture where a lot of them are elevated at the same time, but each on its own is capable. They're all independent, noxious stimuli capable of eliciting this catastrophic consequence, metabolically speaking.
- 1.43** Let's dive into each of them. Let's define them in a little more detail. Now, stress. What do I mean by stress? Stress can be referred to often in a lot of ways, and a lot of people will throw this word around. When I talk about stress as a scientist and one who teaches endocrinology at my university, I look at it through the lens of the two primary stress hormones, epinephrine and cortisol, epinephrine, commonly called adrenaline. Now, these hormones are created—they have almost nothing in common. To a basic student of physiology and anatomy, there's this temptation to say, well, they're both from the adrenal glands. Yeah, but the adrenal glands have very different parts to them, and they're produced in totally different cell types of the adrenal glands.
- 1.44** They're also completely different in the formation, their synthesis. They're different in how they move through the blood. They're different in how they act or signal on other cells. They have, for all intents and purposes, nothing in common except they both

want to increase blood glucose, and do so very, very rapidly, in the case of epinephrine, and a little more slowly, but still substantially in the case of cortisol.

- 1.45** So when any of these stress hormones are elevated, there's this upward pressure of glucose levels within the blood. That, of course, puts them at odds with who? Yeah, the hero/villain of the story, right, which is insulin. Insulin is sensing this upward pressure of the glucose, and no real mind for who's causing it, just sensing this kind of child who's stuck in this custody battle. And insulin is trying now to push it back down. But the more the stress hormones are coordinating to push it up, insulin has to work harder and harder and harder. And as insulin levels go up and it's working harder, that, of course, no surprise, leads to insulin resistance. So stress is a primary cause.
- 1.46** Now, let's just touch on what are some stressors. I mentioned what the stress hormones are, but I wanted to just give you a couple of examples of what might contribute to it in the world in which we're currently living. One is sleep deprivation, something I'm very, very sympathetic to. After all, I am a dad of young kids, and I'm a very, very involved dad. The moment my wife was done nursing our children, I was on night duty, and I reveled every moment. But it basically meant I am chronically sleep-deprived, even now. Even though they don't need me getting up in the night, I still do. So sleep deprivation is something I'm very sensitive to.
- 1.47** When the body has any period of sleep deprivation, the next morning, there is a substantial increase in cortisol, and it's not subtle. It's several times higher than it would have been otherwise. Well, of course, as cortisol goes up, it starts to push the glucose levels up in the body, and the higher the glucose is getting pushed up, of course, the more insulin we need to try to resolve that glucose.
- 1.48** Now, just case in point of just how dramatic this can be, and I'm sympathetic to the fact that I have some students here in the audience now and later. But here in a given night, say an individual, an adult was supposed to get seven or eight hours. One bad night of sleep is capable of increasing insulin resistance the next day via an increase in cortisol by 25%. That's a pretty robust reduction in insulin sensitivity, all from one bad night. Now, thankfully, one bad night can be corrected by one good night. But imagine when one bad night becomes another, and then another, and then another. The insults just keep accruing. The damage gets more and more severe.
- 1.49** Now, even in kids, again, I'm a dad, I'm sensitive to this phenomenon, and I'm sensitive to the metabolic problems that face children these days around the world. But there are distinct time frames of sleep within children. Kids that are 6 to 12 need around 9 to 12 hours of sleep a night. Kids that are 13 to 18 need about 8 to 10 hours a night.
- 1.50** A paper was just published, very, very recently in 2023, that found that if a kid was missing out on at least one hour of sleep a night consistently, over the next two years, they were almost three times more likely to have clinically demonstrated insulin resistance, just from regularly missing one night, one hour, rather, consistently cutting

in to those hours of sleep. It is so important to exercise proper sleep hygiene with kids and to get that habit set for all the reasons in the world, including the child's metabolic health.

- 1.51** Now, let's shift from something depressing, like sleep deprivation, to something a little more exciting, like caffeine. As wonderful as caffeine may be, this most widely consumed drug around the world, caffeine does have a metabolic consequence. Not all bad, but it's a bit of an interesting twist. Caffeine will increase the hormone epinephrine. It will spike adrenaline. Now, one of the reasons athletes like taking caffeine is because of what epinephrine will do to fuel use. Epinephrine will promote the body to use fat for fuel.
- 1.52** Well, it's difficult to use fat for fuel if insulin is up. And so insulin comes down, typically, with caffeine consumption, which helps reduce glucose-burning. But the caffeine is so determined to only burn fat that it makes insulin not work as well. And so there's a transient effect of insulin resistance when it comes to caffeine consumption. So it is something that perhaps should be used a little judiciously.
- 1.53** Now, let's shift from stress into inflammation. And to talk about inflammation, I have to mention these other little players called cytokines. Cytokines are small little proteins that are produced from immune cells, typically, but other cells can make them, too—and I'll introduce some examples later. But these are small proteins that basically will just turn on inflammatory biochemical pathways within the body. Any time the cytokines are elevated in the body, regardless of the cause, they will start to challenge insulin function, and the body will become demonstrably insulin-resistant, indeed quite quickly.
- 1.54** Now, when I mention inflammation, you are tempted to perhaps think this is only relevant to infection and illness, and it certainly is. This is something that's been well documented in this era of time where we are all so focused on our immune health over the past couple of years. It has been remarkable to notice people who aggressively track their glucose levels with continuous glucose monitor or whatever it may be. They can almost predict that they're going to get sick because they will see that their blood sugar levels, which are typically a little static and low, now start getting much more volatile and much higher. This is reflective of the insulin resistance, which is a result of the cytokines, which itself would be a product of infection.
- 1.55** Now, other things can also stimulate this, like, for example, autoimmune disorders. It's a very interesting feature of autoimmune disease, where the autoimmune disease will ebb and flow. That's a natural—there's a natural kind of tendency of autoimmune disease is to be very activated or on and aggressive, or very mild, and it will subside for a period of time. As the autoimmune disease ebbs and flows, so too does the insulin resistance. And one of the citations I have here is documenting this in patients with rheumatoid arthritis, where they found that when the rheumatoid arthritis was very aggressive and the patients reported having very painful joints, they would be

demonstrably more insulin-resistant than they were when the joint pain was a little better.

- 1.56** Now, another one is air pollution, which I'll talk in more detail about in just a moment, as it's becoming more common and has been a substantial problem in various parts of the world for a long time. And then something we'll spend a lot of time talking about later is fat cells themselves. When fat cells get too big, they become their own producers of cytokines for reasons that we'll get into.
- 1.57** Now, with air pollution, I just want to mention this in part because of my own contributions to this with some collaborators. The only really two ways we interact, or something from the environment can get into the bloodstream, and that is either through the skin, which is very, very uncommon. The skin is built to keep things out, but it can get through the skin. Or we ingest it. And when it comes to air pollution, we inhale that very readily. And as the particles, the fine particles that are combusted from something and then inhaled, they will accumulate in the lung and activate an inflammatory response within the lungs. And so the lungs, again, in response to these inhaled particles, will begin to aggressively produce cytokines. And as the cytokine production climbs, insulin resistance starts to get worse, or insulin sensitivity drops.
- 1.58** Now, we talked about stress as a cause. We've talked about inflammation as a cause. Let's go to the elephant in the room, which we will spend the most time talking about. And later in the lectures, when we talk about how to resolve these problems, we'll once again spend the most time talking about this just because of how relevant it is. Any time insulin levels are elevated in the body over an extended period of time, the body will become less and less sensitive to the insulin. At first glance, this seems a little odd, but this is reflective of a fundamental biological principle. Too much of something will result in a resistance to that something. The body will attempt to turn down the reception to a lot of noise.
- 1.59** It happens with insulin. It happens with caffeine. The more someone takes caffeine, the less the body starts to respond to it. The more someone takes an antibiotic, the more the bacteria begin to become resistant to it. And example after example reflect this fundamental biological principle. I would even argue it goes beyond biology. I think there's something so fundamental about this that it begins to impact the soul, the entire aspect of a human, which is if there's a constant stimulus, particularly a noxious stimulus, we need more and more of that stimulus in order to get the same reaction that we had initially. This is just one physical manifestation of that.
- 1.60** Now, one funny little example. I am a family man, as I've alluded to multiple times. I am struck by the family dynamic within my own home. My darling wife is a homemaker. She's a full-time mom, so she is around our kids a lot. And I am interested that when I'm home, my kids are loud. They're running around the house. They're fighting. They're laughing. They're screaming for no reason. And they will be asking mom to do something, and I will hear them. But my wife, who's just wonderful, she doesn't

respond. I can be in another part of the house and hear my child asking mom or screaming for something, and my darling wife, it just doesn't register.

- 1.61** She is so used to that noise that she's reduced her reception just to survive, I'm sure. It's homeostasis. She's got to just adapt. But I'm not around it as much. As involved as I am as a dad, I don't hear that noise as much as she does. And so I'm in another part of the house, the kid's screaming, screaming, and I am hearing it all. And of course, I'm getting angrier and angrier, or whatever the case may be, but I will react. I'm very sensitive to the volume. My wife isn't. She doesn't actually have a hearing problem. She's just adapted to the incessant noise of a busy family.
- 1.62** Now, let's come back to this. Earlier on, initially when we were defining insulin resistance, I presented the view wherein the compromised insulin action is resulting in more insulin production. Now, however, I am bringing this full circle. When there's too much insulin knocking on the door of the cell, the cell just begins to say, basically, this is too much. I can't keep answering the door every time you come knocking because you're knocking too often. So you can start to see how this becomes a vicious cycle.
- 1.63** Whether it is one of the other stimuli we spoke about, say stress or inflammation, that are causing the reduced insulin action, that then leading to the hyperinsulinemia, or whether it is the hyperinsulinemia itself resulting in the reduced action which feeds the hyperinsulinemia, and the problem just keeps going until we break it. Now, here are the three primary causes that we discussed—stress, hyperinsulinemia, and inflammation. And as I noted, each of them on their own is capable of causing insulin resistance in any tissue of the body that is capable of becoming insulin resistant, because some tissues just don't respond the same way. But these ones do, ones we've talked about or I've introduced, and we will spend much more time talking about in the time we have together.
- 1.64** In fact, I want to really dive into the connection between insulin and fat cells. Insulin promotes the growth of the fat cell. In fact, it does it exceptionally well. Indeed, I will impress upon you the relevance of this in just a bit, but it is impossible for a fat cell to grow unless insulin is elevated. It cannot happen. It's impossible. Now, similarly, if insulin is elevated, there is a pressure for fat tissue to start getting bigger. But in all of our wonderful variety, we have two mechanisms of getting fat on the body, of gaining fat mass. We can either get fat through a process called hypertrophy, or we can get fat through a process called hyperplasia.
- 1.65** Now, the difference is obvious now. With hypertrophy, which is how the vast majority of people get fat—as we start to gain fat mass over the years, it's through hypertrophy, primarily. And this is a feature of just becoming an adult. Typically, by the time we become adults, the number of fat cells we have is set until later life. Now, I'm not going to go into much more detail on that aspect of it, the development and the life cycle of a fat cell. Fat cells are not immortal, despite what you might have heard. They have a

long life. They will live about 5 to 10 years. And then, over that time of adulthood, they'll just be replaced by another fat cell.

- 1.66** But again, back to this point, which is, the average individual has a fat cell number that is set. And so as insulin continues to put pressure on the fat tissue to grow, it can't reproduce itself. Each individual fat cell just keeps bearing the metabolic burden of what insulin is telling it to do, getting bigger and bigger and bigger. In some instances in adulthood, some people have the potential to continue to make new fat cells. And in that case, insulin is providing the pressure to the fat tissue, grow, grow, grow. And then as the fat cell gets big, right when it starts to get a little big, it just recruits help. And there's another fat cell, and they all stay much more modestly sized, but much more abundant.
- 1.67** Again, a point I emphasized, and we will not leave this alone yet, fat cells cannot grow unless insulin is elevated. Now, for the first time yet, although there will be several of these, I want to just go on a tangent for a moment and introduce the concept of calories. What about calories? How can I claim to be talking about fat cells growing if I'm not invoking this idea of energy or calories? Calories, after all, are just a measurement of a unit of energy. So let's bring that in for just a bit.
- 1.68** All right, so insulin is promoting the growth of the fat cell. And as I said, when insulin is elevated, or the fat cells can't grow unless insulin is elevated. But this isn't the full story. As much as insulin is telling the fat tissue to grow, there has to be something to put in the fat tissue to promote the growth. And so as much as I'm saying insulin is absolutely essential to fat cell growth, and it is, you also need energy. Calories have to be accounted for here. And so, we have the insulin signaling the growth, and we have the energy in the form of calories providing the fuel for the growth, a difference between the signal telling the cell what to do and the fuel for responding to that signal.
- 1.69** Case in point. Within my laboratory, we grow fat cells. So here we have a petri dish. We grow fat cells in these little—you know what I'm talking about, these clear little petri dishes. And we have, within the petri dish, growing on the bottom of the petri dish, we have this perfect little soup of nutrients. All of the fat and the glucose these fat cells would ever want are there. The fat cells are swimming in a sea of energy, swimming in a sea of calories, and yet they stay very, very small. They don't grow at all, until we introduce one other character.
- 1.70** When we put insulin in that culture medium, all of a sudden, boom, the cells know what to do. In the absence of that signal, the cell simply doesn't know how to respond to all the energy around it. Cells are not autonomous. Little beings, they need to be told what to do. Insulin, in this case, signals the growth that the calories have been waiting to fuel. All right, now, let's just continue this a little further to reconcile some issues. What happens is that there's an elevated insulin to signal the growth, and there's sufficient energy to fuel the growth. And that is a wonderful recipe to promote the growth of fat tissue.

- 1.71** However, what if energy is low, but insulin stayed high? This impossible scenario is indeed impossible because of what would happen. If insulin is elevated, it is so determined to promote the growth of fat tissue that it will start selectively pushing the energy, the calories, into the fat tissue, because that's one of insulin's jobs. It wants to store energy for later use, which is wonderful. That's why we need it. But in this case, if insulin were high, but energy was low, it would be pushing all the energy into the fat cells, resulting in one very energy-hungry tissue, namely the brain, going, well, hungry.
- 1.72** We will revisit this later, but the brain is a high metabolic rate organ. It has a very high metabolic demand. It needs to be fed constantly. And unlike fat tissue, it doesn't have the ability to store energy for later. The brain must be constantly getting its energy from whatever the blood can provide. But if insulin is elevated and it's pushing the energy into the fat cells, then the available circulating energy in the body—by that, I mean how much fats or glucose may be in the bloodstream—gets so low that the brain starts to panic. It would indeed be catastrophic, and you would faint, and then you would die because the brain wouldn't be able to meet its energetic needs, which is why this can't happen.
- 1.73** To further this, every diet or every method of eating that promotes a reduction in energy will indeed improve insulin sensitivity, which is something we'll get to later. I don't want to get ahead of myself, but it must be accompanied by a reduction in insulin. And then the reduced insulin allows the body to use even the modest amount of energy that may be coming in, rather than forcing it all to store, because there's a difference between storing energy and using energy.
- 1.74** All right, now, let's just flip it around again. What if insulin uniquely was low and energy was high? This is something that can, in fact, happen in human conditions, and we'll revisit this. We'll talk about it later. But in this case, the fat cells will shrink, even if the person is eating an abundance of energy. But how could that happen? How can we be shrinking our fat cells, losing fat mass, even with an abundance of energy, with insulin being low? And that is due to two reasons. If insulin is low, there is a substantial change in metabolic rate or what we commonly call energy expenditure. But also at the same time, not only are we burning energy faster, but we begin wasting energy.
- 1.75** Whether it is that the energy is coming out through the urine in the form of glucose or ketones, something we'll talk about later, or whether the person is breathing out ketones, these are calorie-dense. These are calorie-loaded molecules that are coming out. And so the body has a way of reconciling this, that if there is abundant energy coming in, but insulin is low, preventing the storage of that energy, the body will adjust. It will make this metabolic compensation to, again, accelerate the metabolic rate, the rate at which everything is working in the body, and second, by just dumping the energy out back into the environment. And in this case, a low insulin diet is not necessarily a low energy diet, unlike the flip of that.

- 1.76** All right, now let's come back to this. All that was a tangent. When insulin is elevated, it promotes fat tissue growth. I want to highlight the effects of one study that really puts a fine point on this. So there's the title of the study published a number of years ago. And what they did is take adipose biopsies of people who were treated with insulin, so diabetic patients that were taking insulin injections. And we are commonly told, diabetics are told to rotate injection sites. And what they did, they took patients who hadn't been rotating their injection sites. And they did it, so let's say the patient was injecting all of their insulin right here in this abdominal fat pad. So they're putting their insulin right here. They pinch their fat, inject the insulin.
- 1.77** So these scientists took a biopsy of that injection site and then just went a little over, up or down or to the side, in that same region and took another biopsy of the fat. But again, in an area where the insulin injections weren't happening. And then they compared, what is the fat tissue like? Firstly, at a microscopic level, although the image is a little washed out, you can see that, on the left side panel, which was the site of insulin injection, the fat cells with diameter are about three to four times bigger diameter in this two-dimensional image.
- 1.78** But if we were to look at that as a three-dimensional image, it ends up being a roughly 10-fold greater volume. So the fat cells that were seeing this big load of insulin from the syringe of this insulin-treated patient grew to about 10 times larger volumetrically than the fat cells just a few centimeters to the side. So a pretty significant effect at a microscopic level.
- 1.79** But the results are even as noticeable at a macro level as well. This is what it looks like from outside the body, where the patient has only been injecting their insulin at the same sites. The patient is told to rotate injection sites to prevent this exaggerated, admittedly odd-looking method of storing fat. Insulin is so determined to promote the fat storage. This is how good it does it.
- 1.80** All right, now, we introduced the different ways the body can store fat. The majority of us storing fat through hypertrophy, when we start gaining fat mass, and some storing it through hyperplasia, relatively more. Hypertrophic fat storage is an unhealthy method of storing fat. There are metabolic consequences to this. Whereas hyperplastic fat storage, while having other considerations, is a metabolically healthier way. There are fewer metabolic consequences. And let's talk about two reasons why this is the case. Two reasons why the hypertrophic adipocytes are problematic.
- 1.81** The first one is that the hypertrophic fat cells simply run out of room. Now, let's just do a little bit of biochemistry here for a second. Insulin, normally, when it is going through the body, will inhibit fat release from fat cells. So if insulin is high, the breakdown of the stored fat within the fat cell is going to be low. That's a typical response. However, when the fat cells become more and more and more hypertrophic, they reach this point of maximum dimension, a point where they can't go any further. There's no going

past it because the cell simply gets so big that it will start to, it can't hold itself together, and the membranes would start to rip apart.

- 1.82** So in order to prevent further fat growth, when the fat has reached maximum dimension, it becomes resistant to insulin's growth signal. Insulin is continuing to try to tell this fat cell to grow, and the fat cell basically starts to tell insulin, "Hey, look, I can't listen to you anymore. I can't do this anymore. I'm already as big as I'm going to get." And so, despite, against insulin's wishes or desired actions, the hypertrophic fat cell begins leaking fat. That becomes problematic.
- 1.83** Now, in contrast, if fat tissue is capable of undergoing hyperplasia, then as insulin is continuing to promote the storage of the fat, insulin goes up. Well, these are fat cells that can accommodate. And so if the fat cell starts to get a little big, maybe right around the time it would start to become resistant to the growth signal, we can simply adapt, and the hyperplastic fat cells begin to just produce more and more of the fat cells. Any time there's a potential pressure to start resisting, they just accommodate. They want to keep the body happy.
- 1.84** In a way, this is like a train, and we have passengers that are trying to board the train. So insulin is trying to pack passengers into the train cars as they're going to work in the subway. Well, in the case of hyperplastic, the hyperplastic magic train, anytime the train car gets full, not even bursting, we just hook another train car on. And then that train car gets full-ish, but not to bursting, and then we just hook another train car on. And so all the passengers, we continue to be able to accommodate all of the passengers that are attempting to board the bus or the train. And accordingly, everyone's happy because they can all get on, and everyone stays on, and they're perfectly fitted within the space.
- 1.85** In stark contrast, if we have a magic hypertrophic train, well, it's less magical. So in this case, insulin is trying to force the passengers onto this fat train. But we can't add cars to it. And so we have one train car, and insulin just keeps packing the passengers in. And the passengers start to say, "Hey, look, I can't fit anymore. You keep pushing people in." And so what some passengers start to do is, as the train car is loaded, they begin to sneak out, and they start climbing out through the window. Because as the insulin keeps pushing more people in, the people just start climbing out. Climbing, there's simply no room for all of them. Suffice to say, very angry passengers.
- 1.86** So this is, again, analogous to the fat cell that has undergone hypertrophy, failing to respond to insulin's signal to keep it all in, and just letting people go as insulin continues to push people in. Not a very happy situation. So that was the first one. The fat cells just run out of room.
- 1.87** On the second case, the fat cells start to suffocate, or the technical term is, they begin to undergo what we call hypoxia. They just aren't close enough. They can't get enough access to the blood to get the oxygen from the capillary. And the capillary is the

smallest unit of the blood vessel system. That is the unit where we start moving things in and out of the blood to feed the cells with oxygen and nutrients, and to take back the carbon dioxide and the metabolites from the cell, all the waste, if you will, from the cell or whatever products the cell wants to give. So it's the capillary, which is the action part of the circulatory system.

1.88 And in this case, when fat cells are small, they're all very, very tidily close to a capillary. Now, let me just shift this paradigm a little bit. So here at the bottom, we have a capillary, looking at it down the length. And we have some fat cells that are in close proximity. And indeed, a cell needs to be within just a few tens of a micrometer. You know about, immediately touching the capillary, practically, to maybe up to 50 micrometers away. So it needs to be in very, very close proximity for the gas exchange and the nutrients to be exchanged.

1.89 That's all fine and dandy when the cells are small, because you can see, as I've laid it out, not every cell is immediately circumventing the capillary. They may be one away from the capillary, but it's fine. Because the cells are small enough, they can stay in close contact. That is less benign when the fat cells have undergone hypertrophy. So here we have a hypertrophic fat cell, and this fat cell is lucky because, despite its mammoth diameter, itself being about 300 micrometers, it is close enough to the capillary to get what it needs.

1.90 But its neighbor is not so lucky. This neighbor, as they have expanded and expanded, has become pushed so far from the capillary now that it cannot access what the capillary is trying to offer. And so the distance becomes too great to overcome, and that far cell becomes hypoxic. It can't get enough oxygen to survive. So back to this view, we have the hypertrophic fat cells. They are insufficient in the amount of capillaries they have.

1.91 But the fat cell doesn't want to die. Like every cell, it wants to survive and will begin to undergo a process to try to correct this compromised blood flow. And it does so by making its own cytokines. In this case, the fat cell begins to act like an immune cell and starts to release these cytokines, a whole panel of them, a whole bunch of them, because some of them will increase the blood flow. It will correct the blood flow in the area. But others are simply promoting inflammation like cytokines do. But the fat cell doesn't care about that at the moment or at all. It basically just wants to correct the hypoxia.

1.92 So the poor fat cell, right? Let's feel bad for it because it's the two things, the two tools the fat cell has to try to ensure its own survival, end up being this perfect metabolic storm to metabolically wreck the rest of the body. The fat cell is reaching a point beyond which it cannot grow or it will literally break apart, and so it becomes insulin resistant to prevent the growth. As the fat cells continue to grow and they push each other further and further from the capillaries, from the life-giving blood, they become hypoxic.

- 1.93** And in order to try to correct that and create new capillary growth, it basically starts to release this trail of breadcrumbs in the form of cytokines. That when they get to the blood vessel, the blood vessel will follow the breadcrumbs with a new capillary, all to try to correct the blood flow. But at the same time, some of those breadcrumbs that get into the blood just start to promote inflammation throughout the body. And of course, inflammation is a primary cause of insulin resistance. So with the hypertrophic fat cell, with both of these mechanisms working to try to ensure the fat cell survival, the fat cell is leaking fat into the body when it shouldn't, because insulin is high, so fats should be locked in, and they're leaking cytokines, all to try to correct the pathology.
- 1.94** This combination of signal and stimulus end up resulting in the creation of something called ceramides. Ceramides are a type of fat, and they are necessary in the cell. But when ceramides start to get too high, they become problematic, pathogenic, harmful. And here is a different view of some of what insulin wants to do. Ceramides will directly disrupt some of these things. So we've gone right down to the molecular mediator within the cell itself. We've been talking about insulin resistance at a big macro level. Now we're talking about the actual molecular mediator of what's disrupting insulin. Insulin comes and knocks on the door, and there should be a series of actions. Ceramides are stopping the action from happening.
- 1.95** Now, ceramides themselves are produced from a fat called palmitate and an amino acid called serine, each of which on their own is perfectly benign. But when they are forced together, we have the synthesis of ceramides. Now, there are three points I just want to briefly make when it comes to palmitate, which is a saturated fat. We'll talk about saturation and fats later. But for now, palmitate is the most abundant fat within the blood. It's not because we eat a lot of palmitate. In fact, our consumption of saturated fats has been going down for decades. But that is because the liver is so good at making palmitate.
- 1.96** This is the product of a process called de novo lipogenesis, or the synthesis of fats from scratch, from new. But the primary signal for telling the liver to make the palmitate is insulin. When insulin is elevated, the liver is dumping palmitate out into the system, providing all of the building block for ceramides to be created later in the body or elsewhere in the body; and to really make this relevant, what are insulin levels in someone who has insulin resistance? Well, they're elevated, of course. So there's this almost constant pressure on the liver to be releasing these fats.
- 1.97** Now, having the building blocks for ceramides isn't enough. We need to actually turn the process on. And that's what brings us back to these three primary stimuli. When they are elevated, the signal goes from off—so palmitate and serine are just minding their own business— to being turned on. And all of the metabolic machinery or the enzymatic machinery suddenly fires up, and we're making ceramides like gangbusters. And then as this is coming from the fat cell, we begin to spread this throughout the body.

1.98 And we have consequences from top to bottom, whether it's the blood vessels, increasing blood pressure, infertility, compromised cognition, and more, some of which we'll get to later. Now, just to bring this back to these three tissues, because I haven't forgotten about them yet. When the ceramides start to accumulate in the alpha cells of the pancreas and in the liver and the muscles, that's what moves that. Where we had been, the patient had maybe just been insulin resistant, but now they start to get pushed all the way into full type 2 diabetes.

Lecture 2: Understanding Insulin Resistance

At a glance — Understanding insulin resistance and ceramides.

- 2.1** So we've been talking about how ceramides are contributing to insulin resistance across the body and how the fat cell is providing the signals in the substrate, the building blocks, for those ceramides. And as we talk about fat cells, this brings us into another very interesting point that I wanted to make. And I've been lightly touching on this, which is that fat has a finite capacity to store. And this is something referred to as the personal fat threshold. Every body has a level of fat that it wants to get to, or that it doesn't want to go beyond, is how I should say it.
- 2.2** Here we have it indicated on the fellow here. That is the threshold below which the body, if fat storage is below that threshold, metabolically speaking, the person is going to be sound; they're going to be healthy. Of course, we would define that as insulin sensitive, because whatever fat mass that person has, it's below their threshold. However, if a person exceeds the threshold, if the fat storage has gone beyond where that body wants it to be, now we've moved into the realm of insulin resistance for the reasons that we touched on before, which is, this is the point wherein the fat cells have reached their maximum size.
- 2.3** If we've reached that threshold, that's what we've gotten to. Whatever the capacity is to store more fat, we've reached it. Then any pressure to go beyond that is going to start promoting insulin resistance, all in an effort to try to keep the body below or at least at the threshold. But what if we could force fat tissue to store more fat? What if we could flip a switch, if you will, and take this hypertrophic fat and turn it into a more hyperplastic fat? There is, in fact, a mechanism to do this, which we'll talk about later when we discuss solutions to insulin resistance, but it's just appropriate to bring it in here.
- 2.4** So we take the big, excessively hypertrophic fat and just start producing more fat cells. Well, in this case, we don't do anything to changing the person's fat mass, but we just simply increase the threshold. They're as fat as they were before. The amount of fat mass they have on the body didn't change, if you will. Indeed, maybe it even went up, but the threshold went up as well. If they're below the threshold, and I'll just take it back, what might have been above the threshold and insulin resistant before, by increasing the body's potential to store fat, we push it all upwards in what was once an insulin resistant state, despite having the same or even more fat than before, becomes an insulin sensitive state, all because we moved the threshold.
- 2.5** Now, this has a very interesting real-world application. I did my postdoctoral work in Southeast Asia, where there are a lot of South Asians. This study looked at the fat differences between South Asian men and Caucasian men, so men of northern European descent or, again, South Asian descent. And they explored the differences in

fat storage in men who weighed the same. These are men of the same overall size and body type.

- 2.6** And they, firstly, the primary thing that they noticed was looking at the fat cells themselves. And remember, these are men who have the same amount of fat. But look at the difference in fat cell size. The image is a little, maybe too blurred, But you can see on the left, the fat cells from the men of South Asian descent are substantially bigger. They're about 20 or even more, 20 times bigger than the fat cells of their equally fat Caucasian counterpart. So, robust differences in how these two different body types are storing fat.
- 2.7** This might be part of what we see with regards to the prevalence of metabolic syndrome and prediabetes, which is insulin resistance in South Asia. That was actually the first— at the very beginning of this lecture, that first biggest little circle I put on the map was right there. This could very well be part of it. But this is an ethnicity that the South Asian men are tending to store their fat more through hypertrophy, whereas the Caucasian men have a slightly better hyperplastic phenotype or hyperplastic angle or aspect to their fat tissue.
- 2.8** Now, what does this then do to the fat threshold? If we have one population that is primarily just swelling the fat cells through hypertrophy and another who's capable of a relatively higher degree of hyperplasia, that means the South Asian fellow, his threshold is going to be quite modest because the fat cells can only get so big, and the fat tissue doesn't have the ability, obviously, to start stimulating more fat cells. So hyperplasia is clamped down.
- 2.9** In the case of the adipose tissue from the Caucasian subjects, their fat threshold is a little higher because they can produce some more fat cells to carry the metabolic burden. It's not all just being packed into ever-growing hypertrophic adipose cells or adipocytes. So this is the difference in the threshold. And to take that further, as they did in this manuscript, we see that even when the men have the same amount of body fatness, if you are above the threshold, no surprise. They found that the prevalence of insulin resistance was much higher in the South Asian population of the subjects within the study.
- 2.10** In contrast, despite having the same fatness, the Caucasian subjects were still insulin sensitive—again, despite having the exact same amount of fat. So the personal fat threshold is an easy, palatable way of understanding what is the body's potential to store fat. And then lastly, on this point, before we transition, what if the body can't store fat? Imagine a human wherein the fat cells can't be made, or if they were there, they were destroyed through some autoimmune problem. So we take all of those abundant, happy, healthy fat cells and just wipe them off. We just clear them off the table.

- 2.11** Well, if a person has no or very, very little potential to store fat, where would that move the threshold? It doesn't take a lot to get over it, does it? We basically bottom-out the threshold because they have no potential to store fat. After all, the threshold is essentially a way of saying, how much fat can your body store? If you can't make fat cells, you can't really store fat in your fat tissue. So the threshold is very, very low. But what can we do about it? Or what happens, rather, I should say.
- 2.12** This is a condition known as lipodystrophy. It is a real medical phenomenon. People are either born with the inability—it's congenital, they just don't make fat cells—or they develop it due to autoimmune disease. Regardless, what ends up happening is all of the fat on the body is gone. Everything. This includes the buccal fat, so they don't even have fat in their cheeks, so they look very, very sunken cheeks. They're very, very lean all the way through their body. In fact, they look like they are competitive triathletes or something similar. They're so lean. You see all the vessels throughout their body, the blood vessels. You see the striations or the curves of the muscle.
- 2.13** They look like they're athletes, but they're not. They just have an inability to make and store fat tissue. But they eat. They may eat a lot. They may eat foods that are spiking their insulin. They may eat an abundance of energy. But because they can't store the fat in their fat cells—there aren't any—the body is so determined to store the energy in response to insulin's signal, because insulin is so determined to store it, that even in the absence of fat, insulin will find other tissues in which it wants to store the fat.
- 2.14** And so rather than packing the fat into the fat cells, which are designed for fat storage, insulin begins packing the fat into tissues like the pancreas, developing fatty pancreas disease, or more commonly in the liver, developing fatty liver disease, the most common liver problem, which then leads into liver inflammation and then liver scarring and then liver failure. So there are consequences, indeed, catastrophic metabolic consequences, to not being able to store fat in fat tissue. And as much as we curse our fat cells, an absence of the ability of fat cells results in profound insulin resistance. Patients with lipodystrophy have very, very severe insulin resistance, very commonly to the point of true, outright type 2 diabetes.
- 2.15** Now, let's revisit a topic that I brought up earlier, which is this reconciling between insulin and the signal to store fat and calories, the fuel that gets stored as fat within fat tissue. I want to highlight—I feel compelled to highlight two of my heroes. These are scientific giants. They're famous, famous scientists, which means no one's ever heard of them because who knows about scientists, right? But they represent, they're my academic ancestors because of the lessons I've learned from them. They collaborated on a study in the early 1900s, a study of metabolism and severe diabetes.
- 2.16** Elliott P. Joslin is the godfather of modern endocrinology, wrote the textbook for it. The first one, in fact. Then Francis Benedict is the godfather of metabolism. Even now, students in academic settings will use the Benedict equation, an equation of determining metabolic rate based on just the size of the body. So they are legends in

their own spheres. But they came together to study the metabolism of people with diabetes, before it was called type 1 diabetes.

- 2.17** In this disease, people have no insulin, in type 1 diabetes. There is an autoimmune disease that has destroyed the beta cells of the pancreas. No beta cells means no insulin production. And they found that the metabolic rate in these patients was way too high, based on the size of the body, and size determines metabolism. That's one of the myths of metabolism. You'll have some scrawny little dude bragging about how much he can eat because he says he has a high metabolism. That is absolutely not true. That has nothing to do with it. Bigger bodies have a bigger metabolism than a smaller body, so the body size will determine the metabolism.
- 2.18** But in these people of very modest size, they found that the metabolic rate was much, much higher than it should be. And they weren't able to really explain it much because insulin was just becoming a thing around this time. But these studies were, in a way, followed up decades later. In 1984, a group further continued this work by actually, one, confirming that there is indeed a much higher metabolic rate. This is decades later, 60, 70 years later, 60 years later, where we had much better technology to quantify metabolic rate much more accurately than Joslin and Benedict did in the early 1900s. And yet they still came to the same number, roughly. That metabolic rate was about 20% higher than it should be.
- 2.19** Now, however, they had the benefit of treating the patient with insulin. When they gave the patients insulin, they noticed that metabolic rate immediately went to where it should. It slowed down immediately. When I say immediately, I mean it. Where, here on the right, I have the actual recreation of their figure, where the moment they injected the insulin, the metabolic rate started going down, minute over minute over minute. It was this, as I said, immediate effect.
- 2.20** And then it ended up getting the metabolic rate down to where it should have been, based on the size of the person. And this starts to reconcile these ideas, these competing theories, that are out there of obesity—one group saying it's purely endocrine-based, one group saying, no, it's purely calorie-based. We'll revisit that topic later in a bit more detail. But clearly, what I'm hoping to impress upon you is the fact that they're both relevant and that insulin has an energetic component. Insulin determines the degree to which the body is using energy. And insulin is so determined to store energy that it will slow the metabolic rate down. And in insulin's absence, metabolism is basically unchecked. It can just run and run and run, even if it shouldn't be.
- 2.21** All right, now, this phenomenon, that insulin promotes fat storage, is so obvious that it is a point of abuse in some people. Imagine, if you will, a young teen who has just been diagnosed with type 1 diabetes, and they've been finding over the months that they are just losing weight, getting thinner and thinner and thinner. That part might please them because of the societal pressure to be as skinny as possible.

- 2.22** But they will be very, very sick. They will have glucose levels that are through the roof. They will have ketone levels, something we'll explore in more detail later, through the roof. They're urinating all the time because of the high glucose, and they feel miserable, but they're very thin. Then they go to the clinic, having just been diagnosed with type 1 diabetes, and they're given insulin. They're given insulin over a few days in the clinic, perhaps in the hospital, to try to correct everything and get them balanced. They will have gained so much fat during those few days that they can't wear the same clothes that they were wearing when they came into the hospital.
- 2.23** Some diabetics will refer to their insulin shots as their "fat shots" because they learn, due to the obvious changes in their body, that it's the insulin that's making them fat. They develop an eating disorder, some of them, called diabulemia, which is this combination of some obvious terms just reflecting of an eating disorder of diabetes, where they learn that if they can—they can eat whatever they want, a big piece of chocolate cake. They can eat that entire piece of chocolate cake. And as long as they skip their insulin injection or deliberately underdose, they can be as thin as they want.
- 2.24** Now, they will feel miserable. There will be an early death, but they will look great in their coffin, to be crass about it. But they can have complete control. Can you imagine the temptation that would face this young teen? They can eat whatever they want, and all I have to do is not poke myself with this needle? OK, I think maybe I'll do that. And stay as thin as they want and eat whatever they want. So you can imagine the temptation in this instance.
- 2.25** So, in the absence of their insulin, they spike their metabolic rate. They start dumping energy from the body through their urine or through their breath, whether it's glucose and ketones together, even while they're eating plenty. Indeed, they may be eating thousands of calories more than they need to, based on their body size. And yet in the absence of insulin, the body has no choice. Back to the points I was making earlier, if insulin is down, there's no potential to store energy in the body. There's only excreting and wasting. Fat cells must shrink if insulin is low.
- 2.26** All right, now let's revisit inflammation now that we just revisited energy and insulin. I want to introduce a different type of fat. The fat that we had been talking about is a type of fat called white fat. That's what we've been focusing on. The fat that we can, and this comes in different forms, but its primary job is to store energy. It wants to take in fats and glucose, store the fat, or take in glucose and turn the glucose into fat, all to hold on to energy.
- 2.27** Fat cells are—it's like we have energy bars or energy drinks strapped like bandoliers across our body, and they're just waiting to be used, which is a point of interest for me. In the beautiful mountains of Utah, we're going on a hike, and I'll see people who have plenty of fat to burn, and yet they have energy bars or energy drinks. And I think, what? That's what all the fat is for. That fat is just waiting to be used. Well, again, the white

fat is built for storage, but we have a different type of fat on our bodies called brown fat.

- 2.28** They are indeed very different in color for a reason I'll mention in a moment. But in stark contrast to white fat cells, brown fat cells are built to burn fat. In anatomy or in cell biology, they're similar. They're both demonstrably fat cells, and yet in function, they couldn't be more different than each other—one designed to store fat, the other designed to burn whatever fat it's holding on to.
- 2.29** Now, within this realm of white fat, let's go one step further in our definitions and introduce subcutaneous versus visceral fat. Subcutaneous fat is the fat that we pinch and jiggle, the fat that we curse when we're looking in the mirror and we hate it. That is the fat that is just beneath the skin. That's what that word means, subcutaneous fat. In stark contrast to this external insulating fat, it's healthy to store fat there because if it keeps going out, if we keep storing more fat, it just goes out and out and out.
- 2.30** The visceral fat doesn't have that option. The visceral space, which is tucked beneath the subcutaneous, that's a finite cavity. There's a finite volume or space within that area. And so if that fat, which is woven through all of the tissues of this abdominal cavity, and there are a lot of tissues in there, it starts to create a pressure, if you will.
- 2.31** Now, both of these fat depots are capable of undergoing both hypertrophy and hyperplasia. But in the case of the visceral fat cells, there's a problem, which is, again, the volumes. And we have these tissues, that if the visceral fat were expanding, it would physically start to compress our intestines. It would physically start to compress the kidneys, the lungs, the heart. It would make it hard to breathe. It would make it difficult for the heart to beat and move the blood around. And so naturally, it's no surprise that the body wants to try to limit the amount of visceral fat.
- 2.32** Now, that is not to say the visceral fat is all bad. It isn't. It does provide a structure and a form to the viscera, which is a messy area. It does help things stay in their place. It also plays a part in the immunity, surrounding the intestines in order to make sure whatever is coming out of the intestines is cleaned up as needed. There's a functional role for it. But the problem is, of course, when it starts to go too far, where, again, it begins to compress on the neighboring tissues, which are all essential in their function.
- 2.33** Because of its limited space, which type of fat growth do you think it undergoes? A limitless hyperplasia, where it can always make more and more and more fat cells? Or a limited hypertrophy? Yeah, the latter. Visceral adipocytes will selectively grow through hypertrophy because they inherently know, "I can't get too big. If I get too big, I will start to crush the kidneys or I will start to compress the intestines, and now nothing can move through it." And so the visceral adipocytes know, "Whatever pressure I have to grow, I have to grow through hypertrophy because then that will start to limit my growth."

- 2.34** But indeed, we've talked about some of the consequences of that limitation, where it will start to leak out the fats, and it becomes heavily loaded with cells called macrophages. That's the first time I've introduced a macrophage. A macrophage is the poster child of immune cells or white blood cells. We'll talk more about the macrophages in a future lecture, but here I've introduced them. Macrophages are heavily prominent through the visceral adipose for the reasons I just mentioned.
- 2.35** Visceral adipose actually plays a part in kind of making sure things are clean. When you have blood coming from the intestines, what if some of the blood from the intestines have been loaded with bacteria that shouldn't be there? Well, it will flow through the visceral fat, and those macrophages can clean that out. That's a good functional reason for it. But as the fat cells are swelling too much, and the visceral fat's growing too much, we have an overabundance of macrophages which begin contributing to cytokine production, and now inflammation is aggravated as well. Hence, me bringing this up in the inflammation topic.
- 2.36** So, one last point on this. This is the primary reason why waist circumference is a relevant marker. And later on, in the time we have together, we'll come back to some of these points. But measuring the adiposity, or the fatness of that middle area, is a good indicator of how much fat you're storing viscerally. As important as it is, it's just as important that we keep it in check.
- 2.37** Now, I mentioned white fat stores energy, brown fat burns it. It wants to chew through whatever energy it has, all for the purpose of creating heat. And the color is a reflection, well, quite literally, as light is reflected back to us, of what is in the fat cells themselves. White adipose tissue is white because it basically is filled with, like, coconut oil or, like, shortening. You look at, you know, you're cooking with a cooking, kind of physical oil, like a coconut oil, and it's very, very white. White adipose tissue is white because almost all of it is made up of this huge big bubble of fat.
- 2.38** The other organelles of the cell, like the nucleus or the mitochondria, are very rare. There's very little of it. It is literally pushed right up against the edge, just how I have it shown here. Again, the bulk of the cell is made up of this huge, big lipid droplet that looks like a big bubble. Of the fat images I've shown you before, you'll notice that they were big and white and clear; it's because it's just a big bubble of fat. You can barely see anything else.
- 2.39** That is not at all the same as brown adipose tissue. The brown adipocytes are brown because of the heavy, heavy prevalence, particularly of the mitochondria. Mitochondria have inherently a reddish-brownish color to them, and there are so much mitochondria in the brown adipocytes that it reflects that darker color. And indeed, it is a difference. If I had a sample of white adipose from a biopsy or brown, I could hold it in my palms, and you would immediately tell the difference, all because of the varying composition of the cells themselves.

- 2.40** And mitochondria, you're familiar with from just Biology 101. It is the powerhouse of the cell. The mitochondria are the part of the cell that is responsible for taking in nutrients like glucose and fats and turning it into usable energy for the cell. So it's basically, it takes the fuel; it's the engine, allowing the fuel to come in and then providing the workable molecule for the engine, for the cell to do what it needs to.
- 2.41** One further point, though, to make is that, while it has a massively greater amount of mitochondria, these mitochondria are unlike any other mitochondria in the body. They have something on them called UCP1. This is an uncoupling protein. I'll talk about the uncoupling in just a moment. But suffice to say, it separates the function of the mitochondria from its two parts of itself.
- 2.42** Now, to show this in another way, white fat is designed to store energy. So what do you think its metabolic rate is? If it's designed to store it, it doesn't want to burn it. No surprise, white fat cells have the lowest metabolic rate of any cells we've measured in my lab. I'm unaware of any cell that has a lower metabolic rate. It is very, very low. You can barely detect it with the fine instruments that we have. But it is breathing, it's living, it has a metabolic rate, but it's very, very low.
- 2.43** The brown fat cells have a metabolic rate that is easily 10 times higher. Indeed, its metabolic rate is comparable to muscle. It looks a little more like muscle, too. It's really darkish, brownish red, although muscle is just more pure red, but they both have a lot of mitochondria. So a much, much higher metabolic rate. That's not surprising. The brown fat cells are designed to burn energy. And so it's no surprise that they burn energy a lot faster. Indeed, up to 10 times higher.
- 2.44** Now, for many, many years, up until very, very recently, I drove a 5-speed manual transmission, a little old Subaru, and it got me around wonderfully. I would park on the college campus, and I'd never lock my door because no one could steal it because no one can drive stick shift anymore. There I had my three pedals, and this is particularly relevant, where I could be pressing the clutch and/or I could be pressing the gas. I'm using the analogy of the engine here, with the speed of the vehicle. How much work is the vehicle actually doing—that's the movement of it— versus how hard is the engine actually working? Those are not the same thing, particularly in a manual transmission.
- 2.45** So with white fat tissue, these are very coupled processes. It's a very, very low speed. The car isn't really moving. Or in other words, the cell isn't really doing a lot of work, and that is reflected in an engine that is idling very, very low. But again, appreciate the difference. This is the amount of work the engine is doing—how much fuel are we burning as we're revving the engine?—and then how much work are we actually getting? How much are we moving the car, or how much is the cell doing from what it's burning?
- 2.46** Now, on brown fat, it's very different. While the speed of the brown fat, the actual work the brown fat has to accomplish, is very modest, it's not like muscle cells. My

muscle cells are up here moving me around. I'm controlling my breathing, I'm controlling my movement, my articulation and gesturing. But in the fat cell, it doesn't have a lot of work, but it's designed to create heat, which is perfect because the engine, while not moving the car, is very, very active. The RPMs are being revved and revved and revved. In this case of my old manual transmission Subaru, I have my foot on the gas, but I'm also keeping my foot on the clutch. I'm not moving.

- 2.47** The cell isn't working, if you will, but it's sure burning a lot of energy to get me nowhere. That's what the brown fat is doing, and it can do so because of the uncoupling protein. The uncoupling protein has taken the mitochondria and told the mitochondria, "I want you to burn energy, but I don't need you to create any workable molecules that the cell needs. I don't need any of that. I just want you to burn. Just burn, burn, burn. We don't need to go anywhere. We don't need to get any work done."
- 2.48** That means that this engine, the metabolic engine of the white fat, needs very, very little fuel. It's not working a lot. It's not burning a lot of energy. The RPMs of the engine are very low. It's a very low idle. In contrast, the metabolic engine of the brown fat is revving, revving, revving. So we got to fill that engine up all the time. Even though the car is not really moving, that's not the point. We just want the engine to heat up, and that's a great way to do it.
- 2.49** Now, of course, we aren't engines in this sense, and so we don't use fuel from a jerry can or a gas can. We have the metabolic fuels, like glucose and fat, and this is generally reflective of what would be happening. The very, very low metabolic demand of white adipose is reflected in a very low demand for glucose and fat. It needs very, very little energy to do what it needs to do to survive. In contrast, the brown fat tissue has such a high metabolic rate that it needs a lot of energy. And in particular, as it's burning a lot of glucose, just to create heat from it, the glucose levels can come down and so too can insulin. So, it's no surprise that brown fat tissue can be helpful with regards to metabolic issues.
- 2.50** Now, let's just create a bit of an interesting story here and view. This brown fat is, as I jokingly call it, calling it in the tangent topic, a hot fat. It creates a lot of heat. And that is in response to temperature, of course. It's no surprise that if the body is starting to get cold, brown fat wants to try to warm the body up. It is interesting to note the changes in fat mass, white fat and brown fat, in the human life as we get older. When we are babies, most of the adipose tissue we have is, in fact, a browner adipose tissue.
- 2.51** And that's because when the baby gets cold, baby has so little muscle, the baby can't really shiver very well. It's mostly just a big, beautiful ball of chubby fat. There's not a lot of muscle there. Again, if baby's cold, there can be no shivering-induced thermogenesis or the production of heat through shivering. When you're working a muscle, it heats up the body. As we get older, of course, we have more muscle, and because of that, we need less brown fat. Because if we get cold, we shiver and it

increases our body temperature very well and very quickly. Baby doesn't have that option.

- 2.52** Now, as a metabolic scientist and father, I have been delighted to pay particular attention to that. That when my darling little cherubs would be coming out of the bathtub, I noticed they wouldn't shiver when they were chubby little butterballs. Then as they would get older, maybe even one year old or older, they would start to shiver. As their muscles would get bigger, then the fat, the brown fat, would start to subside.
- 2.53** Now, that is not to say we don't have any, despite what my overly simplistic view suggests on the bottom. In adults, we have pockets of brown fat. It's primarily relegated to the thoracic cavity. The reasons for that aren't known. Why store fat through the rib cage up into the clavicles? It's a weird area to store this very metabolically active fat. It could be simply, as everywhere else can shiver to stay warm, the brain can't shiver. And so if you have a bunch of hot fat surrounding blood vessels that are about to go up to the brain, it's a good way of making sure that the blood going to the brain stays warm, in case there is some thermal challenge to the body, if the body is starting to get cold.
- 2.54** And given how metabolically active brown fat is, and how readily it chews through glucose as a fuel in particular, it is no surprise that people, adults who have more brown fat are more resistant to metabolic disorders like insulin resistance. They will have an easier time resisting weight gain and resisting this progression towards diabetes or just metabolic disarray.
- 2.55** All right, now another tangent. This will be the last one for a minute, and that is to look at fat as an endocrine organ. But it's a tangent, but it's still related to insulin resistance, of course. Fat is a very active endocrine organ. It produces dozens of hormones. Once upon a time, we would have only looked at organs like the pancreas or the thyroid or the gonads, like the testes and ovaries, as endocrine organs. But now we know that the fat is a very active endocrine organ. It's producing dozens of hormones and releasing them into the blood at every moment of the day.
- 2.56** And these hormones have myriad effects influencing blood flow and inflammation, as I've touched on, bone growth, mitochondria, signaling to the body when it's time to stop eating in the form of a satiety signal, and even directly influencing insulin sensitivity. And that hormone is primarily a hormone called adiponectin, derived from the term adipose. It is made from adipose tissue, from fat cells.
- 2.57** Adiponectin acts by attacking the attacker. So if a body, if a signal within a cell is stimulating the cell to start creating ceramides, adiponectin will come in and knock the ceramides back. It will literally activate enzymes to degrade the ceramides, to remove them from the cell, thereby enhancing insulin sensitivity. It's a very direct effect. But tragically, as fat cells start to get bigger and bigger and bigger, the production of

adiponectin goes further and further down, amplifying the insulin resistance that is caused by the hypertrophic adipocyte.

- 2.58** Now, in addition to adiponectin, there's another hormone, one that you have heard of, I'm sure. No one's heard of adiponectin. Most people have heard of leptin. Leptin is a hormone that induces satiety. That's what made it famous. Although there's a very interesting backstory to that. But leptin will, is one of the signals, one of several, that will tell the brain, "I'm done eating. I'm going to stop eating now." It also induces the synthesis of mitochondria. That's one of leptin's lesser-known effects. It will signal to cells of the body, "Hey, make new mitochondria to just burn energy."
- 2.59** But while leptin production, in contrast to adiponectin, adiponectin production will drop as the adipose tissue is growing. Leptin production will go up. As a person has more adipose, they have more leptin. And as the person keeps accruing adipose tissue, they keep accruing leptin. And so the body is flush with leptin, a lot of leptin. Well, we've talked about, in the time we've had together, an instance where a lot of a hormone will result in a resistance to that hormone, namely insulin. Do you think it's any different in this case? No.
- 2.60** Just like we have insulin resistance, which can be caused by chronically elevated insulin, we have leptin resistance that can be caused by chronically elevated leptin. So whatever benefit the leptin might have been providing with regards to satiety or mitochondria, that benefit starts to go away. To make it even worse, there is an interplay. There's a bit of an abusive relationship between leptin and insulin. When leptin is being produced from a fat cell, it will signal to the beta cells of the pancreas to produce less insulin. That's nice when insulin is working well, right? If insulin is working well, it helps keep insulin down, and the body can be burning fat, and it helps us feel more full.
- 2.61** However, as the signal starts to break down, as we have too much leptin as a result of ever-growing fat tissue, what was once an inhibitory signal, leptin attempting to reduce insulin production, becomes an inert signal. Insulin, in the absence of this inhibitor, this bully who is trying to keep it in check, it begins to have its day in the sun. It has lost this inhibiting signal, and insulin levels will start to climb, of course, aggravating insulin resistance.
- 2.62** OK, now, up until this point, with a few little tangents, we've been talking about the primary causes of insulin resistance. These fundamental noxious stimuli that challenge what the body wants to do—how much insulin it's making, how well the insulin is working. There are secondary causes of insulin resistance that I want to lightly touch on. And these are ones, I call them secondary because, despite the known contribution, they aren't validated across all the models that I discussed. I don't want to imply that that makes them less relevant. Indeed, I think they're both very relevant. It's just a different level of scientific support for them.

- 2.63** So I feel compelled to mention them, but I also feel compelled to put them one step down. And that is uric acid, which is a molecule the body will make that we'll talk about. It's primarily a problem— we typically think of it as only being a problem in gout, when a person starts to have an inflamed and swollen toe, as the uric acid crystals will accumulate in different spots around the body, causing an inflammatory response and causing great pain at joints. And then lastly, or the second part of this, is the seed oils, the refined seed oils. We'll define each of them and explain them in a little more detail.
- 2.64** Now, let's come back to this view, the hyperplastic fat cells versus the hypertrophic fat cells. What if we took a body of fat that was perfectly happy with the amount of fat it had because it could undergo hyperplasia and create new fat cells, but forced it to go through hypertrophy? You could imagine that would have metabolic consequences, and they wouldn't be good. This is what refined seed oils are capable of doing and why I believe they're very relevant when we talk about insulin resistance, because the fat cell is just so fundamental to the metabolic descent, into the metabolic madness of the body.
- 2.65** In this case, when refined seed oils, when we consume these seed oils, and I'll talk about them, I'll define them in just a second, the primary fat we're actually getting is a type of fat called linoleic acid. Now, linoleic acid has been a champion of conventional nutrition, the conventional dogmatic dietitian's approach, because it's a polyunsaturated fat, something we'll get into more detail later. But in this case, the linoleic acid has two fates.
- 2.66** A modest fate is that, like every fat, it has the potential to be burned. And so a fat cell or any cell can burn the linoleic acid, this primary fat that is in these refined seed oils. But the body will more often convert this linoleic acid into a molecule that we'll just call 4HNE. This is what's called a lipid peroxide. But when you think of—you've heard of oxidative stress. This is basically the bastard child of an unholy union, where you take linoleic acid and oxidative stress, and they give birth to this rotten little molecule called 4HNE.
- 2.67** And among 4HNE's horrific consequences is this forced pressure. It limits the fat tissue's ability to make new fat cells, which sounds nice, doesn't it? You know better. If we have a limited potential to make fat cells, and yet there's a pressure to store fat in fat cells, then we're forced to go through hypertrophy. And these seed oils, it might be—we might be inclined to think, well, it's not a problem for us, but these are what they are. And we eat them remarkably common, as I'll show you more in a moment. But things like corn oil, soybean oil, canola, what are commonly called vegetable oil. There's nothing vegetable about them. Strictly speaking, there's no such thing as an oil from a vegetable. These are seeds, and they have a direct effect at the fat cells' ability to store fat.
- 2.68** Now, what about uric acid? Uric acid has become a problem primarily because of our consumption of a type of carbohydrate called fructose. Now, as the name suggests, this

is a natural carbohydrate that comes from fruit. Basically, if you taste something sweet in nature, it's the fructose that gives it the sweet taste. But whenever we consume fructose, it is a carbohydrate, it is an energy source, and the body can burn it for energy. But every time we send the fructose down that biochemical pathway—here's the fructose, and here's the end of it being, I'm burning you for energy—there's going to be a byproduct, and that byproduct is uric acid.

- 2.69** Every fructose molecule that we're burning, we're making a uric acid molecule. Then the uric acid accumulates throughout the body, and it promotes inflammation aggressively. We're doing further studies on this in my lab even now to better understand what is the mechanism whereby uric acid promotes inflammation. But we know that it does. It's been very well documented, and we make it worse because of how we eat.
- 2.70** Now, indeed, as we're nearing the end of this time together in this distinct lecture, why has this become so common? We've defined what insulin resistance is, and we just got done thoroughly exploring its origins. So what is the backstory? Where does it come from? Now, let's just, as we near the end, talk about why it's become so common. In order to explore the commonality, the prevalence of it, I want to highlight these distinct causes again—the hyperinsulinemia, the refined seed oils, the stress in the form of sleep deprivation in particular, and then the uric acid. Why have all of these—why do I believe all of these are such massive contributors to the insulin resistance and the metabolic storm that we're suffering around the world?
- 2.71** So with hyperinsulinemia, let's just look at what we eat. Within the human diet, there are three main parts or three big parts of the diet, or macro parts, what we call macronutrients. And this is fats, proteins, and carbohydrates. Let's look at what's happened over time first. This was a study that explored eating patterns in the amount of macros people were eating from this period of time, '71 to 2000. They found, over these few decades, that the consumption of fat trended to go down a little. It had been around almost 40%, and by the end of it all, it was a little closer to 35 or even 30. So, this downward trend of the amount of fat people were consuming as a percent of their calories.
- 2.72** And then they looked at the amount of protein people were eating as a percent of their calories, and that went down a little as well. But they didn't all go down, in large part due to the food guide pyramid and the dietary guidelines that we've been given around the world—it's all reflected in this—we ate more and more carbohydrates than ever before, very likely. So the actual amount of the diet that was constituted by carbohydrate just climbed the entire period.
- 2.73** That has a relevance to the topic of hyperinsulinemia, because when we look at what happens to insulin in response to these macronutrients, we can start to get a better idea of just why this has become such an issue. If a human eats pure fat, which few people are inclined to do, but it happens nowadays, where people will add some kind

of oil or some fat into a drink like coffee or something, there will be no effect on insulin. This is a direct reproduction of this old open-access study. So no effect on insulin. At no point did it reach a degree of statistical significance.

2.74 If someone eats pure protein, which people are very inclined to do nowadays, they may have a modest effect. At no point did it reach statistical significance, but some people can have a bigger insulin effect than others in response to protein. If someone eats pure carbohydrate, which is not hard to do at all, it's very easy to do, most carbohydrates in nature come as pure carbohydrate, then we have a robust insulin effect. You'll notice there are two phases. We'll revisit that later. But suffice to say, insulin was low and it will easily go up 10 times. Easily. That would be even in a normal.

2.75 Indeed, this would be the insulin response to carbohydrate, to a glucose load, in a healthy insulin sensitive person, but it doesn't look quite as good in an insulin resistant person. So this, again, that lower line is a person who has insulin levels or insulin that works very well. Glucose has come up, insulin will come up, it will resolve the glucose, and insulin will come back down. But if you have a body where insulin doesn't work very well, then you take what was a modest curve, and now you not only spike the glucose higher than before, or the insulin, sorry, but it'll stay higher for much longer.

2.76 It took the healthy person about three hours for their insulin to come down. An insulin resistant person can easily take up to five hours, from that point of consuming the glucose to the point of the insulin getting back down to where it was. In other words, back to the very beginning of our time together today, the time spent in sugar-burning, the fed state, can be almost twice as long as it is in a healthy insulin sensitive person. So that much longer before they can enter the fasted state and get back to fat-burning.

2.77 So to show this another way, the insulin effect would look a little something like this, but not all carbohydrates are created equal. The study that I showed you used pure glucose, almost like a piece of bread. Well, of course, you can have more complicated things. Eating a piece of bread is not the same as eating a plate of broccoli, which has very little digestible carbohydrate. So not all carbohydrates are created equal. But just to further impress upon you how relevant this is, the issue of hyperinsulinemia—because it might be tempting. Someone might be tempted to think, "OK, thanks, Ben, hyperinsulinemia causes insulin resistance, but I don't have hyperinsulinemia." You probably do.

2.78 This is what an average person would—in fact, an uncommon person, their levels, their insulin levels would look like this. I say uncommon because one, this assumes the person is insulin sensitive, and two, it assumes the person is only eating three times a day. Well, we live in this insane culture where we think it's appropriate to eat every minute of the day. And so while a person eats some junk breakfast, which is pure sugar and starch, two hours later, they think they need a mid-morning snack.

- 2.79** And so then, right as the insulin would be about to come down, they spike it right back up. And then after their mid-morning snack, as it's coming down, they do it again for lunch. Then they do it for their afternoon snack and then their evening snack. And so even an insulin sensitive person is essentially spending every waking moment in a state of elevated insulin.
- 2.80** Now, what might this look like in an insulin resistant person? Well, it's a lot worse. It goes much higher even, and it stays higher throughout the day because it doesn't even have time. Insulin doesn't even have time to start curving over the top, over the spike, and coming down into the valley of the insulin, off the peak. It is basically peak all day long, further compounding the problem.
- 2.81** And there was a study that was done a number of years ago that had healthy subjects just eat a lot of carbohydrates for one week. Now, I'm showing you here the glucose level. It's not in scale to insulin, but over the course of this overeating carbs for one week, you'll see the glucose levels stayed the same. These are fasting levels. So they'd be eating a lot of carbs during the day. And then at the point of the blood tests, they were fasted that next morning.
- 2.82** Now, however, you are clever enough to know that just looking at glucose doesn't tell the full story. We need to look at insulin. And when they looked at fasting insulin levels each morning in these patients, you could see over the week, from day zero to day six, that insulin levels were 2.5 times higher in a fasted state. This, of course, right here, this endpoint is insulin resistance. Not type 2 diabetes, because remember, type 2 diabetes is high glucose and high insulin. Insulin resistance, which is the state that precedes type 2 diabetes, is high insulin, 2.5 times higher or more, but normal glucose.
- 2.83** So within just a week of overeating carbs, we'd made these healthy insulin sensitive people 2.5 times more insulin resistant than they were at the beginning. Now, what does this do to fuel use? If every moment now, even in a fasted state, their insulin levels are 2.5 times higher, well, that, of course, will influence the fuel the body burns.
- 2.84** Now, I alluded to this, but I just wanted to put a fine point on this with a funny little analogy, and that is a fuel truck. So here we have this big fuel truck rumbling down the road, and there's a lot of mass to this thing. And so it has a big engine, and it is a hungry engine. Now, you look at the fuel truck, and what fuel—where's the fuel coming from that is feeding that engine? Well, it's coming from this little tank right here. That's the tank that is feeding that engine.
- 2.85** But ironically, much of the fuel needed, much of the mass that this thing is driving down the road, is this huge tank, which is filled with fuel, but it can't use it. This truck is forced to rely on using this small little tank of fuel, all in an effort, in part, to haul this massive amount of fuel around, but it isn't able to tap it. Imagine, wouldn't it be genius if this truck could just shift from this fuel source to this fuel source, and then it could

drive for days, for weeks. It'd never run out. But as it is, only relying on that small tank means that the truck has to stop every few hours and fill back up.

- 2.86** You can see where I'm going with this. So in this case, we have the small tank, but we have that big one that's just waiting to be used. This is analogous to the human body, where we may have, at any moment, about 2,000-ish calories stored as glucose within the body. Now, the liver can make glucose. We'll revisit that later. But we have at any moment, right now, we've got about 2,000 calories of glucose. In contrast, we can easily have 100 times more, even in a lean person, have 100 times more fuel on their body stored as fat. We got a lot of fat on us, even lean people, and it's capable of driving us down the road for miles.
- 2.87** We just never tap it. Because of our high insulin, the chronically elevated insulin means we're always relying on the tank that we have to stop and fill up. So we have breakfast and then we snack and then we have lunch and then we snack, because we always feel like we're running out of energy. Imagine, though, the metabolic advantage, if we can lower that insulin through some dietary changes, then all of a sudden we replumb the fuel lines.
- 2.88** And what was once a metabolic engine that was only taking the fuel from the glucose, now can begin tapping into fat. That, after all, is what it's there for. These are energy bars and energy drinks just waiting for us to open the package and consume them. But we can only do it when insulin is low. And of course, when we do so, the fat cells will shrink.
- 2.89** Now, let's talk about seed oils very briefly. It might be tempting, just like people would say naively, well, "Hyperinsulinemia isn't a problem for me," some others may be tempted equally in their ignorance to say, "Well, I don't eat seed oils. I don't eat refined seed oils." Yes, you do. This is a study that was published through the NIH and that looked at the consumption of fats through almost a century, almost 100 years, within the United States. And they found that pork and beef, these much maligned, vilified fats in the human diet—we shouldn't eat these; they're red meat—you can see that the consumption of pork, which used to be the main source of fat, it dropped the entire time. I mean, a temporary little bump, but it went down.
- 2.90** Beef, as much as people are now afraid of beef and telling us we should never touch it, well, we've responded. Sure, it went up for a little while in the early '70s, and then the food guide pyramid came out and boy, boom, it dropped like a rock. You can see, over this roughly 100 years, as I've reproduced this, we're actually a little lower than we were about 100 years ago. So what has made up for it? Well, one is shortening. And shortening is a bit of a hybrid fat, where it is a cooking fat that has been hydrogenated to be more solid. It used to be primarily made of lard, but around the mid-1900s, when the fear of saturated fat came to life, then it started to be replaced with refined seed oils, like cottonseed oil in particular.

- 2.91** You can see that beef and pork used to be higher consumed than shortening, which itself was from a different source. Now, shortening has beat them both out by the end of the century. Now, even more so, we look at soybean oil. Soybean oil was a new product that used to be used completely as an engine oil, if at all, a lubricant industrially, and it went from literally being nothing in our diet. We didn't eat soybean oil at all to being the number-one consumed fat. Indeed, soybean oil is the most consumed fat in our diet. We eat more of our fat calories from soybean oil than any other fat in the diet.
- 2.92** So, don't be too smug if you think you avoid it. Basically, if you're eating foods that come in bags and boxes with bar codes, if these are crispy, crunchy carbs or sweet and gooey things, very, very likely the fat in it is going to be from soybean oil. Now, uric acid. I'd already mentioned to you whereby the metabolism of fructose increases uric acid. Every time we burn a fructose molecule, we produce a uric acid molecule as a byproduct. Fructose is one of those things that, like soybean oil, we used to have little or none of.
- 2.93** Well, then I'm only looking at the fructose in the form of high-fructose corn syrup, which we've all heard of. It's just a slight difference of sugar, where there's relatively more fructose than there is glucose, the other part of a sugar molecule. So this is a form of a sugar molecule, but more fructose than normal. Of course, as humans, we've been eating fructose in the form of fruit since time immemorial, maybe even indeed from the beginning. But an apple at the beginning isn't what an apple is now.
- 2.94** We look, in our hubris, in our genius, we've taken something that used to be a small, little bitter fruit and turned it into this big sugar bomb. And they're delicious, but we've changed these plants to enhance the fructose in them. So as much as I could attempt to look at the amount of fructose across all fruits, I'm not going to. We'll just look at high-fructose corn syrup, which used to be absent in the diet to now. At least 20 years ago, it's only gone up. We were consuming up to 60 pounds a year, and indeed, that has gone up. So uric acid is a problem in large part because of our fructose consumption.
- 2.95** Now, sleep deprivation. I have alluded to this. We all know, you know of my sympathy for people who sleep terribly, but we all know how difficult it can be to get a good night of sleep. And for you younger people in the audience, if you don't know yet, you will probably. So there are two things I wanted to highlight with sleep deprivation. Both of them you've heard of, to varying degrees. One is blue light and the metabolic impact of that, and then the other one is hyperglycemia or elevated glucose levels.
- 2.96** Now, with regards to blue light, blue light is a natural part of a full spectrum. So when you're out looking at the sun, when the sun rises, you went from a blue light-absent environment to one that is filled with blue light. And then later, but nowadays, we have an additional source. It's not just the sun, we also have screens. Electronics will emit with a blue light. However, the electronics don't play quite as nicely as the sun does, because whereas the blue light from the sun will contribute to help us stay awake, and

it is a very good thing to help us wake up and have a normal rhythm to our days and nights, when the sun goes down, and it is replaced with a much dimmer reflective light of the moon, then we don't have that blue light stimulus.

- 2.97** But of course, the cell phone doesn't operate by the physics and astronomy of the world. In the universe, the sun has gone down, but the cell phone has come right up into the face, and that is capable of stimulating. It's eliciting such a powerful stimulus of blue light that we can quantify the metabolic effect. And indeed, the blue light from electronic devices will stimulate insulin resistance. This is something that can be tracked, and it can be thankfully quite remedied with any blue light blocking.
- 2.98** If you need to look at a screen, and we'll touch on the solutions to insulin resistance more later, but just as an obvious relevance here, blue light blocking will completely mitigate the insulin resistance that can be caused by blue light exposure in the evening and night. Now, with hyperglycemia, this was the average consumption of a person. Remember, I was tracking insulin here, but the insulin spike, the cause of it is glucose consumption or carbohydrate. So the person is eating all these carbs all day. But let's focus at the end of the day, which unfortunately tends to be a time where people indulge the most.
- 2.99** They're home, they're back from work. For me, the house has finally gotten all tidied up. The kids are all in bed. And that is like the witching hour for me, and I'm sure it is for most people. I could look at an indulgence, a salty, crunchy or a sweet and gooey thing all day, and it won't tempt me even the littlest bit until that moment of time, those couple hours before bed. And then all of a sudden, it is a Herculean effort for me not to just indulge. Even though I know, as much as I know, I still have to battle this temptation, but everyone will face it to some degree.
- 2.100** But what happens in the evening, of course, we indulge in something salty and crunchy or sweet and gooey, which means it's going to have a lot of carbohydrate in it. And we go to bed in a state where not only is our insulin elevated, but our glucose is elevated. In this case of sleep deprivation, it's the high glucose that I want to point out. The most common cause of insomnia is body temperature. The person is too hot. That is the leading cause. There are a lot of causes. That's the most common.
- 2.101** It is a fact of physiology that when blood glucose levels are really high, it activates what's called the sympathetic nervous system, something we'll define in more detail later. But that turns the body on. That's the fight-or-flight response. We don't want a fight-or-flight response when we're going to bed. One of the consequences beyond the beating of the heart being faster and harder will be a higher body temperature. Glucose levels spike body— when glucose comes up, body temperature does, too. This is a very well documented phenomenon. Happens in diabetics and happens in non-diabetics.
- 2.102** Now, the final point that I want to make, as we wrap up with this lecture, the last few minutes, I just want to mention that there is another side to this. As much as I have

been talking about insulin resistance as a true villain of human health, and it is— it is one that we should be wary of and we should do our best to challenge in combat— there are instances when insulin resistance isn't so bad. The insulin resistance that we've been talking about, we could call pathological insulin resistance. It's pathological. It hurts the body. It's what's causing all of these problems that we're going to get into in more detail later.

- 2.103** But there are instances throughout human development, in all of us and in some of us even more frequently, called physiological insulin resistance. And this is when the body has become insulin resistant for a purpose. It is helping the body do something, and that something is, uniformly, growth. Any of the periods of human development where there is substantial growth, there will be insulin resistance, in part to fuel that growth and to help the growth work and to occur in the proper way.
- 2.104** This, no surprise, is in large part influenced by the hormone growth hormone. Growth hormone is the most famous of all the growth signals. But what people don't appreciate is that growth hormone, one of its actions is to increase glucose. It wants to increase blood glucose so that it can then direct it to fuel growth. But while growth hormone is trying to increase glucose, once again, who is trying to push it down? Insulin is trying to combat that glucose stimulus from growth hormone, and so the two end up at odds. And so, if you have a stage of human development where growth hormone is very, very high, it will contribute to insulin resistance.
- 2.105** But the combination, remember, this is not a pathology. This is not a problem, because that combination of the hyperinsulinemia of insulin resistance with the growth hormone results in substantial growth, because remember, not all tissues become insulin resistant in an insulin resistant state. Some stay insulin sensitive, but you now have the hyperinsulinemia of the insulin resistance, combined with the growth hormone, which reflects the substantial growth during this period, and you do indeed get a wickedly wonderful combination in this case of promoting substantial growth.
- 2.106** There are two times in the life of a human where this becomes very, very significant, and they are the two Ps of physiological insulin resistance—puberty and pregnancy. Of course, puberty applies to both sexes equally, pregnancy is unique to one. With that idea in mind, but regardless, in both instances, these work together to promote growth. It is a physiological insulin resistance. But relative to the fairer sex, we include progesterone. Because progesterone also, particularly during pregnancy, of course, it is after all the hormone of gestation, of helping that body grow another new little body, but even in puberty, of course, in females, where progesterone becomes part of a normal ovulatory cycle.
- 2.107** That's what I want to end with now, this last tangent of the day, of this lecture, which is looking at progesterone and its effects metabolically outside of those two periods of physiological insulin resistance, outside of puberty and pregnancy, because progesterone is a part of the normal ovulatory cycle, where here we have the cycle

outlined on the X axis in days, and we have the point of ovulation right in the middle. Right after ovulation, we have this big spike in progesterone, whereas it had been very, very low up until then.

- 2.108** It goes up multiples, significantly up, during, right after, at the end of this, in this last phase of ovulation. During this same time, we can track the insulin resistance in this woman's body, and it will follow that progesterone curve. I'm making it a little dramatic here just to show the effect, but she would have been perfectly healthy, insulin sensitive. And then for a few days, as progesterone is peaking, we would say, whoa, gosh, you're a lot more insulin resistant than you were before. And then as the progesterone goes away, so too does the insulin resistance.
- 2.109** And then one last little interesting aspect to this is what happens to hunger. Because beyond progesterone's ability to change how the body uses energy, reflected in insulin resistance, progesterone has a central effect at stimulating hunger. Just as these are peaking, so too is hunger. And so during the ovulatory cycle, at this period of time where the body was saying, "Oh, wait, hey, did we get pregnant? Do we need to start growing something? Let's start eating just in case. Oh, we didn't. All right, well, then let's go back to normal." And hunger will follow that.
- 2.110** So it's just part of the myriad effects that progesterone has, metabolically, and even with regards to satiety signals, complicating the wonderful thing that we call the human body. So in the time we've spent together in this lecture, we defined—it was really, the big topic was, what is insulin resistance? We spent time discussing where it comes from, what are the origins. Why has it become so prevalent? Not only here in the US, at home, but even abroad, where it's even worse in many countries around the world. Then lastly, is it always a bad thing? Well, there are some times when it isn't. Thanks for listening. I'd be happy to answer any questions.
- 2.111** What determines whether fat grows hypertrophically, through hypertrophy, or through hyperplasia?
- 2.112** Yeah. Isn't it fun? We got to try to turn it into an adverb? I appreciate the attempt there. There is two factors, genetic and nutrition. I mentioned one nutrition factor. Do you remember? It was the seed oils. When the linoleic acid gets converted to 4HNE, it will promote, it basically forces hypertrophy. Second, though, that might not touch on the differences between that one study, South Asian men and Caucasian men, although nutrition does play a part there.
- 2.113** But genetically, we know that some ethnicities, and even within ethnicities across individuals, there are varying levels of the expression of a transcription factor within the genes, within the cells, called PPAR gamma. PPAR gamma is one of several master gene regulators. When PPAR gamma is turned on, and it's working well, it will promote what we call adipogenesis, the synthesis of new fat cells—in other words, hyperplasia.

- 2.114** If someone has a variant of PPAR gamma, where PPAR gamma isn't working as well, then they can't promote adipogenesis, the synthesis of new fat cells. Thus, if the fat cells are getting pressure to grow from insulin and fuel to grow in the form of calories, then they're only left to grow through hypertrophy. That's a really good question. Again, part nutrition and then part genetics.
- 2.115** I'd like to start by thanking you for making this intelligible for me because, before I was here, I thought insulin was something diabetics needed, and it had nothing to do with me. That was like two hours ago. Obviously, I had no idea what I'm doing. And so I preface that by saying that if this is an improperly formulated question due to my ignorance, I apologize.
- 2.116** Remember, I'm a college professor. No question is too dumb.
- 2.117** Thank you.
- 2.118** I've heard them all. The threshold is very high, rather.
- 2.119** I'll try to stay above the bar.
- 2.120** Yeah, OK, good.
- 2.121** My wife's family, it seems the women on the side of that family have a gluten intolerance. Pretty rampant. Her mother has celiac. It's recently been diagnosed. In response to this, her aunt decided to cut out carbohydrates, pretty much all together in a gluten-free diet. In doing so now, she's coming up as a prediabetic. My thought was, if I've understood this correctly, she could engineer a part of her diet, say, to reintegrate carbohydrates before she takes the gluten test, to pass to see if her insulin is working properly or if it's not a novelty diagnosis, she actually does have an insulin problem. Is that founded properly? Does that make sense? Have you seen that before?
- 2.122** Yeah. Just to clarify, is she eating—not that I'm giving any medical advice at all. Is she eating a carnivore diet, basically? Almost pure meat?
- 2.123** I think it's closer to paleo, but I don't know entirely.
- 2.124** OK.
- 2.125** I couldn't tell you.
- 2.126** Yeah, so, prediabetic. I strongly suspect, for reasons we'll get into more, but I've touched on them even now, that would be based on a glucose measurement, because we don't measure insulin. Insulin isn't on the radar of most, of conventional clinical care. And so your aunt or whoever it was, in-law, would have been diagnosed based on a glucose level that might have been hovering above 100 consistently.
- 2.127** I mentioned earlier, when I showed to you guys the insulin curve in response to macronutrients, I'd mentioned that protein can have varying levels of stimulus in people. That goes with glucose as well. For example, people with type 2 diabetes, they eat protein, pure protein. And, like, a non-diabetic, you guys go into the gym, you eat

your way, and your glucose won't do anything. If a type 2 diabetic ate that, there will be a substantial insulin effect. That's, I think, partly one reason why we should never eat pure protein. In nature, protein doesn't come alone. And we'll talk about that more in the solution lecture on the last one.

- 2.128** But even still, some people have an exaggerated response to protein, where they eat a protein load, and it stimulates a lot of what's called gluconeogenesis. In other words, the liver takes those amino acids from the digested proteins and starts turning it into glucose. That can happen. We're actually doing a study on this very topic in my lab now, where we found that people who have, for lack of a better word, a hyperglycemic response to a protein load, it's because they have too high of a hormone called glucagon.
- 2.129** Glucagon is not a hormone we've talked about yet. It is insulin's opposite, the yang to the yin. Whereas insulin wants to lower glucose, glucagon wants to increase it, not unlike some of the other hormones we've already talked about—epinephrine, cortisol, and growth hormone. Insulin has a lot of enemies, but it can handle them because it is so powerful in its own right that it takes a small mob to try to fight what insulin is doing. Anyway, glucagon is one of them. So when some people eat protein, they have an exaggerated glucagon response. And if glucagon is going too high, blood glucose goes too high.
- 2.130** So it could be that, as she has adopted a relatively carbohydrate deficient diet, which is perfectly sustainable, there's no such thing as an essential carbohydrate, but I don't want to get ahead of myself. We unpack that a lot later. But it could be that she's just eating enough protein that it's stimulating glucose response. I would say, I don't think that's a bad thing. But in addition to measuring her glucose, she should get her insulin measured to actually know where she's at. Because if her insulin is really low, then I'd give your aunt-in-law a high-five and say, keep going.
- 2.131** Say you're an athlete or you're just working out a lot.
- 2.132** Which I do.
- 2.133** You're fueling up on carbohydrates. Your glucose is pretty high, so your insulin would, say, probably be pretty high the whole time. How is that bad? Does that lead to insulin resistance? What's the connection for athletes who are working out?
- 2.134** Very, very good question. Are you asking it, during the exercise itself? Is that the time frame you mean?
- 2.135** Both, probably. During the exercise, and also just during the day.
- 2.136** OK, so let's go with both. Let's just go with both. Exercise is a very unique metabolic state. What do you think? I mean, overall metabolism, the word metabolism can be summed up as the balance of two biochemical processes. Anabolism, which is the process of storing energy, and ketabolism, which is the process of breaking things

down. In exercise, which one are we doing? It's the latter. We're breaking things down because we need all that energy. And what do you think insulin promotes? It promotes anabolism.

- 2.137** And so when we exercise, what do you think happens to insulin levels? It drops. Now, so that's a very good thing. Insulin levels come down. However, what happens to the glucose? Why doesn't the athlete become hyperglycemic, even if they're drinking Gatorade or some other sugar starchy things? They don't indeed. But in the—exercise is such a demand for the muscle. The muscle becomes so hungry that actually we're going to talk about this later. But as the muscle is contracting and relaxing, at rest, the muscle needs insulin to come knock on the door and open the glucose doors for the glucose to come in.
- 2.138** Exercising, the muscle says, insulin, I can't wait for you. I'm going to open the doors on my own. And so there is an insulin-independent method of opening the glucose doors. And so an athlete who engages in frequent exercise absolutely has a higher threshold for carbohydrate consumption. That that big curve I showed, it would be much flatter for significant periods of time, most certainly overnight, because they just consume so much of it into their muscles.
- 2.139** My question is about the extreme of what you have been talking about when there's a lot of fat. When someone goes through an extreme food deprivation, to the point that the person looks like skin and bones, what happens to the fat cells? Are they recoverable? Are they affected?
- 2.140** Yeah. What a wonderful question. So you guys are going to still—if you're all coming for the next two days, you'll love it because I'll be visiting a lot of the topics you're bringing up. If someone has gone from, say, having 100 pounds of fat to having 10 pounds of fat, like a 2% body fat, those fat cells are there, but they might not get replaced. But a fat cell will live for about 5 to 10 years. It will just have gone from this size to this size. That will be reflected in the significant reduction in fat mass.
- 2.141** So commonly, during adulthood, until about the '70s, we will not lose fat cells. They stay the same. Whether we're getting fatter or leaner, fat cell number stays the same. It's mostly just a matter of fat cells growing and shrinking that will determine the weight gain or the weight loss. The fat cells are there. But just to add another dimension to this, later on, we're going to talk about, I'm going to mention liposuction, which is a process whereby we actually remove fat cells from the body.
- 2.142** Unfortunately, where do we remove those fat cells from? When you're undergoing liposuction, you have two different fat depots. You have subcutaneous fat, the pinch-and-jiggle fat, or you have visceral fat, the fat that is tucked and locked away within the organs or around the organs. Which one does liposuction take? It takes the subcutaneous, the pinchable, jiggable. Well, remember, I said that's a good fat. And so, what happens when people undergo liposuction, they remove their subcutaneous fat,

which is where the body is designed to store fat. And if the person hasn't changed the way they're eating, and they never do, because they've used the liposuction as just an easy way out, right? You can't fake the metabolism of the body.

- 2.143** And so, if they're continuing to eat, and they have that insulin and the fuel to fuel the growth, they now start storing more fat in their visceral space. Or if this was a study, particularly was done on women, if a woman has had liposuction of her butt and hips, now she starts storing it more in her belly. The body will not be tricked. You can't beat it. If you remove fat cells, the body will just say, fine, I'm going to store it somewhere else. Yeah, and you won't like it. You're going to regret it.
- 2.144** So anyway, to answer the question, fat cells just shrink. Then around the age of 70, then the fat cell number starts to go down. Some people will say, well, then why am I not getting leaner? I'm 75 now. That's because it's actually a terrible combination. While fat cell number is going down, if the person is continuing to eat the same amount, in the same way that put them at that fat mass in the first place, fewer fat cells now have to carry the metabolic burden. So what starts to happen to the size of the fat cell? They undergo hypertrophy. That can amplify the metabolic problems, the insulin resistance in older age. Paradoxically, it's because they start running out of fat cells to store their fat in.
- 2.145** I wanted to ask, within the context of the slide where you were referencing the progesterone in women, as many of us are aware, most female birth control methods are hormonal. In the context of a plateaued state of progesterone, what implications would that have on insulin in general and insulin resistance?
- 2.146** That's a very wonderful question. A lot of hormone-based contraceptives are progesterone-based. And that's for the reason of—there's a good reason for that, which is estrogen-based birth control or oral contraceptives have other consequences like possible cancer risks that progesterone just doesn't have. But weight gain is a likely consequence of progesterone-based, for the reasons that I just mentioned. I was this close to explicitly talking about birth controls, but I thought, I don't want to get into it. But you brought it up anyway.
- 2.147** So yeah, the consequence of basically tricking ovulation, because progesterone is the hormone of gestation. And so, in normal female physiology, the progesterone is going to ensure she doesn't go through another ovulatory cycle. Basically, progesterone is saying, "Eggs, stay in the ovaries. We're already gestating here." That's normal pregnancy, why a pregnant woman won't get pregnant again, because the progesterone prevents another ovulation during the gestation period.
- 2.148** But if she's hacking that and taking progesterone, then it still has the same effect as the pregnancy did on ovulation, but it also has part of the same effect that pregnancy has on metabolism, which is, let's get a little fatter. That's a normal process of pregnancy. Every woman should gain fat from day zero to day month nine over the course of her

pregnancy, and you can track the insulin resistance perfectly over the same period of time, which is designed, because her high insulin levels and her insulin resistance not only is helping mommy's body get a little fatter, but it's helping the placenta get big, the uterus get big, the mammary gland tissue preparing for nursing to get big.

2.149 And providing a little baby with all of this growth stimulus because it helps baby get big, too. That's why it is so physiological. But again, when you've removed the gestation but you've kept the progesterone, you get some of the metabolic consequences. Not as dramatic as pregnancy, but it will make it more difficult for a woman to balance her fat mass.

2.150 My doubt is about white fat and brown fat, because I can understand that are the same tissue but with different function, right? But can we transform white fat in brown fat, or are always different kinds of tissues? Maybe adding mitochondrias to white fat, we can transform that.

2.151 It's a very, very good question. You guys are just getting ahead of me. I'm going to talk about this, but yes, you can. In fact, just by way of an obvious one, so white fat is this insulating fat with a very low metabolic rate. Can you imagine a stimulus to the body that would tell this white fat, "Hey, we need you to work better and make heat"? Can you imagine a stimulus? When would we want all this white fat to start acting a little more like brown fat? Cold immersion. So this has been very well documented, and we've done some versions of this in my own lab, and I'll talk to that later. I think, not tomorrow, but on the next day.

2.152 But you can take a fat biopsy of someone's white adipose tissue, their belly fat, which we do, measure the metabolic rate, quantify the mitochondria, and say, boy, this thing is burning slow, and there's very little mitochondria. And then, do cold therapy for a week, where it's cold immersion. They're sitting in a nice bath up to their collarbone, and then do a fat biopsy at the end, and it's much higher. Now, it's not as high as brown fat.

2.153 But what happens is, this conversion that we say, well, it's not white fat anymore, but it's not brown fat. We call it beige fat. That's the word for it. Isn't that funny? But it is a process that we call "beiging," where you are beiging the white fat because it has a little more mitochondria in it. So when you take the biopsy, you can see. This one was really yellowish-white before. Now it's a little bit tanner than it was before, a little more beiged. But again, not like brown fat. But what we found is that you can take the metabolic rate of white fat and increase it by three times. That's what we did with other studies, which I'll mention, through ketones. But I'll get to that later. So it doesn't get to the 10 times higher, but it starts to get in the middle. Good question.

2.154 All right, thanks, guys.

Lecture 3: Insulin Resistance: Beyond the Basics

At a glance — Insulin resistance beyond the basics.

- 3.1** It is wonderful to do this again. So last time, the first lecture, I set the stage. I began the introduction to this horror story, and we became familiar with the villain of the story, insulin resistance. Unfortunately, as we move into chapter 2, the story actually gets a little worse, because now we have entered the crime scene, if you will, and now we're seeing the damage. Last time, we were just telling the story of insulin resistance is, how it became the villain that it is. Now we look at the crimes. So we move into what does insulin resistance do? Why do we care? Why would I devote my career to understanding it?
- 3.2** Beyond the prevalence, as I showed you last time, affecting so many people around the world, now we talk about how relevant it is with regards to the chronic diseases. So as we go through the lecture today and discuss why insulin resistance matters, I'll bring your attention to two points. One we'll go through quite quickly, which is that it is, I'm reminding you, that it is the most common disorder, and I will add a little more context to explain why it has become so common.
- 3.3** Then second, we will then dive into the diseases, some of them at least, with the time we have, to explore how insulin resistance plays into the ideology or the origins of the disease itself. By way of reminder, 88% of adults in the United States are considered metabolically unhealthy. Last time, I showed you the scope of it, how we are far from the worst here in the United States. We're not even the worst within North America. But what all that metabolic health stems from is insulin resistance.
- 3.4** Insulin resistance has become such a problem, I submit, because we don't detect it. We're looking at the wrong thing. That leads us to treating the wrong thing, which we'll get to later. But for now, we'll just focus on how missing the mark has led to this giant growing right behind our back. Now, this is the traditional view. I presented the view earlier, and will more in a moment, that type 2 diabetes or diabetes, just to say, generally, is derivative of insulin resistance. Unfortunately, as also was reflected in some of the questions yesterday, many people hear the word insulin, and they immediately believe that insulin resistance is itself just encompassed within what we call diabetes, that it has no relevance outside that scope.
- 3.5** That's because, again, we're looking at the wrong thing. Traditionally, the conventional dogmatic view is, if you're talking about anything with insulin resistance, or insulin at all, then as you are looking around to try to find the problem, we end up looking only at glucose. Now, despite my criticism of that perspective, that narrow-minded view, I am empathetic to it. I understand it. I can understand why someone would want to look at glucose, why they think it would be valuable. That's because there's two very relevant

issues here. There's a precedent here, and that is a historic justification and a scientific one.

- 3.6** Now, historically, from ancient Egypt, all through the current time, what we refer to as diabetes, there was one key symptom. In fact, I'll give you the hint. Diabetes means to flow through. It's no surprise that the most common feature of diabetes is the polyuria, or the excessive production of urine. That is what the word diabetes means. Diabetes means an individual is producing a great deal of urine. And that urine is driven, I won't get into the mechanism, but it's driven by the high glucose. Briefly, when glucose levels get too high in the blood, it overwhelms the kidney's ability to filter that glucose and to keep it in the body.
- 3.7** And so the glucose spills into the kidneys. And as the glucose overflows from the blood into the kidneys, it pulls water with it, causing the patient or the individual to produce a phenomenal amount of urine. And of course, they'll be dehydrated. But here we are. This is the most common symptom, and it has been since time immemorial. And these early physician-scientists noted, beyond just the amount of urine being produced, there was a particular attraction of the urine with regards to dogs and flies.
- 3.8** It has been noted, again, for centuries, that the urine that is produced from patients with diabetes, dogs would come and lick it, and flies would come to it as if they were coming to honey. And that's what the term "mellitus" means. When we begin to talk about diabetes, it's diabetes mellitus. Basically, a lot of urine being produced and it has a lot of sugar in it. And so it will attract animals to come and lick it, or the insects will come and buzz around it. Now, that's the historic precedent.
- 3.9** The scientific precedent is a little more obvious and easy, which is, glucose is very easy to measure scientifically. This is the case with other similar nutrients like ketones and lactate, but it is easy to measure glucose. In fact, so easy that you can just put a little drop of glucose on a little glucose meter, a glucometer, and it has that, within it is so well condensed, the technology is so simplified, that you can measure it then. We've gone even beyond that, where you can put a little patch on someone's arm or on their belly, and it will measure their glucose levels constantly. Glucose is easy to measure.
- 3.10** Unfortunately, insulin is not. There is not a precedent for measuring insulin. Historically, it was very difficult to measure, but we can. And I just have to acknowledge the contributions of Rosalyn Yalow. One of my heroes, again scientifically, I mentioned Joslin and Benedict last time, and I'm compelled to mention Dr. Yalow. I admire her. She is the one who pioneered, and ultimately winning the Nobel Prize, for measuring insulin. I also greatly respect her devotion to family and faith, something that I try to emulate.
- 3.11** Nevertheless, thanks in large part to her contributions and those who have followed, we can shift the paradigm. We can move the magnifying glass away from just doggedly pursuing glucose as the marker to measuring insulin. Because after all, we're talking

about a problem called insulin resistance. We should, don't you think, consider insulin. And as a reminder from last time, insulin can be elevated decades before the glucose ever starts to change. This—high insulin, low glucose or normal glucose— is insulin resistance. But our failure to measure insulin and only looking at glucose means we fail to detect the problem. We only start to detect the problem once we've become so resistant to insulin that now the glucose levels start to climb. We got it now. Well, we could have had it when it was here.

- 3.12** All right. Now, that is some of the justification for how insulin resistance has become so common, because we just don't measure it. We're looking at the wrong things. Now, let's transition and spend the rest of our time talking about just why it's so catastrophic. Here's the wheel of misfortune, and I'm your demented game show host, as we go through all of these problems. Now, we're not going to go through all of them. With the time we have, I want to focus on some of the big ones. Last time I took off obesity. We spoke before about how insulin interacts with the fat cell, and we explored the dynamic between insulin and energy. So I consider that topic covered.
- 3.13** Let's talk about the big three—diabetes, heart disease, and cancer—and discuss infertility as it's becoming more common in men and women. And then we'll also speak about Alzheimer's disease, the most common form of dementia. Now, with diabetes, we'll spend most of the time talking about type 2, because you cannot understand type 2 diabetes without appreciating insulin resistance. There is no type 2 diabetes without insulin resistance. And then I'll mention type 1 diabetes, although it's a totally different beast.
- 3.14** Indeed, what these have in common is only the high glucose. I actually think it's a bit of a tragedy that we consider these part of the same family, because while they do have one symptom that is the same, namely the high glucose, the actual origin of the problem couldn't be more different. Whereas type 2 diabetes is a disease of too much insulin. There's too much insulin. The body is becoming ever more resistant to it. In type 1, it is a disease of a deficiency of insulin. So again, I think it's tragic that they're put in the same family; they shouldn't be. They are totally opposites. We wouldn't do this in any other situation. They just have one symptom that is the same.
- 3.15** Nevertheless, again, type 2 diabetes is a disease of insulin resistance. It's the foundation upon which type 2 diabetes is built. This is a familiar paradigm where we looked at the progression through insulin resistance and into type 2 diabetes. And I introduced these relevant tissues, presenting to you guys the idea that adipose tissue is the first domino to fall. And then once it falls, it tips into the other tissues. Well, we spoke a lot about the fat cells last time, and that was relevant for the origin of insulin resistance.
- 3.16** But now we want to see what flips the switch. How does the patient, how does an individual go from insulin resistance or prediabetes into full-blown type 2? It's when the dominoes of these three tissues fall. Let's start with skeletal muscle. That's

appropriate because, by mass, most people are made of skeletal muscle. That's what constitutes the majority of what is our body, by way of tissue weight. And it's no surprise, given the, not only mass of the skeletal muscle, but just how busy it is, how it animates the physical form, that the muscle consumes the lion's share of glucose.

3.17 If we consume something that is sugary and starchy, and it increases our blood glucose, most of the blood glucose will go into the muscle. So when we see glucose levels come up and then they come down, up to 80% of what's driving the glucose back down is by what's being consumed by the muscle. Now, insulin opens those doors, something I alluded to in the previous lectures. When insulin comes and knocks on the door of the muscle cell, one of the effects, one of the many actions, will be an opening of the channels, these transporters or doors for the glucose to come in.

3.18 However, if insulin can't work well, well, the consequences become quite obvious, don't they? You've now lost access to the largest glucose consumer in the body. And so as the muscle becomes ever more insulin resistant, less glucose can come in, and the glucose is left to simply accumulate in the bloodstream, and thus glucose levels start to climb. And if we go back to this little gauge, the person maybe only had fat cells that were insulin resistant, so that had moved them to the cusp. But once the muscle becomes insulin resistant, the individual will absolutely have a harder time controlling glucose. Blood sugar levels will start to spike.

3.19 Now, let's move from the muscle to the liver. The liver is one of the unsung metabolic heroes of the body, beyond metabolism, in fact. It is one of just the unsung heroes of the body. It's interesting for me to note that in the Western cultures, we will talk about—a term of affection will be that you're my sweetheart. That's a term of affection, a term of endearment for us. But in certain Persian cultures, they will say golden liver. "You're my golden liver." And I tried to use that term of endearment on my wife, and it didn't work very well.

3.20 But I appreciate the sentiment, which is that the liver does so much more. Now, both are essential. You can't live with either of them. But the heart just does one thing. My darling soccer-mom wife does all kinds of things. So she's much more like the liver to me than the heart. But be that as it may, I appreciate the sentiment where the liver is such an important part of the body, and it is particularly relevant with regards to nutrient metabolism. The liver knows what to do with all the nutrients.

3.21 But I want to focus on lipids and glucose, or fats and glucose. It's two of the most prominent tricks that the liver has. On one hand, lipogenesis, as the name suggests, is the production of fat. The liver can take in almost anything, any carbons from a nutrient. And if the overall milieu, the metabolic milieu as such, is correct, it can turn it into fat. That is called lipogenesis, the synthesis of fat. At the same time, the liver, being the ultimate soccer mom, wants to make sure that it has snacks and everything ready for the family. Now, not only can it do that by making and storing fat, but it can also do that by taking in glucose.

- 3.22** Now, unlike the selfish muscle, the muscle will take in the glucose, and once the glucose has gone in, it will never go out. The muscle will use it, all of it. It's very selfish. The liver, it's selfless. It will take in glucose and it will say, the body might need this later, so I'm going to hold on to it. And it converts it into something that we called glycogen. So the liver will activate lipogenesis to store fat for later use when the body may need it. And it will activate glycogenesis, the synthesis of glycogen, which sounds really compelling and remarkable, but it is simply the storage of glucose. That's how the body stores glucose.
- 3.23** Now, insulin is a signal of the body to store. As we mentioned last time, insulin abhors wasting things. So it forces the liver into this hyperproduction, into this exaggerated mode of storing, storing, storing. And so, no surprise that the insulin stimulates both the synthesis of fat. Because fat is a wonderful way. Just the molecule of fat itself is a very, very efficient way to store energy, nutrient energy. But at the same time, it will promote the storage of glycogen.
- 3.24** When the liver becomes insulin resistant, things get a little weird. Now, if the entire, if every biochemical effect, if every event within the liver became insulin resistant, then both of these would be compromised, both the synthesis of fat and the synthesis of glycogen. But that doesn't happen. The insulin's ability to stimulate lipogenesis is uncompromised, even in the most profound insulin resistant liver. But it's the glycogen that's broken. The liver doesn't remember how to make glycogen from glucose coming in. Insulin is not able to stimulate that.
- 3.25** As a result, rather than storing glycogen, storing the glucose as glycogen and pulling it in from the blood, lowering blood glucose, the liver is now doing the opposite. Insulin is trying to tell it to hold on to it, but now it's just breaking it down. So the liver is breaking down the glycogen, giving the glucose, opening up the bags of snacks for the kids, if you will, and dumping them all over the car. Don't get me started on that. But then the glucose is now dumped into the bloodstream, and blood glucose levels start to climb as a result—once again, taking the patient who might have been flirting with higher glucose levels, pushing them even deeper into this mounting hyperglycemia.
- 3.26** Now, lastly, we have the pancreas. I mentioned last time, we've discussed the beta cells before. The beta cells have become famous in the story we've been telling because it produces insulin. But right next door, literally right next to it, within these little eyelets or these pockets of cells within the liver, we have cells called the alpha cells. And the alpha cells are the opposite of insulin, or the opposite of the beta cells. Where the beta cells produce insulin, the alpha cells produce glucagon, and they don't get along. They have opposite effects. Insulin wants to lower glucose; glucagon wants to increase it.
- 3.27** Now, neither is bad. We need both to maintain homeostasis. Normal glucose levels are a balance of all of these hormones that we've talked about. But this one is particularly relevant because they are literally next-door neighbors, but they don't have a great relationship. Not only are they fighting over glucose levels, they're fighting over the

fence line, but also insulin takes things a little more seriously. Insulin climbs the fence and beats down the alpha cell, its ability to produce glucagon.

- 3.28** So, all things considered, in a one-to-one battle, insulin wins that battle. Insulin is so effective at lowering blood glucose, and is so determined, that it will directly inhibit the alpha cells' ability to produce glucagon. And by inhibiting glucagon, insulin further lowers glucose level, which is one of its primary functions. However, what happens if the alpha cell becomes insulin resistant? Now that signal is lost. Insulin's ability to suppress glucagon production is now compromised. It's broken. And so now the alpha cell is thinking, "Hey, my abusive neighbor is not here to tell me what to do anymore. I'm just going to start dumping out the glucagon."
- 3.29** And as glucagon gets released, it actually goes to the liver and tells the liver to dump even more glucose. That's the connection there. Altogether, though, suffice to say, we have pushed the patient to full-blown, severe type 2 diabetes, and they will have raging, out-of-control hyperglycemia. Now, to really put a fine point on this and to stack it all up, from last lecture to now, we have insulin resistance beginning in the fat cell. Then as it progresses and goes through the stack of tissues, we have this further, further progression into type 2 diabetes. And of course, the molecular mediator that's making it all happen is that vicious little fat molecule called ceramides, which we discussed before.
- 3.30** Now, I don't want to leave type 1 diabetes untouched. Type 1 diabetes is a completely different beast, as I noted. In this case, the person has an autoimmune disease that is destroying their beta cells. And as the beta cells are getting decimated, insulin production gets decimated. So this is the overall phenotype or form within the type 1 diabetic, low insulin leading to uncontrolled hyperglycemia.
- 3.31** Now, tragically, one of the greatest mistakes of type 1 diabetes is that the type 1 diabetic patient is told, "Eat whatever you want and just cover it with insulin." In other words, go eat those starchy, sugary junk food treats, but just give yourself your insulin injection and you're fine. The problem with that view is, the more glucose is climbing, the more insulin the diabetic needs to treat themselves with.
- 3.32** And what does high insulin do to insulin resistance? Yes, you can see the problem. Chronically elevated insulin, even if it's coming from a syringe, will induce insulin resistance. And this has led, this phenomenon is growing in prevalence, where we call it a double diabetes. Basically, it's a type 1 diabetic who's now brought on also type 2 diabetes.
- 3.33** All right, now let's leave diabetes behind and talk about the leading cause of death within the United States and much of the world at this point. Heart disease is the main cause of death, and insulin resistance is absolutely fundamental to it. The general problem with what we call heart disease, which is an umbrella term, is the formation of plaques within arteries, particularly arteries of the heart, coronary arteries. This is a

phenomenon referred to as atherosclerosis. The person has their arteries that are feeding the heart muscle itself to enable the heart to engage in its busy work, but the arteries begin to get occluded or blocked through the formation of plaque, which I'll elaborate on more in just a second.

- 3.34** But I also want to, in the time we spend on this, we're going to discuss briefly about hypertension, which is when blood pressure starts to get high. That is, in its own right, considered a form of heart disease, but it directly contributes to atherosclerosis because of the pressure damaging the blood vessels. Then we'll also introduce this concept of dyslipidemia. In fact, in short order. The sentiment is that disordered lipids, blood lipids, cholesterol levels being out of whack, that contributes to the atherosclerosis.
- 3.35** Now, with regards to atherosclerosis, we're looking down the length of an artery of the vessel. With atherosclerosis, the idea is that we have these plaques forming within the blood vessel wall, narrowing and narrowing the lumen or the tube of the blood vessel, making it harder and harder for blood to get through and to nourish the tissue. Dyslipidemia is likely a contributor here. Now, I say likely not because I am unfamiliar with the topic. It is simply, what actually drives the formation of plaque is strictly within the realm of theory. We don't know, despite the fact that it is, maybe, it's the leading cause of death because we don't know.
- 3.36** But despite all the attention, the precise series of events that leads to the formation of a plaque is truly unknown. But the consensus view, which always should be challenged, is that it's a result of lipid. There are some holes in that story, and we'll poke some of those holes, make them a little bigger. But I'll also disclose information where it might just be some error in the way people are looking at the data.
- 3.37** Now, LDL cholesterol, which we'll talk a lot about, is scrutinized so heavily because it is thought to contribute to the synthesis of the plaque, atherosclerosis. And that is through the accumulation of cells called foam cells. Now, what's a foam cell? A foam cell is a macrophage, which is the poster child of the immune system. When you think about white blood cells, which very few people do, it's the macrophage that is at the pinnacle of all of them. The macrophage is the primary cell that's involved in immunity and in inflammation and in healing and recovery and infections and everything.
- 3.38** When a macrophage engulfs a lot of LDL cholesterol, it becomes the foam cell. And then the foam cell, again, is really the core of the plaque. That's what's accumulating within the vascular wall. Now, what is LDL? We are obsessed with it. Clinical care is obsessed with it. We measure it every time, and we've got a lot of drugs that will very effectively lower LDL. So what is it? Well, to demystify it, LDL is basically a big ball of fat that is not a singular glob of fat, but a bunch of fats, thousands of fats lumped in a little ball with a few little proteins.

- 3.39** It's the proteins that allow us to define it as LDL. LDL means low-density lipoprotein, after all. And the lipoprotein refers to lipids, fats, that is, and proteins. Another way of describing this, and pardon me for making you feel like kids, is like a bus. LDL is like a bus that's transporting a lot of fat and some proteins through the blood. That's it. That's what it is. And the fats, some of the fats, are cholesterol itself. There's no difference between what we call LDL cholesterol and HDL cholesterol. There's no difference in the cholesterol. The difference is the bus, but the cholesterol is the same. If one of those passengers on this fatty protein bus is cholesterol, it's the same, regardless of which bus is carrying it.
- 3.40** When we say LDL versus HDL, cholesterol, we're talking about what's actually transporting it. We're really mostly talking about the proteins on them. Regardless, it's like a bus. It's transporting things through the bloodstream. But the type of fats that this bus is carrying really determine the destination. And this is something we'll revisit later, where we start to tease apart why might LDL sometimes be a marker of disease, and sometimes it's not, indeed. And other times, LDL appears to be very, very beneficial. It could just be, it could depend on what fats the LDL is carrying.
- 3.41** All right. Now, for the next while, I want to go over a handful of questions challenging the idea of a cholesterol being relevant in heart disease, making sure that we never forget that insulin resistance is really the basis of the story, the foundation. And then we'll talk about, does LDL even matter? Maybe we're looking at it the wrong way. What about saturated fat? Because that's certainly been demonized. I think when Isaiah talked about "the whore of all the Earth" in the Old Testament, everyone thinks he's talking about saturated fat. We will blow that apart, and then we'll look at, is LDL relevant at plaques or why is it at the plaques at all?
- 3.42** OK, so the first one. I'm just going to highlight a handful of studies here, and I actually have trimmed this down from what was once, in my own lecture, an extraordinarily long list. I'll just highlight a few studies. So this is all addressing, does cholesterol actually kill people? Does it predict mortality? This was a study done out of UCLA medical school, where they looked at over 130,000 patients who had presented with a heart attack, and then they bell-curved it, and it created this beautiful bell curve, basically just looking at what is the level of cholesterol in these people.
- 3.43** And they found that over half of the people who came in with a heart attack had cholesterol levels that were below the threshold of what's considered good and bad. So how could LDL cholesterol be a good predictor of this disease? It wasn't. It was a terrible predictor. Again, just as many people came in having had a heart attack with cholesterol levels that were low or considered good, compared to those who came in with a cholesterol level that was high. That's a terrible way to base a predictive model. You cannot say—mind you, that's just a correlational study anyway, where you're just looking at coincidence, but that's a terrible coincidence. Indeed, it can't explain it whatsoever.

- 3.44 This study published just a few years ago challenges that even more. This study found, looking at population surveys, that the people with the lowest death had a general LDL cholesterol level of 140 milligrams per deciliter, about 40% higher than what is considered the upper end of optimal. That's laughable that, in the midst of all of this data, supporting LDL might not be a problem, we still try to push it down as low as possible. Even further, this study found that as people, either through drug or just through lifestyle, pushed their LDL cholesterol, their total cholesterol down, the death rate went up.
- 3.45 And then they found, I highlight this conclusion, for every one millimolar reduction, which, in the previous slide, I was presenting LDL in milligrams per deciliter, it just depends on what units we use, that would be about 30 milligrams per deciliter, but there was a 15%—so for every one millimolar increase in cholesterol, the risk of death, just total mortality, dying from anything, went down by 15% for every one point. Isn't that funny? It's tragic, really.
- 3.46 This study continues the same idea, which is that they found that the people with the lowest cholesterol had the highest death. What I found amusing as a scientist who has to write peer-reviewed papers was how they began their conclusion, which is we can't explain it. Well, you can't, if you believe that cholesterol is "the whore of all the Earth" and the root of all disease, as so many people do. If cholesterol is the great killer, it's impossible. Indeed, there is no explanation, as they say. Another view would just simply to be, cholesterol is clearly not the villain we thought it was, because people with the highest cholesterol had the lowest mortality.
- 3.47 And then one last study along this vein of does cholesterol predict mortality is showing some studies out of Shanghai, finding that the people with the lowest LDL had the highest risk of developing Alzheimer's disease. Now, we're going to talk about Alzheimer's disease later. But let me just say that another way. People with the highest cholesterol had the lowest risk of developing Alzheimer's disease. Well, that seems pretty appealing to me. I'll take it, personally.
- 3.48 All right. Now, we've been talking about, does cholesterol predict mortality? And now let's discuss, does LDL even matter at all? Well, the answer is, it might. Maybe it does. This is me being very honest and admitting, again, repeating the sentiment from earlier, which is, we don't really know how all of these things connect. It is strictly within the realm of theory. But it might. LDL might matter if we look at it the right way. LDL actually comes in two forms. They're not totally different in form, but they're different in size and mass and density. What we call patterns.
- 3.49 LDL can come in what can be measured as a pattern A. LDL, and remember, LDL is just a big ball of fats and proteins, or a bus carrying some fats and proteins through the blood. But we have pattern A and pattern B. Pattern A is a larger, more buoyant, less dense version, due to the composition of proteins versus fats. Pattern B is a smaller, more dense. It has a relatively higher mass to its volume. Now, the relevance of LDL is,

as the theory goes, the LDL must get into the vessel wall. If LDL is just flowing through the blood, it's perfectly benign. The theory is that, with regards to the progression of the plaque, the LDL must penetrate the vessel wall.

3.50 Well, which one of these do you think penetrates it better than the other? Here's a metaphor. Imagine we're standing over a bridge, over a stream. There's a big busy stream flowing beneath us. And on one hand, I'm holding a beach ball, and in the other, I'm holding a golf ball, much to every golfer's dismay. They do not have the same buoyancy. If I drop the beach ball, it's just going to go right with the flow. It's merrily just bouncing down the river. When I drop the golf ball, because of its density, it will sink right to the bottom and be bouncing along the rocks, along the stream bed.

3.51 That has a direct relevance in the progression of atherosclerosis. Not all LDL is going to have the same propensity to penetrate, to bump into the vascular wall and actually get into it, to initiate the series of events that could likely cause atherosclerosis. It's no surprise, given that silly metaphor, but your rational understanding of these patterns, that if you can confirm a patient has LDL-B rather than a pattern A, I should say a pattern B LDL versus a pattern A, it's no surprise that the risk of an actual heart attack goes up by more than three times if the patient has confirmed pattern B LDL—smaller, denser, more likely to bump into the vascular wall.

3.52 It is no surprise now. It's been a little too long. I haven't mentioned insulin resistance, so I have to get right back to it because I'm uncomfortable if I can't mention it every few minutes. Here, I take my sigh of relief. Insulin resistance promotes a pattern B LDL. It will change the way—and something I haven't mentioned is that the liver is the source of all of the lipids that we measure in the blood. When we have a blood test, we measure LDL. You will always measure HDL cholesterol, and you will always get the triglycerides measured as well. All of those are products of the liver. Once again, the liver being the hero of all nutrient metabolism and many, many other things as well.

3.53 Insulin resistance will increase pattern B, but it also changes HDL and triglycerides in a very interesting way. Insulin will stimulate the synthesis of triglycerides, which is basically the main form of fat within the body. And then insulin resistance will reduce the synthesis of HDL cholesterol. Now, these on their own, I believe, should be viewed as problematic, but it's particularly interesting to note that, as insulin resistance elicits these changes in the levels, they themselves can become an indicator of LDL cholesterol pattern.

3.54 Unfortunately, most people will not be able to get their pattern measured. It's a more advanced clinical test. You either have to pay more, insurance won't pay for it or whatever. It can be done, but it's harder to get done. But there is a remarkably accurate surrogate, which is that the same overall change of insulin resistance within the body that increases pattern B will also increase triglycerides and reduce HDL cholesterol. And so that ratio can become very indicative of the pattern. If an individual

has a higher triglyceride to HDL ratio, that reflects a greater pattern B. In contrast, if a person has a lower triglyceride to HDL ratio, it reflects the other pattern, pattern A.

- 3.55** Overall, the sum of several studies, which I cite here, is that, as insulin goes up in insulin resistance, this body, the pattern or the profile of LDL, shifts to B, the smaller, denser, more atherogenic. As insulin resistance comes down, and it can, it is correctable, this has been measured, and I cite it here, you can take individuals who have a primarily pattern B LDL cholesterol and shift their diet around in such a way that insulin resistance is resolved and insulin comes down. They shift their pattern of LDL from a B to an A, from the golf ball to the beach ball.
- 3.56** Now, this is the elephant in the room, saturated fat. The reason we fear saturated fat is —not all of us, though, and you won't either, soon enough. But the reason saturated fat is so feared is because of its potential effect on cholesterol. Now, that whole idea is getting challenged more and more with every passing year. I highlight this article in part not only due to the content, noting that there is no beneficial effects of reducing saturated fats, and indeed, saturated fats might be beneficial because it increases the pattern A, but it's the source. So beyond the content, beyond the statement, it's the source.
- 3.57** This was a review article published by the Journal of the American College of Cardiology. So the American College of Cardiology, this collection of cardiologists, these individuals who should be more concerned about cholesterol than anyone else on the planet—they are, after all, experts— they came out with this paper, basically dispelling the entire myth surrounding the fear of saturated fat. So again, I highlight it not only because of their content, their conclusions based on evidence, but the source, from the Journal of the American College of Cardiology. So, pretty reliable.
- 3.58** And these are guys who, if anyone wanted to espouse an idea of cholesterol being bad, it would be them. Now, with regard to saturated fat, I appreciate the fact that not everyone might appreciate the difference in fat type when it comes to saturation index. So, across the span of fats, we have fats that are purely saturated based on the chemical structure. Every bond is filled, every carbon bond is filled, if you will. And then, not if you will, that is exactly what it is.
- 3.59** And then we can have one of those little bonds in this molecule of fat be unsaturated. That's called a monounsaturated fatty acid. Or we can have several of these bonds being not fully bound up, if you will, chemically speaking. And that is called a polyunsaturated fatty acid. And I highlight the examples on the ends. An example of a saturated fat is coconut oil. It is the most pure form of saturated fat in nature. As much as we scrutinize animal fats like tallow or lard or butter, those are only partially saturated fat. They have a lot of monounsaturated and some polyunsaturated. But coconut oil is the most singular concentrated source of saturated fat.

- 3.60** Monounsaturated is something like olive oil. And then lastly, on the end, there is no natural source of polyunsaturated fat. That is only from industrially refined sources in large numbers, like what we call seed oils, which we've discussed previously. Those are the refined seed oils that have had added—the actual process of producing the refined seed oils is a feat of chemistry and technology, where, by the time, in addition to all of the chemicals that have been added to it to make it work, to extract an oil from a pulpy mass of seed, one of the most prominent additions is a deodorizer, because by the time you've taken it from the shelf, it has already gone rancid.
- 3.61** That rancidity actually becomes part of the pathology here that I won't get into because it's a little more chemistry, but it partly explains why polyunsaturated fatty acids are so problematic. They undergo this process of rotting, essentially. But there are deodorizers added to the so-called vegetable oil so that you don't notice the rancidity. Seems pretty underhanded, doesn't it?
- 3.62** Now, regardless, when someone consumes these fats, different things will happen to LDL cholesterol levels. Saturated fat will either have no effect on changing LDL levels or it may increase it. It's variable. It depends on the person. Monounsaturated fats typically are inert, have no effect. Polyunsaturated fats will tend to push LDL down. That explains, that singularly explains the love affair we've had with polyunsaturated fats since around the '60s.
- 3.63** When the fear of saturated fat came to life around that time because of its ability to increase LDL cholesterol, that was the reason saturated fats were vilified. It's that same reason that explains why polyunsaturated fats and the technology now enabled its production from seed oils, where we became enamored with the polyunsaturated fats because of their tendency to lower LDL cholesterol.
- 3.64** Now, I challenged that. We just got done talking about this cholesterol matter. So we've challenged that idea, the overall benefit. Should we be worrying about lowering cholesterol and thus dropping the saturated fats in favor of the polyunsaturated fats? Well, we've challenged that. And of course, the consensus from the data I showed you would be a resounding no. But even when saturated fats may increase LDL, as I mentioned from that statement from the Journal of the American College of Cardiology, it is more of a pattern A, which is a benign, even beneficial, type of cholesterol carrying.
- 3.65** There have been, to my knowledge, and I will show them both to you, two randomized clinical studies. Now, as you know—if you didn't, now you will—a randomized clinical study is the gold standard of confirming a hypothesis, of determining the validity of a view. There are two very, very large studies that both ran multiple years, and we'll look at both of them, in two totally different parts of the world. And they compared—you couldn't do this study nowadays because they did it in people who were living in institutions, where they had two cafeterias.

- 3.66** One cafeteria ate all their meals. That included saturated fats in all of the baking and cooking. The other group had their fats primarily coming from polyunsaturated fatty acids, from these newfangled seed oils. Two studies. Let's start with the first one. This is the Sydney Diet Heart Study. Now, I note here, from the title, recovered data. You're going to see the same thing on the next study, which was done in the United States. In both instances, these were studies conducted in the '60s, and the conclusions were so troublesome, they were not published at the time. They were deliberately withheld from publication.
- 3.67** It was only as scientists, more recently, within the last 10 years, got access to all of those literal old files—there's no hard drive with it from the '60s— and reevaluated the studies that they found out why they had been shelved. Because in this first case, a 7-year study, they found that in the group that swapped out the saturated fats for the refined seed oils, they died more, from everything. All-cause mortality skyrocketed. It was significantly higher. That's pretty damning.
- 3.68** But it didn't fit the narrative, which is very sobering and scary. And it should plant within all of us a deep seed of cynicism and being skeptical of what our experts are telling us. Case in point. And this was similar results conducted around the same time, once again from recovered data, literally recovered from the basement of one of these scientists, where the guy, Christopher Ramsden at the NIH, he's the one who published this. He went to the son of the main scientist, and the son said, "Oh, yeah, we have all these old boxes." And it was from just laboriously going through the old data, the actual hard copy data, that they were able to find this, which is something I've already alluded to.
- 3.69** For every drop in cholesterol, there was a 22%—for every 30 milligram per deciliter drop in LDL cholesterol, there was this 22% greater risk of just dying. It didn't matter what. Well, that's not a great trade-off. Now, how might we bring all this together? Saturated fats, polyunsaturated fats, cholesterol. Well, to make it very clear, and perhaps a little silly, here are the two buses that are carrying their fatty passengers. And on one hand, the LDL cholesterol that's carrying the saturated fats in itself, it has the saturated fatty acid little chubby students, it's just taking them to school and dropping them off. It's just every day. It's just a normal Monday on the way to school. But, and it works.
- 3.70** And this would be like just a normal macrophage. If this macrophage encounters this LDL with the saturated fats, it's just happy to be there. There's no issue. There's no contention. If, however, the bus is carrying these rotten little psychopaths that are polyunsaturated fatty acids, it's not taking the kids to school, it's taking them to prison. These little juvies. And so it drops the kids off. And rather than having this nice learning environment at the school, it becomes the foam cell.
- 3.71** If a macrophage encounters LDL carrying just native saturated fats, it doesn't turn bad. If the macrophage encounters LDL molecules that are carrying these seed oils, which

have undergone a process called peroxidation, which is the rancidity of the oil, they're not benign. The macrophage senses, the macrophage knows, "You're carrying a bunch of little delinquents. I need to stop you," and it engulfs it. And as it engulfs more and more of these fatty molecules, it becomes the infamous foam cell. And then, as a reminder, it's the foam cell with the LDL consumed, perhaps, that is thought to cause the atherosclerotic plaque.

- 3.72** Now, a final sentiment with regards to LDL and plaque. Let me just bring up this metaphor. Why is a fire truck at a burning house? Or you drive by the fire. You see the fire and you see the fire truck. If you are new to this planet, you would look at that burning house and you would look at that fire truck. Then sometime later, you'd see another burning house and you'd see a fire truck. You might be led, and you see it again. Boy, every time you see a fire, you see a fire truck there. You don't know what the fire truck is doing. You're new to the planet. You might be tempted to come to an erroneous conclusion with regards to what that fire truck is doing.
- 3.73** This might be relevant to LDL cholesterol being at a plaque. Now, I'm going to introduce a brand new idea, which is that atherosclerosis could be the result of infections of the blood vessel. This is known, that if you actually root out an atherosclerotic plaque, you find dozens of infectious bacteria within that plaque. That's long been known. And so with this is growing, this idea that the plaque itself might be the result of an infection. And perhaps LDL is there because LDL has this remarkable ability to actually bind. As the bus is driving through the blood, if there's a bacteria there, it has the ability to reach out and grab the bacteria and bring it into itself and toss it into the back seat, all to take to the liver and have the liver dump it through the biliary duct into the intestines. It's a mechanism of clearing the body of pathogens.
- 3.74** So much to LDL's frustration, it has been viewed exclusively as a mechanism of carrying fats and then contributing to atherosclerosis. But one of its unsung effects is the ability to neutralize and bind pathogens, infectious agents like bacteria, even parasites and viruses. This is why, this explains in large part why people with the lowest LDL levels are so much more likely to die from infection. That's particularly relevant in this day and age over the past few years when everyone's become obsessed with immunity. Perhaps our efforts, this unending war on cholesterol and LDL in particular, has put us in a situation of compromised immunity.
- 3.75** Because if we get infected with a harmful pathogen, we have less of this neutralizing agent, this thing that is attempting to go through the bloodstream and scrub the blood of these harmful pathogens. Beyond the infection, the increased risk of fever and sepsis—sepsis is severe infection—comes with it also a 15-fold greater risk of dying from cancer. That is a remarkable statistic, and it touches on even beyond what I was talking about with LDL's role in infection, but something else. Perhaps LDL has an effect in cancer biology as well.

- 3.76** Now, as a brief tangent, because we haven't had one yet in this lecture so far, so I want to bring in a tangent, which is, where might the infection begin? There's this strong idea and strong evidence that the infection might start in the mouth. Periodontal disease is an infection of the gums. And people with higher incidence and frequency of periodontitis have a much, much greater risk of heart disease. And the thought is that the infection starts in the mouth, but in short order, it starts to move through the body, ultimately affecting the arteries of the heart, leading to atherosclerosis in the foam cell.
- 3.77** Because remember, what was the origin of the foam cell? What was it before it became a foam cell? It was a macrophage. And a macrophage wants to neutralize pathogens, infections, these bacteria from the mouth; the macrophage wants to consume them. So if the bacteria go from the mouth into the wall of the artery, well, then the macrophage wants to go there, too. And as it keeps eating all of these bacteria, it just takes up residence in that spot and becomes the foam cell.
- 3.78** A paper was just published, as we do this in the year of our lord 2023, where they found that, if people frequently, daily, were brushing and flossing, their risk of heart disease went down by half. Interestingly, not mouthwash. Mouthwash had no effect. And I'm not trying to sound like I'm giving you some advertisement for proper oral hygiene, but the evidence is, in fact, quite overwhelming. All of it feeding into this idea that an alternative view is, perhaps this isn't that the atherosclerotic plaque is less a matter of all these dangerous cholesterol molecules just accumulating, and perhaps simply a matter of an infection.
- 3.79** Let's come back to this view then. Why is the fire truck at the burning house? Because this is what we see. We see at a plaque that there's LDL cholesterol there. And the assumption has been, well, just like it might be naively saying, "Every time I see a fire, there's a fire truck. The fire truck must be causing the fire." We know that's not true. We can state that with confidence because we see what happens. Well, with admittedly a little less confidence, we could state perhaps LDL, cholesterol, is at the site of the atherosclerotic plaque because it's attempting to correct the issue. Maybe it's coming by to just pick up the pathogens or to just help that part of the body recover and heal, which does involve an immune or an inflammatory reaction.
- 3.80** OK, now let's leave the dyslipidemia part and then briefly discuss the effects of hypertension. Hypertension is the most common heart disease concern. So many people have it. That's because it is so heavily driven by insulin resistance. In fact, it's almost impossible that a person would have hypertension, consistent chronic hypertension, without them having insulin resistance as the cause.
- 3.81** Insulin resistance influences blood pressure and causes hypertension through three distinct mechanisms. It impacts the nervous system; it impacts how the kidneys handle salt; and it impacts the production of a molecule called nitric oxide. In each of these cases, it's a balance of these two aspects of insulin resistance. In some instances, it's because of the high insulin. In others, it's because of the compromised insulin signaling.

- 3.82** Now, let's talk about the sympathetic nervous system first, because that's what's turned on by chronically elevated insulin. We have two competing nervous systems. Now, mind you, they're overlapping, so the way I'm showing it here is a bit silly. But on one hand, certain times of the day or life, we have the primary activation or the predominance of the parasympathetic nervous system. This is what's commonly referred to as the rest and digest. When you're calm, you're digesting food, literally, or you're just, you're calmly sitting and reading. That's the parasympathetic nervous system. It helps us be calm.
- 3.83** But there are other times when we can't be calm, that for whatever reason we need the body to be ready to go. That is the fight-or-flight aspect of what is the sympathetic nervous system. Well, insulin turns that on. In a normal body, when insulin levels are at a normal level, the blood vessel itself has a normal caliber. It's nice and dilated, which is something we'll revisit in a moment. And the heart is beating at a normal intensity. Not too fast, not too hard, just nice and normal. And we have a normal blood pressure as a result.
- 3.84** However, with insulin resistance, the higher levels of insulin start to turn everything on. It not only narrowed the blood vessel wall, narrowing the blood vessel, which increases the pressure, but also it increases the depth, the power with which the heart is beating as well. And as the heart is beating harder and faster, all of this contributes to an increased blood pressure. This is so important because this is part of what I touched on last time, where when we go to bed, we set ourselves up for failure. When we're indulging and snacking and eating garbage at night, we go to bed and we wonder why we feel so anxious and we're so hot. We're not anxious about anything. We've just set our nervous system to work against us because of what we just ate. And this is part of it.
- 3.85** Now, let's shift the topic onto nitric oxide, but still talking about hypertension. Nitric oxide is a molecule that, when it's in the blood and insulin will help produce nitric oxide, it will induce the relaxation of the muscle around blood vessels. And as the blood vessel can relax, blood pressure will go down. So insulin, in a normal insulin sensitive body, actually helps contribute to lowering blood pressure because of the synthesis of nitric oxide, when the endothelial cells are responding well.
- 3.86** In other words, if the endothelium, the inner lining of the blood vessel, is insulin sensitive, then it works. Insulin will come and knock on the endothelial cell and tell it to produce nitric oxide, which will then permeate through the vessel and cause it to relax. This is actually one of the reasons why, this effect of nitric oxide, where if someone presents to the clinic with a heart attack, they will give them nitroglycerin and just give them a little pill to absorb through the body. It's because that will convert into nitric oxide and help relax the blood vessels.
- 3.87** Insulin can do that, too, if it's working. However, if we take that same blood vessel in a person who is insulin resistant, now insulin is coming and attempting to produce the

nitric oxide, but it doesn't work. There's a compromised production of the nitric oxide. And with less nitric oxide comes a narrower blood vessel. And as the blood vessel starts to compress on the blood, pressure starts to climb, and then we have hypertension.

- 3.88** Now, lastly, let's spend a few minutes talking about salt. Salt is a molecule that is utterly essential for life. We must have it. And the kidneys are very, very good at regulating it, making sure that we have it and it's staying within a normal range. And if salt is going too high for the body to handle, well, then the kidney has a nice way of getting rid of it. But we sometimes, by just dumping it into the urine. But short of that, the kidney will say, hey, blood, you need a little more salt. And so it will push the salt back into the blood, and water will come with it.
- 3.89** And so that gives us a normal amount of water within the blood, what we call plasma. What is called plasma is mostly water. Nevertheless, here the kidneys will sample the blood and the urine and they'll say, hey, blood, you don't have quite enough salt or water. Here's a little salt, and the water will follow the salt because it wants to stay with this concentration of the molecule. That's just a little bit of the physics within the chemistry here. So where the sodium, the main part of salt, is going, the water will follow it.
- 3.90** But as the body becomes insulin resistant, and insulin levels start to climb, insulin will force the kidney to put more sodium back into the blood. And with that sodium will come more water, and with the water comes a higher plasma volume. And as the volume of water keeps climbing in the blood, the pressure will go up as well. And the kidney, in this case, is helpless. It may very well know we shouldn't be dumping more sodium back into the blood. I need to let it dump into the urine. If insulin is elevated, it disrupts the process. It forces the kidney to disobey, even against it, what it knows to be ideal.
- 3.91** Now, so I've presented the paradigm wherein salt is problematic because insulin disrupts it. But not everybody is insulin resistant. And this leads to some of these national surveys finding that, as they look across all of the population, all of this focus on salt doesn't really do much because sometimes salt is perfectly benign. You can eat as much of it as you want and your body just knows how to handle it.
- 3.92** This was very well established in this study here. This study took people and put them into four groups, quartiles, based on how much salt they ate. They found that, across the quartiles, the second quartile was eating about 50% more than the control group, if you will. Then the next group was eating about double what the control group, group one. Then the last group was eating about four times more salt. Then they looked at various health outcomes, and I'm highlighting all the cardiovascular-related ones. They found that stroke, no effect. Cardiovascular and mortality, actually dying from heart disease, no effect across these salt groups. All dying from anything, no effect. Salt had nothing to do with this.

- 3.93** Interestingly, these were normal weight subjects. When they looked at the same set of outcomes in the same rough quartiles of salt consumption in the overweight subjects, every single one of them was correlated to mortality. Now, all of a sudden, the salt appeared to matter. Well, what's the difference between—one difference—between the non-overweight and the overweight subjects? Of course, it would be their insulin resistance. This is something that plays out consistently.
- 3.94** Now, the paradigm that I've shown you is that, as insulin goes up, it forces the kidneys to retain more salt. But what happens then if we just restrict salt? We tell the people eat as less, the smallest amount of salt as you can. Well, there is, in fact, an unintended consequence. One of insulin's many actions is to ensure that the body has adequate salt. And here, if insulin is going up, or rather—let me flip that around—if salt is being pushed down, that makes insulin actually go up higher and higher. Because, again, one of insulin's actions is to try to retain salt.
- 3.95** And if it senses a reduction in salt coming in, insulin starts to panic and says, "Hey, I got to conserve all the salt I can. I got to help the kidneys do this." And so, unwittingly, by restricting salt, we increase insulin, which will increase, of course, insulin resistance. And there are studies that have now confirmed that if you just take normal people and put them on very low-salt diets, you make them more insulin resistant than they were before you actually intervened.

Lecture 4: Challenging the Origins of Chronic Disease

At a glance — Challenging the origins of chronic disease.

- 4.1 All right, now let's move from the heart and go up to the brain. We've spent a lot of time talking about how heart disease is a consequence of insulin resistance. Now I want to teach you guys some principles about the leading form of dementia, namely Alzheimer's disease. In addition to being the main form of dementia—dementia is a broad term of neurological disorder— but Alzheimer's disease is a top-10 killer. In its own right, it is a very lethal problem that just wastes the body. It leads to just broad degeneration. And then not only is life miserable up until death, but it is a leading cause of death.
- 4.2 The classic view of Alzheimer's disease is that it is the result of these squiggly little messy plaques that begin to accumulate in the brain. Then as the plaques accumulate in the brain, they disrupt the signal of neurons. One part of the brain can't communicate with another part of the brain, leading to the prototypical signs of Alzheimer's, the memory loss and the motor or movement function that gets compromised as well.
- 4.3 However, kind of analogous to LDL being at a plaque, why is LDL at a plaque? Is it causing the problem or is it a consequence, and possible solution to the problem? Well, why is LDL in the brain of someone? Is it causing the disease or is it just there—and we happen to see it more as we're looking at the brain more? That's the question I want to ask as I'm planting this little seed of doubt in your mind. So is it caused by plaques? Well, maybe, but there are two real problems that have to be reconciled before we can confidently state that, that it is a plaque-based disorder.
- 4.4 The first being that many people with plaques confirmed at the point of death because there's no other way to confirm it—when you actually do an autopsy or a biopsy, or sample the brain of people who have died, it is the—the plaque is no predictor, is no indicator of who died with confirmed Alzheimer's disease and who didn't. So many people, as these studies, found that older people who would die with no hint of cognitive decline, their brains were perfectly well, and yet they had plaques in their brain. Similarly, people who died with confirmed Alzheimer's may have no plaques in their brain. That clearly is a problem with this plaque-based view of Alzheimer's disease.
- 4.5 Then the second issue is that we have drugs that reduce plaques, and they work very, very well at reducing plaques. But Alzheimer's disease, we don't care. We don't look at our beloved grandpa and say, "Oh, grandpa has a lot of plaques in his brain." No, we say, "Grandpa can't remember. He can't speak well. He can't move well because of the Alzheimer's disease." We care about the disease and its impact on life. We don't care about the plaques.

- 4.6 Now, let's start with this first one, though. How many people will die with perfectly healthy cognition and yet have plaques in their brains? That was this study here, where they noted this, doing biopsies, post-mortem, detecting plaques in people across the whole spectrum of cognition. The plaques simply didn't follow the cognitive decline. Now, this seems a little bit funny, but we do have plaques. We have plaque-reducing drugs. They're available and heavily prescribed, in fact, at a premium. But I want to kind of tell you what I jokingly refer to as a love story with an anti-plaque drug. This is actually quite relevant.
- 4.7 It was just a couple of years ago, in 2020, when the FDA put together a special panel, as commonly happens. The FDA will say, "Here's a drug that we want reviewed. Special panel of scientists and authorities on this topic, please review the drug and give us your recommendation." Well, the panel, this group of true experts, authorities in the field, they reviewed the drug and found that there was no evidence to suggest that it actually improved cognition, which is the only thing we actually care about, right?
- 4.8 And so they said that, given the incredible cost, because the drug is very, very expensive, they said this should be rejected. However, the FDA overruled the panel because they noted that it is so good at reducing plaques. And then we think, they said, it's "likely" to help cognition. Although there was no evidence to support that, they said it should be approved. In fact, the deeper part of this story is that part of the justification for the FDA was, apparently, that they didn't want to deny this drug because they didn't want to discourage other drug manufacturers from continuing to pursue Alzheimer's disease as a therapeutic target.
- 4.9 To hell with the patients and their limited budget, and to hell with the fact that it might not actually help them; there was no evidence to suggest it would. They said, well, it does reduce plaques, and so we should approve it. And again, they overruled the panel of authorities, of experts on that topic. That's not great, is it?
- 4.10 Another problem with the plaque-based theory of Alzheimer's disease is that it all might be based on a lie. When this report came out just a few months ago, it was at the mid, end of 2022, there was this big bombshell report in Science magazine. They found that the initial evidence upon which the plaque-based theory of Alzheimer's disease was built was falsified. Now, that was particularly validating for some of us who have been cynical of the plaque-based origins of Alzheimer's disease for a long time, because it immediately goes right to the foundation upon which the whole, as I noted, the plaque-based theory of Alzheimer's is built. "Was it all built on a lie?" is the question that appears to be answered with a yes, it probably was.
- 4.11 Now, however, there is a player within Alzheimer's disease that has a lot of attention. It's famous, and for good reason. That is this molecule called APOE4. So if plaques aren't causing the Alzheimer's disease, why is it that people who have the APOE4 gene do have a two or three times greater likelihood of developing Alzheimer's disease? That

is a true statistic that bears out across the population. It's never really been refuted. It's a very, very strong line of evidence.

- 4.12** So briefly, APOE, among its many effects, this enzyme, this molecule, has the ability to clear out plaques, in addition to doing all kinds of other things. APOE4 is a variant. It's not a mutation, it's a variant. It occurs within the population. If someone has an APOE4 gene, that just means this gene isn't working quite as well at clearing out the plaques, and the brain may have more plaques. Now, again, if Alzheimer's disease were purely a plaque-based problem, that would explain it. But I just showed you evidence wherein the scientists found that if you reduce plaques, it doesn't improve cognition.
- 4.13** Perhaps it's not the plaque effect of APOE4 explaining their greater risk of Alzheimer's disease, it could be a metabolic issue. The brain is an insulin-responding tissue. Every tissue responds to insulin. Every cell of the body has those insulin receptors. But the brain relies on insulin in part to get its glucose. Now, it has an insulin-independent mechanism as well. But if here's the brain, and glucose is coming into the brain from two sides, one side is dictated by insulin. Insulin must come to the brain and knock on the door, open the doors for the glucose to come in.
- 4.14** But like so many other tissues of the body, the brain can become insulin resistant. And as the brain becomes insulin resistant, then that responsiveness to the brain becomes compromised. And while insulin is trying to open that glucose door, it is less able to do so. Then the brain starts to go a little hungry. That is an issue because the brain is a high metabolic rate organ. It is among this trinity of high metabolic rate organs that are separated by any other tissue by multiples, a great deal.
- 4.15** The kidney has the absolute highest metabolic rate, and then the heart, and then followed very, very closely by the brain. The brain is just a very busy organ, hopefully very busy now as you're learning and understanding so much, more than when you're sleeping, for example. But even then, it's still busy and has a very quantifiably high metabolic rate. That simply means that the brain is very sensitive to nutrient deprivation. If it can't get what it needs, it starts to work poorly. Its function will be compromised.
- 4.16** So this is kind of like what might be happening. Whether the person has APOE4 or whether they just can't use glucose through any other means, basically, the brain starts to run out of fuel. That's not good for a high metabolic rate organ. It needs fuel to work well. Now, thankfully, the brain is so hungry that it doesn't rely completely on glucose. The brain is itself a hybrid, but it doesn't use fat. It uses a product of fat called ketones. Ketones are the alternative fuel for the brain. And the brain will readily use either, whatever is available.
- 4.17** Now, let's just take a little break and define what ketones are. They're a loaded term. People will utter ketones as either a curse, or they will talk about it in these glowing, almost holy references. A ketone is simply a product of fat-burning. When insulin is low

for an extended period of time, not even that long, about 12 to 16 hours after eating, insulin will have been sufficiently low that fat-burning will have been sufficiently high.

- 4.18** When insulin is low for this period of time, it's almost as if the body can't stop burning fat, even though maybe here is the amount of energy my body needs. I need this much energy to work. But if insulin is low, I keep burning fat, keep burning fat, keep burning fat, and now I go beyond what my body actually needs. But I can't stop burning fat because insulin won't allow the fat-burning to stop. This excess, if you will, in this overly silly comparison or analogy, this excess is what becomes ketones. Ketones are simply a product of a lot of fat-burning. That is the very origin of the molecule.
- 4.19** And you can never guess which tissue makes the ketones. Who is the unsung hero of all nutrients? Yeah, it's the liver. The liver is the one who's making the ketones. It starts burning fat when insulin is low, and if insulin stays low, it just can't stop. And so it starts burning the excess into ketones or turning the excess into ketones, which is wonderful because the brain is so hungry for ketones. There's no tissue in the body that will consume ketones as readily as the brain does. The moment the ketones hit the bloodstream, the brain starts taking them in.
- 4.20** In a fed state, if you have a person who eats this, they go eat a sandwich, then after that, we know that the brain will use 100% glucose because there's just so much of it there. However, a few hours later and further, the brain starts to shift. As we enter the fasted state, like 12 or 16 hours or so, or heaven forbid, an actual 24-hour fast, then the brain has shifted so that the vast majority of the fuel that it's getting is coming from ketones.
- 4.21** What's so important, even though the brain is getting 65, 70% of its energy from ketones, the actual ratio of glucose to ketones is not even. You would look at this transition and think, well, here, we ate, we had a sandwich, our insulin went up, and if insulin goes up, ketone production stops. And so the brain has no choice. That's true. All of that is accurate. There are simply no ketones there for the brain to eat.
- 4.22** But in a fasted state, glucose could be about 5 millimolar. Ketones may only be about 2 millimolar. Even still, the brain is using most of its energy from the ketone. So people will commonly make these ridiculous statements like the brain needs glucose or the brain prefers glucose. That is demonstrably false, the latter statement, particularly. We cannot say the brain prefers glucose, because if glucose is here and ketones are here, if the brain preferred glucose, it would only be getting the glucose because there's plenty. But the fact is, the brain prefers ketones, because even though ketones may be less than half of what the glucose is, it is using 2 to 1, almost 3 to 1, more ketone than glucose.
- 4.23** Thankfully, so the brain is hungry, it has a high metabolic rate, and we have two nutrients that can fuel that growth. But not surprisingly, as this topic focuses on insulin resistance so much, in Alzheimer's disease, one side of this becomes broken. This is

something that doesn't just happen in Alzheimer's disease. Even in early-stage Alzheimer's, what we would just call early cognitive decline, we can detect that the actual amount of glucose the brain is using is down by 13%, which seems small, right? It would be small in its own.

- 4.24 But if this is your sole source of nutrition for the brain, because you don't have any ketones, because most people don't, then that becomes catastrophic. And most people don't have ketones. We would be tempted to say, well, brain, just eat the ketones. But remember, from last time, the average individual is living in a constant state of elevated insulin. Every waking moment and maybe hours into their sleeping moments is a state of elevated insulin. Anytime insulin is high, ketone production is turned down, because insulin turns off fat-burning, and fat-burning is a necessary prerequisite to ketone production.
- 4.25 Now, however, in these brains that aren't metabolizing glucose well, thankfully, they have no compromised ability to burn ketones. So ketone-burning is working perfectly. The tragedy is most people just don't have any ketones for the brain to eat. This was a paper from my own lab conducted by my PhD student, Erin. In this paper, we had access to postmortem brain samples of people who had died with confirmed Alzheimer's disease versus people who had died with no evidence. They were matched for ethnicity, for age, for sex, for lifestyle habits, for obesity. Everything was matched. The only difference between them was some group had Alzheimer's, some group didn't.
- 4.26 And we quantified the expression of genes related to glucose metabolism versus ketone metabolism. And we found that literally every single gene involved in the brain's ability to burn glucose was significantly stunted in the brains that had Alzheimer's disease. In contrast, quite stark, the genes involved in the burning of ketones, those enzymes, were unaffected. They were normal.
- 4.27 This further plants, this builds this foundation upon which the previous study is built, finding the actual differences in metabolic rate of the glucose versus ketones. We went one step deeper to find that the insulin resistance becomes so profound that it directly affects gene expression of those enzymes involved in glycolysis, but not the burning, so the genes involved in glucose-burning, but not ketone-burning.
- 4.28 Now, another paper was published a number of years ago from Finland. This is a paper from Finland, and it is a correlational study, which means you can only ever determine a coincidence, but the results are just so compelling; I have to mention them. They had this cross-sectional study of a large, large population, and they found, looking, they were trying to find all of the variables that could have helped predict Alzheimer's disease in people. And they found there were eight. Eight variables stood out.
- 4.29 The first three aren't surprising at all. Age is a predictor. The older you get, the more likely you are to have Alzheimer's disease. Education, interestingly, much to my

personal delight, was also an indicator of reducing the risk of Alzheimer's disease. But that just reflects the idea of challenging your brain. The brain that gets challenged more often will maintain cognition longer. Then lastly, the APOE4 genotype. If someone had the APOE4 gene, that was the strongest variable among all of them, likely for reasons I mentioned, probably not plaques. Very probably that the APOE4 will directly disrupt the insulin receptor, preventing insulin from allowing glucose to come in.

4.30 But the other five variables were all metabolic-related. All of them. These were the only eight statistically significant variables. There were no more than these, despite a very, very long list. It was these three stereotypical ones, and then it was all these markers of metabolism. Markers of glucose handling—how well does the body use glucose? How does it respond to a glucose challenge? Which itself are all just measurements of insulin sensitivity, insulin responsiveness. And that's what the latter point, the latter two points, were also directly addressing.

4.31 So the people with compromised metabolism were among the most likely to develop Alzheimer's disease, independent of any other variables. This is why some physicians and scientists refer to Alzheimer's disease as type 3 diabetes. I don't like that term. I don't like to call it that because that sounds like we're complicating it. We should just say Alzheimer's disease is an insulin resistance of the brain. And that explains the compromised glucose metabolism.

4.32 Because if insulin isn't working, and APOE4 makes that worse, but even independent of APOE, if insulin isn't working, glucose can't get in. And that just increases the need for this very energy-hungry brain to rely on the other fuel that it has, namely ketones. And so let's use this as another brief tangent. The role, the interaction with ketones and cognition. Fats can have two fates. A fat can be stored in fat cells or it can be burned. And as I noted that when fat is burned at a high rate, it will begin to produce ketones. But not all fats have the same effect. Some fats burn better than others.

4.33 And to highlight this point, I want to split the fats up into these three canonical groups, based on how big the fat is. We have long-chain fats, which are indeed long. It's a long carbon chain. Some are in a family that are a little shorter, that we call medium-chain fatty acids or fats. Then the shortest of all, going to the very, very short, which is the short-chain fats. These all do exist in nature, but the vast majority of what we eat is long-chain fats, which is fine. Long-chain fats can be stored, which is not a bad thing. We need fat in fat cells for reasons we've discussed previously, and they can also be burned very, very well. Long-chain fats. So either of those can be the outcome. And insulin is the sole determinant of which one occurs. If insulin is elevated, it will direct the long-chain fats to storage. If insulin is down, then it will direct the fats to being burned.

4.34 Medium-chain fats don't really have a regulator. There is very, very little potential to store a medium-chain fat. Most of the medium-chain fats in that small little family have

no option. They cannot be stored. They only can be burned, but not all of them. So I have a little bit of fat storage, but a lot of fat-burning.

- 4.35** And then lastly, the short-chain fats have no potential for storage. They have to be burned. Even if insulin is elevated in both of these instances, some of the fats simply can't be stored. And so the body will burn them. They have to burn them. There's no alternative. And the more you're burning fat, the more you're making ketones. So these shorter fats are more ketogenic than long-chain fats, simply because they have no other option. They must be burned.
- 4.36** Now, where do we get them from? Briefly, to finish this tangent, long-chain fats are anything that you're eating, basically. All of the natural fats, whether it's coming from fruit fats like coconuts, avocados, olives, there can be a little bit of long-chain fat in all of those. Meat, dairy, animal fats, you're going to get long-chain fats. Less common in nature are the medium-chain fats. Coconut oil is the main source of medium-chain fats, but also any goat dairy. In fact, the main medium-chain fats are named after the goat—caprylic, capric, caproic. They're all derivative of the word for goat.
- 4.37** Then lastly, the short-chain fats are products of fermentation. When bacteria are eating starches and sugars, they produce these short-chain fats. That's why, when you're eating, you drink dairy and it's not sour, but when you eat fermented milk, sour milk, or yogurt, it's sour actually because of the short-chain fats. Short-chain fats, when they hit the mouth, activate these sour taste receptors. That's how you can tell. That's the difference between just a bowl of cabbage versus some kimchi, among the other things added to it. But part of the flavor, the bitterness and the sourness, will come from the short-chain fats.
- 4.38** So short-chain fats are products of fermentation, which used to be very prevalent in our culture. Before the modern age of packaging foods and preserving them in refrigerators to keep things lasting long, much of the dairy we would eat or even some of the plants we would eat would have been fermented. So we would have had a lot more frequency, a lot more consumption, of short-chain fats than we do now.
- 4.39** Now, let's talk about coconut oil for just a moment because there are actual case studies published, documenting improvements in cognition in patients with Alzheimer's disease by doing nothing more than giving them tablespoons of coconut oil, maybe mixed in a drink to make it a little more palatable. But regardless, in this case study, the only intervention was giving the patient with Alzheimer's disease coconut oil. It's actually a very compelling case study.
- 4.40** They had the patient undergo several cognitive tests, one of which was drawing out the face of a clock, an analog clock, one, two, three, four, and drawing the hands. They have the patient with Alzheimer's try to do that because that's a complicated feature. And they documented the absolute chaos of attempting to draw out the clock face and then repeating it after interventions with coconut oil. And it wasn't perfect, far from it,

but it was significantly better. And they reported that the patient was able to get himself dressed as well. So meaningful improvements. Now, I'm—and no one else is making the claim that this is actually reversing the Alzheimer's disease, but it is improving it, not curing it.

- 4.41** Another intervention, more and more as we've begun appreciating the direct value of ketones to the hungry brain, is just giving a person ketones. This has become its own supplement family. Just because we know the benefit of ketones, particularly on brain function, is so substantial, even here, giving a patient with Alzheimer's disease a ketone supplement will immediately begin to improve cognition. Not cure the problem. Please don't think I'm being that bold, but it will improve it.
- 4.42** All right, now, as we end this discussion of the brain, I just want to have one last note, which is the role of migraines. I won't say any more than this, but migraines are not the same pathology as Alzheimer's disease, but they have one thing in common, which is what's called brain glucose hypometabolism. Both of these disorders have confirmed quantifiable objective reductions in the brain's ability to metabolize glucose. It's no surprise that migraine sufferers will experience dramatic improvements when they have higher ketone levels, in a state that's commonly called ketosis.
- 4.43** That actually is an old area of research. The first paper I ever saw published was a paper from 1929, documenting this list of about a dozen people who suffer from migraines. And it's a beautiful study, where the physician who was doing this goes through a table, through column after column, noting the direct improvements per patient, giving a little personal note, which is not something you find anymore in science manuscripts. But the results were remarkable, and it was followed up with more reports in the 1930s. So that idea of the ketone fueling the hungry brain in the case of migraines is very long. It's a long story, a long history to it.
- 4.44** All right. The final sentiment on dementia and cognition is this tragic irony that the brain is swimming in a sea of glucose that it can't use. And maybe because I like to pretend I'm a little literary from time to time, when I thought about this, I thought about some bizarre version of the Rime of the Ancient Mariner, which is an old poem/story by Coleridge. And I thought, well, it's not the—because the sailor, he's becoming dehydrated. He's dying in the midst of a body of water. And he bemoans, "Water, water everywhere, nor any drop to drink." It's like the brain saying, "I'm swimming in a sea of glucose. Glucose, glucose everywhere, but I can't get it." And so it's getting hungry. It's drowning in this sea of glucose. It can't use it.
- 4.45** But there's a savior, and basically the ketone throws in the life preserver, allowing the brain to become buoyant and come back up to full function. Now, as a point of transition, we look at the compromised brain, its disability to use glucose because of its growing insulin resistance. We typically think that this is something only relevant in middle-aged or older people. This is something that has, in fact, even been quantified in young women in their 20s.

- 4.46 In this study, they found that women with higher insulin resistance, these young women, had a 14% lower glucose consumption of their brain. That was actually that same range of people who had confirmed cognitive decline in older age. Now, they were still cognitively sound. There was no deficit in their thinking whatsoever. But even still, it's clear that this is something that begins happening early in life. Now, they weren't looking at this, these young women, because of their cognitive decline. There was no concern about loss of memory or function. It was because they had a fertility problem called polycystic ovary syndrome, or PCOS, which is a disease of insulin resistance.
- 4.47 Let's transition into fertility problems. And I want to discuss, in the time we have, the two most common forms of infertility, polycystic ovary syndrome in women and erectile dysfunction in men. With PCOS, we'll just focus on PCOS, but with erectile dysfunction, I want to introduce another concept of testosterone metabolism as well. That's also relevant in PCOS.
- 4.48 Now, in both of these instances, they are a consequence of insulin resistance, but they're a consequence of the different aspects of insulin resistance. Whereas erectile dysfunction is a problem of insulin not working well, PCOS is a problem of too much insulin. Remember, they are the same problem. They are the two sides of the same coin that we just call insulin resistance.
- 4.49 Now, when it comes to PCOS, it is defined clinically by meeting the so-called Rotterdam criteria. A woman who has higher than expected levels of androgens, that's testosterone. Testosterone is the poster child of the androgens, the so-called male hormones, but both sexes have them. Also, she has infrequent menstrual cycle, which is known as oligomenorrhea. Then lastly, the problem that gives the disease its name, the polycystic ovaries. And we'll talk about the origins of that one in particular.
- 4.50 What all of these have in common is this looming, underlying insulin resistance. The elevated insulin is disrupting all of it. Now, this was a study that looked at insulin resistance in women, matched in every metric, same body weight, same age, everything else is the same. But some women had PCOS, some didn't. They found that both of them, when they were looking at whole-body metrics of insulin resistance, they were the same. We'll talk about some of those metrics later, in the next lecture. But for now, we would know that the whole-body measurement of insulin resistance was the same. They appeared to both be very insulin sensitive groups.
- 4.51 That doesn't jive with what I'm saying, which is that PCOS is an insulin resistance problem. But when they actually looked at the degree to which the insulin resistance went right to the origin, the fat cell—remember, the fat cell's the first domino that fall into the others. That's where it first begins. When they looked at a marker of fat cell insulin resistance, they found that the fat cells were significantly, manifested a significant insulin resistance. It just hadn't impacted the whole body yet.

- 4.52** Now, let's zoom down to the ovary level itself. Within every ovary, there are eggs, ova, that are waiting to grow and become full mature eggs through this process of maturation. Through every ovulatory cycle, as estrogens and other hormones are going through their cycle, as the estrogens in particular are climbing, they begin to promote the maturation of the follicle. And then one follicle, one of these budding eggs, becomes the egg. It actually goes fully mature. It beats the other ones to maturation and then gets released, and down into the fallopian tubes to be fertilized or just passed.
- 4.53** Regardless of what happens downstream of that, once that one egg has matured and the ovulation has occurred, then everything resets. Estrogens and other hormones come back down to normal. And that signal then, as the estrogens and others go away, all of these other growing eggs, these follicles, will degrade into nothing unless you don't have an ovulation event. If there's no estrogen spike, then the eggs are maturing, but you don't have one that becomes dominant and actually ovulates, and so they all stick around. Then the next cycle comes, and you have more ova develop and mature, but no actual ovulation. You need the ovulation to occur for all of the other follicles to go away.
- 4.54** And so as this happens month after month, cycle after cycle, what was once a small, tidy little ovary gets bigger and bigger and bigger with all of these cysts. And every cyst is simply the remnant of follicles or developing eggs that never went away. It is the remnant. They can't go away until ovulation occurs. Then they get the signal, OK, we didn't need you after all. Retreat into the background.
- 4.55** Now, let's compound this. Let's start talking about how insulin comes into this. The normal ovary has a very, very high expression, a healthy ovary, of an enzyme called aromatase. Aromatase is a fascinating enzyme because its job is to take androgens, like testosterone, it being the most famous, and actually converting the testosterone into the estrogens. Now, I'm saying plural because estrogen—there's no singular hormone called estrogen. It's a family of hormones, so we'll just say estrogens, and aromatase is responsible for them. So no surprise that the ovary has a lot of aromatase.
- 4.56** But it is a little-known fact that all estrogens in men and women, because both sexes have estrogens, were once testosterone. And whether it's through the testes or the ovaries, there is a conversion into estrogens. Of course, the ovary does it much more aggressively than the testes do because of the relevance of estrogens within female physiology. Regardless, it all goes through aromatase. And wouldn't you know it, insulin who insists on having its way and being heard and being the life of every party, or the wrecker of every party, it inhibits that enzyme.
- 4.57** As insulin levels are climbing in the insulin resistant body, it directly inhibits aromatase's actions. And now aromatase is attempting to take the testosterone and release it as estrogens, but it can't. The result of this is a body, a female body, where the ovaries are producing up to 13 times more testosterone than they should. But that

comes at the expense of the estrogens because all of that production that should have been happening stops.

4.58 So the high insulin levels inhibit aromatase, leading to ever-higher testosterone, that's the hyperandrogenism, and ever-lower estrogens. That combination, that disrupted mix of sex hormones, prevents ovulation. And it's the preventing of ovulation that leads to the polycystic ovaries, which is the fundamental feature of the disease.

4.59 All right, now let's transition to the testes to talk about erectile dysfunction. There are two aspects to this I want to mention. One is insulin, and then one is the fat cells. I didn't have to invoke the fat cells when it came to PCOS, but we kind of do to have to understand what happens in the man who is experiencing metabolic disruption and fertility problems. So, firstly, with insulin, no surprise, testes don't operate the same way that ovaries do. Typically, of course, the testes are very active at releasing testosterone, defining to a degree, allowing the infant to become male in utero and then allowing the boy to become a man.

4.60 Insulin stops that. When a man has high insulin levels, it inhibits the testes' ability to produce testosterone. Testosterone production is compromised. It goes down when the male body has hyperinsulinemia. It's a direct effect at the testes. But we add to this an insult to injury, which is by introducing the fat cell. When the fat cells in a man get bigger and bigger and bigger, they start to act like an ovary. It's as if there's an ovary in the adipose, where the fat cells themselves begin expressing the enzyme aromatase, just like the testes and the ovaries— the ovaries especially, testes very little.

4.61 So they begin to acting like the ovaries do. And so not only are his testes, perhaps because of the high insulin, producing very little testosterone, but what little is being produced literally comes into the fat cell and gets released as estrogens. As I said, adding insult to injury. So his fat cells begin working against him, increasing the release of estrogens while decreasing the free circulating amount of testosterone.

4.62 This, I submit to you, is the singular cause of why we have this now epidemic of men with low testosterone. This is clearly not something that is suitable to evolution or survival of the species. How can men within a generation suddenly have gone from being perfectly virile to being low? Everyone's low T, everyone's on testosterone nowadays. It's not because the testes are broken, it's because the metabolism, namely the high insulin and the insulin resistance, isn't allowing sex hormone synthesis to occur. Again, it's not a problem of the testes. Those are fine. They're waiting to do their job. It's the insulin that's disrupting it.

4.63 So rather than going on all this supplemental testosterone, maybe the fellow should consider changing his diet through strategies we'll get to next time. One final comment on men and testosterone is that the male body appears to be remarkably sensitive to dietary fat. That could be for many, many reasons, but we know from multiple studies that the more a man goes on to a low-fat diet, the lower his testosterone gets and the

lower his sperm production gets as well. So I like to joke that, if you're a man and you're trying to follow a low-fat or vegan diet, it's a self-correcting problem because you most certainly won't pass it on to the next generation because you won't have a next generation. So in men, that just really, really appears to be a little problematic.

4.64 Then one final problem is the erectile dysfunction itself. So up until this point, I was talking about testosterone, which is very relevant to normal erectile function and fertility. But with the actual erectile dysfunction, we can go to the blood vessel. I'm not repeating the animation that I showed you earlier with regards to hypertension, but the same thing happens here. In a man, normal erectile function is facilitated by vasodilation. He has a dramatic increase in the blood vessel size, which enables normal erectile function.

4.65 If his blood vessels have become insulin resistant, those blood vessels cannot produce as much nitric oxide, and they stay constricted when they should have been dilated, leading to compromised blood flow, which is manifested with erectile dysfunction. This is, in fact, considered by some to be one of the earliest signs of insulin resistance in otherwise healthy men. If you have a young man who comes to the clinic with complaints of erectile function, it could be that he actually has insulin resistance, and that might have just been the very first sign of it.

4.66 All right, let's move on to one of the most sobering of all. In the list of diseases we've been talking about, Alzheimer's disease maybe trumps them all. In some people's minds, cancer may even beat that as well. It's a sobering disease that we understand very, very poorly, despite the fantastic amount of resources that have been devoted to trying to elucidate its origins, its ideology, and understanding where the cancer comes from, and then with that, knowing how best to treat it. It could be a result of looking at it wrong.

4.67 So how does cancer start? The primary theory is that it's purely a genetic disease, that there is some mutation within the genome, within the nucleus, that results in this aberrant production of proteins, which end up resulting in an aberrant growth of the cell. Cancer is a disease of growth. Cells are growing uncontrollably. And again, to state that in reverse, because of mutations in the genome, in genes within the nucleus. So there is this genetic theory of cancer, and there is a relatively younger theory, but not as young as most people think, namely the metabolic theory. Each of these have a representative organelle within a cell.

4.68 The genetic theory is based on problems within the nucleus. The metabolic theory is based on problems within the mitochondria. I want to share with you a series of mind-blowing experiments that were done by the Seyfried lab at Boston a number of years ago. By way of background, and you need to orient yourself with these cell types as we go through this story of mine—on one hand, we have a tumor cell, and a tumor cell will beget tumor cells. As that cell is growing and dividing, its daughter cells will be tumor cells, which then further grow and divide.

- 4.69 A normal healthy cell will give birth to normal healthy daughter cells as it undergoes cellular division. So they stay in their own respective lanes, one being sick, cancerous tumor cell, one being healthy in a normal cell. Now, the beauty and the genius in the model of what they did is that these scientists took a part of each of these cells. They took the mitochondria from the tumor cell and the nucleus from the normal cell. OK, so far, so good? So that was a sick mitochondria, and it was a healthy nucleus. And then in another set of cells, they did the opposite. They took the healthy mitochondria, but they took the tumor nucleus.
- 4.70 And what do you think happened? It depends on the theory, right? But if the genetic theory holds that it all starts because of bad mutations within the genome of the nucleus, then this one should be the problem one, the one with the tumor nucleus. Assuming then in that theory that the mitochondria, the source of all metabolism, if you will, has nothing to do with it. That's not how it played out. Those cells were normal. They manifested with perfectly normal growth in response to growth inhibition. In other words, there was no more tumor, no more cancer.
- 4.71 In contrast, the cell types that had the normal nucleus, but the cancer mitochondria, were aberrant in their growth. Those became cancer cells. That directly challenges the idea that this is purely a genetic or nuclear origin. Now, however, the lab where this work was done, they actually posited a different paradigm, not stating that the nucleus was irrelevant. Their theory goes, with evidence to support it, some of which I've shown you, is that the fundamental problem occurs in the mitochondria. And the disrupted mitochondria starts to produce oxidative stress, which is something everyone's heard of.
- 4.72 Strictly speaking, it produces things called reactive oxygen species, but we just can refer to that as oxidative stress. Then the oxidative stress, having its origins in the mitochondria, then disrupt the nucleus, leading to the mutations within the nuclear genome, causing the cancer as we know it. The theory, their theory, the metabolic theory of cancer, isn't that the nucleus isn't relevant, but it is that our ongoing effort to just address the nucleus won't work because by trying to fix the nucleus, we're ignoring what is the actual origin of the disease—again, as the theory goes, which is as good as any theory that exists in cancer biology today.
- 4.73 Now, as I noted, this is not new. The first evidence of this has existed for almost 100 years, thanks in large part to another hero of mine, Otto Warburg. His contributions to science were so remarkable that beyond winning the Nobel Prize, he was allowed to maintain his lab as a Jewish man in Nazi Germany. He was considered so remarkable that they didn't dare mess with him. But he's a remarkable character and worth anyone's time to look into. But he was the one who gave birth to this idea, noting that when it comes to metabolism in cancer cells, they don't have the ability to burn nutrients like normal cells do.

- 4.74 They're stuck only fermenting sugar. And notice the word fermentation. That is a process of burning sugar for fuel outside of the mitochondria. It's a way that every cell has of just using sugar in a quick and dirty way. Sugar can also, glucose can also be burned through the mitochondria, but any other fuel, whether it's fats or ketones or lactate, any other fuel must be burned by the mitochondria. So his observation was, and that led to Dr. Seyfried's work later, decades later, was that the cancer cell has broken metabolism, where it's stuck burning sugar outside the mitochondria through fermenting it just like bacteria do.
- 4.75 And so this is basically what the Warburg effect noticed. Healthy cells can burn anything for fuel. Cancer cells are completely reliant on glucose, and indeed, they burn glucose at about 200 times the rate of normal cells do. They are fantastically glucose-dependent. And wouldn't you know it, we put the cancer cell in this perfect metabolic milieu. The average individual who has hyperinsulinemia and insulin resistance has a cancer cell that is swimming in a sea of insulin, and insulin signals growth. To every cell, cancer cells, especially, as we'll get into in just a moment. But the growth alone wouldn't be enough. There must be fuel.
- 4.76 In this same body, as I said, with this optimal metabolic milieu, we have hyperglycemia or this constant consumption of glucose, which can match the growth signal of insulin by providing the cell with all of the fuel it needs. It's basically like a fertilizer for the cancer cell. The insulin resistance, promoting the growth, and the hyperglycemia, providing the fuel for that growth. It's a wicked combination.
- 4.77 Now, let's just talk about these two cancers specifically. Breast cancer is the most common cancer in women. Prostate cancer is the most common cancer in men. It's interesting to note the mechanism whereby we can determine where cancer cells are. If we have a patient who is suspected of having cancer or we've confirmed a mass somewhere, a tumor, we can actually inject them with a radio-labeled glucose. This is a glucose molecule that will give out a subtle radioactive symbol, a signal. It's not harmful. The person is not going to get radiation poisoning. It's so subtle. But it's enough of a flare that we can then image where the glucose went in what's called a PET scan.
- 4.78 We can measure the radiation of that glucose, and we will see wherever all of that glucose goes, whether it's a breast tumor or a prostate cancer, that's where the cancer is. It's just an interesting diagnostic leverage of cancer biology. We know that the cancer cell is so hungry and so totally reliant on glucose that we can use that glucose to see where the problem is. Unfortunately, that hasn't led to a solution to the problem, but it's part of it, at least knowing where the cancer is.
- 4.79 Now, in both of these cancers, and I highlight them because they are the most common, they are the most common. The fact that they have a metabolic origin is just icing on the cake for a guy like me, who believes cancer is the root of all evil with regards to the health of the body. In breast tumors, they are known to have seven

times more insulin receptors than normal breast tissue. So if you just take a biopsy within the breast, here's a tumor, here's the normal breast tissue, the tumor side of the breast will have seven times more insulin receptors. That's part of its mutation. It's basically the cancer's way of saying, hey, insulin will help me grow better, and that's my singular reason for existing, so I'm going to enhance my ability to respond to the insulin.

4.80 But we see similar effects with prostate tumor cells, where prostate cancer also, part of its mutation is to over-express insulin receptors, making it more responsive to hyperinsulinemia, facilitating a more exaggerated growth. Then lastly, we know that men with insulin resistance are three times more likely to die from cancer. It increases the mortality of the cancer significantly. Similar statistics are seen in women with breast cancer, independent of any variable, obesity or heart disease or smoking habits, it doesn't matter. If she's insulin resistant, her risk of developing breast cancer and dying from it goes up significantly.

4.81 Now, I'm cautious when I speak about cancer. My last note on it— cancer is, among all the diseases I've discussed, the most poorly understood. I am not claiming that insulin resistance is the cause. I don't believe that it is. What I believe happens is that there is some event that initiates the cancer, whatever it may be. I presented evidence that it could be mitochondria, compelling evidence. Regardless of the origin, the insulin resistance clearly exacerbates the condition. It maybe didn't light the fire, but it sure as hell is pouring gasoline on it.

4.82 All right, so in the time we've had together, we very briefly at the beginning, we discussed why, some of the reasons why insulin resistance has gotten so bad. I emphasized the fact that we might not be looking at it the right way. We aren't detecting it as early as we could or ever, accurately, because of our overlooking insulin itself. Then we talked about all the chronic diseases, well, in these ones here, that are derivative of insulin resistance. We went through diabetes and heart disease. We then moved into dementia, Alzheimer's disease being the main form of dementia. I teased you with that mention of migraine headaches as well.

4.83 Then we went to infertility, the most common forms in men and women, erectile dysfunction and PCOS, respectively, and finally, cancer, one of the most poorly understood of all the diseases, but nevertheless impacted by this most common of all problems, namely insulin resistance. Thanks for listening, and I'd be happy to answer any questions on this.

4.84 OK, so yesterday we talked about protein's potential to increase insulin, and I was wondering whether whey protein and, what is it, the other protein I mentioned.

4.85 Well, plant protein.

4.86 Plant protein. The difference between those two, whether or not it had an effect on insulin.

- 4.87 Yeah, that's a great question. So the insulin effect depends on the amino acids consumed. And so whether it is a protein coming from whey—and the nice thing about any animal protein, any animal protein, is that you know it will have all of the amino acids. The problem with plant proteins is that you become less and less confident. But let's go with soy, which is the most complete of any of the plant-based proteins. Again, the insulinogenic effect of the protein is determined by the amino acids contained in that protein.
- 4.88 So both of them, if we matched it to amino acid content, both of them would have an equal effect on insulin. If any, remember, not all body types will have an effect. A young healthy guy may ingest a significant amount of both of these proteins sources and have no effect whatsoever. Giving a type 2 diabetic that much protein would very likely induce an insulin response to both. So it's not the protein per se, the source, it's the amino acids themselves. And then it would be the deeper question of, what's the composition of amino acids from the animal protein versus the plant protein?
- 4.89 Again, animal proteins are superior by every conceivable metric. Plant proteins are a little problematic. A little. One, because they might not contain the full complement of amino acids. And two, we do know that the more you refine a pea, for example, to get proteins from it, a pea is so deficient in amino acids that you have to get a thousand peas and refine it to get the amino acid. It's the amino acids that you want, to get that pea protein, but you get some of what you don't want, including things called anti-nutrients. These are actual honest-to-goodness molecules— phytic acids, tannins, trypsin inhibitors—that will prevent the body from digesting the protein as it comes in.
- 4.90 And another reason I encourage people to be careful with plant protein, especially if they're relying on it as their singular source, you're going to have a very hard time getting enough of what you actually want. Once again, there's a little bit of hubris here. If I just riff on that for one more moment, which is, once upon a time, if our earlier humans wanted to get proteins from plants, they fermented it. I'm such a fan of fermentation because fermentation, like, say, old soy, fermented soy, the fermentation allows the degradation of those anti-nutrients, which enhances absorption.
- 4.91 In a patient with insulin resistance, we will try to improve the metabolic flexibility and the capacity to use fat for food. But can we obtain benefits of the administration of some external ketones, or could that intervention interrupt in some way the natural ketogenesis?
- 4.92 Yeah. Oh, what a wonderful question. So if we, looking at someone who has metabolic syndrome, they're insulin resistant, they have that poor metabolic flexibility. So the metabolic flexibility, as a reminder, was the ability to eat, burn sugar, not eat, and then transition to fat. If you can't transition to burning fat, you're not going to make ketones, which we discussed does have explicit benefit in the case of the brain. That has given birth to these exogenous ketones. So the question, would taking ketones help?

- 4.93 Taking ketones would definitely help if you need ketones. So if this were a person, if we were talking about an individual who was worried about cognitive decline and they had a hard time getting into ketosis because of their insulin resistance, which can happen, then I would say a resounding yes. The exogenous ketone, the ketone supplement, would absolutely be beneficial. If we were talking about an average individual who's overweight and diabetic and they just want to lose weight, then they adopt a diet, as we'll talk about next time, where insulin comes down and that might lead to sufficiently high fat-burning that they have higher ketones.
- 4.94 The ketones are just a beneficial byproduct in that person. It's not the ketones that are providing the insulin sensitivity benefit, the reduction in diabetes, the improved any of these diseases other than Alzheimer's disease. All of those diseases are improved and the weight loss by lowering insulin. So in his body, the exogenous ketones wouldn't be really as beneficial as otherwise. So in my mind, exogenous ketones have an obvious role in people who suffer from cognitive decline or endurance athletes. Endurance athletes have been shown to be able to perform better with this extra fuel that they're just kicking. It's like a little additional rocket fuel, like, spitting nitro into the combustion.
- 4.95 I have noticed so far, we've talked a lot about the physiological consequences of insulin resistance, but I'm curious about the thoughts on psychological consequences of insulin resistance. Most especially, it seems like we are aligning the overdiagnosis of these problems with anxiety and depressive disorders in young people today. Do you think insulin resistance plays a part in that, especially given the note we made about how the sympathetic nervous system is overstimulated in those times?
- 4.96 Yeah, great question. Indeed, this has been getting a lot of attention. There is a psychologist who just published a book called Brain Energy. His name is Chris Palmer. The whole book is devoted entirely to documenting the sheer number of neurological disorders that are derivative of metabolic problems. I highlighted Alzheimer's disease and migraines because they are so obvious, but he has focused on less obvious ones like schizophrenia, bipolar disorders, anxiety, and depression. I'm familiar with some of that research. The short answer would be, yes, that's a known effect. If the brain is hungry, problems ensue.
- 4.97 You'd mentioned earlier that insulin can be elevated decades before glucose. Is the baseline level for measuring elevation different for different ethnicities or body types, given our conversation yesterday?
- 4.98 Great question. No. No. Thankfully, insulin levels are insulin. Insulin will be what it's going to be across the body. It should be at a low level. I'll give you some ranges tomorrow in the solutions part, what we want to monitor and target. But yeah, regardless, around the world, we want insulin levels to be in the same range, thankfully. It makes it tidy.

- 4.99** Hi. I wanted to ask, with regards to the context of cancer and the study with regards to the mitochondria and the nucleus. I'm not going to ask you whether, I mean, unless you know the answer, because I feel like it's like a which came first, the chicken or the egg, with regards to epigenetics of whether the mitochondria influences the genetic expression of the nucleus or if it's the nucleus that affects the mitochondria?
- 4.100** Yeah, I don't know. I don't know for the record. That's a good question.
- 4.101** But on that note, I was going to ask if, well, I don't know if you're going to cover this tomorrow, if there was any way to correct, I guess, the improper function of the damaged mitochondria, in those damaged cells.
- 4.102** Yeah, that's a great question. I can't answer it in as detail as you might like. I can cite Dr. Seyfried's work. Rather than trying to directly correct the mitochondria, what they have found in certain cancers, it has been just remarkably effective. What they basically try to do is starve the cells of glucose. They have those patients go into very deep ketosis, and they have what they've called the glucose ketone index. I've mentioned some levels earlier, how normal fasting glucose right now, about 5 millimolar, whereas in the average person, ketones will be zero.
- 4.103** What they try to do is get that even. They want the person to be avoiding carbohydrates so much that their glucose levels go down to around 3.5, and they want them eating such a low-carb or ketogenic diet and fasting that their ketone levels are at that level or higher. Basically creating a menu of nutrients that the cancer cell can't really rely on. And they have found that to be fantastically effective in certain cancers, like glioblastoma, which is an inoperable brain tumor. They can stop it in its tracks and even shrink it.
- 4.104** But again, to state the weakness in my answer, they're not trying to directly fix the mitochondria. They rather have just said, OK, fine, regardless of where it started, we're just going to starve the hell out of the cancer cells by not allowing it to have the one fuel that it's crying out for. So forcing it. If you force a cancer cell to rely on its mitochondria, which is what it has to use if it wants to burn ketones, then it can't survive.

Lecture 5: Resisting Insulin Resistance

At a glance — Resisting insulin resistance.

- 5.1 Wonderful. Well, I am smiling because this is the part of the story that I like to tell, which is the happy ending. That the time we've had together, we've been discussing the origin story of a villain, how the villain became so bad, and then all the damage that the villain has been doing, all of the grisly murder scenes, if you will. This is the resolution, where we end with the good news, the gospel of metabolism, if you will, is that it is very rapidly resolved. It can be corrected very quickly, and well, not permanently, because as someone returns to bad habits, of course, the consequences come back. But now we'll explore the consequences of better decisions, of interventions.
- 5.2 So as we discuss how we can improve insulin resistance, before we get into that, I want to start with some ideas on how to know where you're at at the moment. When you have been going through this course, hopefully this has been planting some ideas in your mind, and now you ask yourself, well, how am I doing metabolically? Where am I with regards to the insulin resistance and these metabolic consequences? So to measure insulin resistance as thoroughly a way as possible, I want to introduce you to one dynamic test, which I'll explain, and then several static tests, which are much simpler to get done. All of these are clinical and medical related, although I will end with some easy ones that you can do at home.
- 5.3 The first one, the dynamic test, this is the gold standard. It is also, unfortunately, the most complicated, where it involves you doing what's called an oral glucose tolerance test. So you'll notice on this timeline on the bottom, where I have it from zero to 120 minutes, that there's a first point of drink some glucose. Any woman who was pregnant who had to do this test knows exactly what I'm talking about, where you go into the clinic, you drink this glucose-rich solution, and then you have to get your blood drawn. Typically, that test will only measure the glucose levels. And so let's start with the glucose.
- 5.4 Ideally, by the end of the 2 hours, the glucose level has come back to around 100 milligrams per deciliter. Ideally, it's come below that level. That's a good sign. If you do this 2-hour oral glucose tolerance test, and at 2 hours, you're below 100, that's a good sign. You handle glucose well, and that likely means that you're more insulin sensitive. But if the opportunity allows itself, we go a little further than just measuring glucose. We want to measure insulin.
- 5.5 And then if we look at the insulin curves, there are several options, of course, in all of the variety of human bodies. But the key is the peak. This is the simplest way of determining with this gold standard test. If the insulin peak is at right about 30 minutes, as you're measuring this every 30 minutes, then that's a good sign. If the peak is

happening at that 30-minute blood test, and at every other insulin test, insulin measurement is lower than that one, that's a good sign that your body is insulin sensitive.

- 5.6** The body was challenged with the glucose load. Insulin's most famous job, but remember, not its most important. Everything insulin does is vitally important. But its most famous job is to lower the glucose. So if insulin is able to work sufficiently well, that that peak was at 30 minutes, that means indeed insulin is working quite well. If the peak is at any other point, if the peak is at 45—there is no 45-minute here in a 30-minute span—if it's at 60 or 90 minutes, that's a warning that something's wrong. Your body isn't responding very well to insulin. Thus, you would likely have insulin resistance.
- 5.7** Now, let's go over to the static tests. The static tests are much simpler because it can be determined on just a one-time blood test. Firstly, we'll measure insulin and C-peptide. Now, C-peptide is a molecule you've not heard of yet. This, however, is a paradigm you have seen, which is, we orient ourselves back to the beta cell of the pancreas, the beta cell being the producer of insulin. When glucose levels climb, the beta cell gets that signal and thinks, "Hey, I can solve this problem by dumping insulin into the blood."
- 5.8** Well, one of the features of insulin is that, for every one molecule of insulin that gets made, there is another molecule of C-peptide. This is part of the physiology of the hormone insulin, that when insulin is being formed, created from its base molecules within the beta cell, it is created as this little duo. It's basically a twin, where there's this molecule connected to it in a pro-insulin form. So this is insulin in its juvenile state, insulin coupled to this other little molecule. And then as insulin fully matures, right as it's about to be released into the bloodstream, it gets split apart. So you have one molecule of insulin and one molecule of C-peptide.
- 5.9** However, C-peptide is a marker of insulin production, not insulin levels. Indeed, C-peptide is an order of magnitude or so higher in the blood than insulin levels are. And so sometimes within my space, I hear people talking about how C-peptide can be a good indicator of insulin levels. That is not true. They get metabolized and cleared from the blood at very, very different rates. Even though they're being produced at the same rate, insulin gets cleared so much faster, so that the actual amount of insulin in the blood is just a small fraction of how much C-peptide is in the blood.
- 5.10** Now, why would I bother mentioning it? It's because one of my hopes at the end of all of this is that there may be some type 2 diabetics, people with type 2 diabetes, who have been treated with insulin. They're on an insulin therapy. Well, if I tell them, well, just measure your insulin. And if it's under 6 micro units per mL, you're golden. Just start going on with the plan as we'll lay out in a bit. But if they were to get their insulin measured, they're measuring—a lot of what they're measuring is the insulin that's coming from their syringe, if they're on an insulin therapy.

- 5.11** And so for them, measuring insulin is pointless. If you're putting a lot of insulin into your body, you don't know what your insulin levels are. And in that case, C-peptide becomes a little more helpful, where the person can just be confident that they are indeed making insulin. And thus, they can begin engaging in some of these lifestyle changes that will very likely have them reducing their dose of insulin, working with their clinician.
- 5.12** OK, so these are important measurements. Insulin and C-peptide. Insulin should be about 6 micro units per mL. I say about because insulin is a hormone with a rhythm. It has a circadian rhythm to it, and it's possible someone might measure it at a point that's a little higher than that, and yet they're still actually really insulin sensitive. But all the more reason to look at additional markers, as we'll get to.
- 5.13** So the next one is a test called the HOMA score. Homa stands for homeostatic model assessment. HOMA. HOMA is clever because it accounts for both insulin and glucose. And it is divisible, then you divide it by this random number, all just to create a nice range. But in the HOMA range, insulin times glucose, divided by 405, if your HOMA score is less than 1.5, that's a very good sign that you're insulin sensitive. If it's 1.6 up to 3.9, that's a warning that you are insulin resistant, and then anything above that likely means that you have type 2 diabetes. At that point, the glucose is likely going up quite a bit.
- 5.14** Regardless, these are the cutoffs. This is the test. I like it a lot. The nice thing about this test is you always measure glucose. If you are measuring insulin, to get your fasting insulin, to see where you are with this test, then you can also get your HOMA score. It starts to just become one more piece of this growing puzzle, the collective of which will be an image that allows you to know where you're at, which is the point of what I'm talking about now.
- 5.15** Another test is the triglyceride to HDL ratio, which is not an unfamiliar one because we talked about it yesterday, in the previous lecture, I should say. In that previous lecture, we described how insulin resistance affects the lipids, the fats that the liver produces. When the body is insulin resistant, the liver begins overproducing triglycerides and underproducing, or more rapidly clearing, HDL cholesterol, the so-called good cholesterol. Well, that's not a favorable dynamic. We take these two numbers then and create this triglyceride to HDL ratio by simply dividing triglycerides by HDL.
- 5.16** Now, there are ethnic differences with regards to the natural release of lipids from the liver. I have here indicated the general cutoffs, where with black populations, typically around 1 is the optimal cutoff. With Caucasians and Asians, it's typically closer to around that midpoint of 1.5. And then Hispanic and Asian Indian populations, that's closer to two naturally. That's just part of the variety of the human body, once again, reflected in these subtle differences in the basal state of how much the liver is handling and changing fat levels in the blood. But this is a very, very good indicator of insulin resistance.

- 5.17** But interestingly, when we spoke about it in the previous lecture, I emphasized a great deal how it is also relevant to heart disease because it is a good indicator of the LDL pattern. It's the poor man's method of knowing, are you a pattern-A LDL, which you'll recall is a larger, buoyant. That's the beach ball form of LDL, which just happily flows through the blood. Or are you a pattern-B? If the triglyceride to HDL ratio is above those ranges, not only does it reflect insulin resistance, but also adjusting a narrower, denser form of LDL. In other words, the more pattern B, the LDL will be. That is, of course, more atherogenic, so implications in both insulin resistance and atherosclerosis.
- 5.18** Now, this test is my favorite one, the adipose insulin resistance score. I want you guys to note that you put a star here because if I could give you a pop quiz, this is the one I would talk about, although we will end with some homework today. In this test, what I love about it is it reflects the beautiful nuance of biochemistry and endocrinology, hormones dictating the fuel use in the body. But it also allows us to determine how well the fat cells are doing. Because remember, the fat cell is the first tissue type to become insulin resistant. That is the domino that will fall before the others.
- 5.19** When insulin is elevated in the bloodstream, it inhibits a process called lipolysis, which is simply the breakdown of the fat that is stored in the fat cells. Those are the fat cells in the middle. So insulin inhibits the release of the fatty acids or just the fats, which is what you have when you're breaking the fats off that is stored within a fat cell. So normally, because of this, if insulin is high, free fatty acids, which are what is released from the fat cell when it's breaking down fat, should be low. That's reflective of the fed state. The body doesn't want to be burning fat in the fed state. It wants to be burning sugar, so it turns off fat breakdown. So high insulin will normally lead to low free fatty acids.
- 5.20** Sometimes these free fatty acids are called non-esterified fatty acids. That's just a complicated way of saying the same thing, free fatty acids. Now, in contrast, if insulin levels are low, like during a fast or through carbohydrate restriction, then the lipolysis, the fatty acid breakdown or release, is disinhibited. It is allowed to occur now. So high insulin should connect to low free fatty acids. Low insulin should result in higher free fatty acids, unless the fat cells are insulin resistant.
- 5.21** If the fat cells are insulin resistant due to their excessive hypertrophy—remember, when the fat cell has undergone such hypertrophy, it can reach a point of maximum dimension. It begins to tell insulin, "Insulin, you bully, you're trying to force me to continue to grow. I know that I can't do that anymore. So even though you're high and you're pounding on the doors of the cell, I'm not going to listen to you and I'm going to start releasing fats." So we have this unique metabolic scenario in which both insulin is high and free fatty acids. That shouldn't happen. It reflects a fundamental breaking of metabolism, if you will, within the fat cell.

- 5.22** And so this formula, as you see at the bottom, is derived from just multiplying insulin by free fatty acids. That will give you a number, but it's not going to be the same across the sexes because, no surprise, there are profound differences between males and females. One of the many differences, more subtle than some of the obvious ones, is how we metabolize fat. Insulin, within the fat cells of a female, estradiol, the primary sex hormone that we spoke about last time—we spoke about estrogens—estradiol is the main one. It's the mother of the estrogens, if you will. It promotes greater lipolysis from female fat tissue at any moment, particularly at the buttocks and hip areas.
- 5.23** So at any moment, a woman will have higher free fatty acids than her male counterpart, reflective of the differences in sex hormones, enabling a greater lipolysis. In fact, it's not even subtle. At any moment, women are burning on average about 40% more fat for energy than men. Now, the gal listening may be thinking, "Well, then, why am I fatter than my male counterpart?" That's because there are other hormones that counter that. Basically, what the female fat cell has going on is a lot more turnover. It's a lot busier than the male fat cells, if you will, driven largely by the sex hormones, where, in the female fat cell, while her fat cells are breaking down fat more rapidly, they're also putting more in.
- 5.24** And so the collective effect will be more fat in the fat cells by design. That is supposed to happen for reasons that I won't elaborate now because that's a little outside the scope of the time we have. But suffice to say, female fat is breaking down fat faster, and that is reflected in this adipose insulin resistance score. She should have a higher level. And so in the males, it should be less than around 5. Females, it should be less than around 8. If they're below that, that's a good sign that the fat cells are insulin sensitive. And then as it goes higher, then the warning just gets bigger and bigger, that the fat cells are problematic.
- 5.25** This is relevant because, remember, I highlighted a study in the previous lecture where we discussed women with PCOS and how one of the studies wasn't able to detect differences in insulin resistance with the HOMA score, the one I just showed you, but they did detect differences here. This is where it starts. Now, all the tests we've been talking about are the more complicated tests that require you going in and getting a blood test and you asking the clinician or asking the phlebotomy lab to do this measurement for you. Let's just highlight two that you can do at home.
- 5.26** The first one is done just with the eye, looking at the skin. The skin is a window to the metabolic soul, where if someone has one or both of these disorders, it is almost proof-positive that they have insulin resistance. Blood tests can just be put on hold or not even done. I'm so confident in that. The first one is a condition called acanthosis nigricans, which is usually at a skin fold, but commonly, most easily seen at the neck. But there are a lot of skin folds in the body, like for example, armpits or around the groin, where you can have these same things happening. The neck is just the most obvious.

- 5.27 Around the neck, the acanthosis nigricans is a darkening of the pigment of the skin, and as well as the texture of the skin getting a little crinkly, like tissue paper. Sometimes people will think it's just like a leftover from their collar or it's some dirt that they can't get off, but it is not. It is a fundamental skin problem, reflective of the insulin resistance impacting the skin. Now, of course, in all of the variety of human bodies, there are substantial differences in pigment. The darker a person's skin pigment naturally is, the harder it would be to see the pigment changes or that becomes irrelevant, but the texture changes, the touch of it being rougher and more flaky and tissue-papery will be present across any ethnicity, even if you can't see the pigment differences.
- 5.28 Then the second part of this is the aberrant growth of the skin as a result of insulin resistance, creating what is commonly called skin tags. Now, a skin tag is not a mole. It's not a bump that comes up like a little mound. It's like a little stock, like a little mushroom, bulging up. Interestingly, both of these are rapidly resolved as insulin resistance is resolved. The skin tags just shrink back into the skin and the acanthosis nigricans, the pigment resolves and the texture resolves. So both of these are very rapidly fixable without any external dermatological intervention. That is itself reflective of an internal metabolic issue, and resolving the internal metabolic issue resolves the external manifestation.
- 5.29 Now, the last one is another simple one. You'll recall from the previous lectures that we've spoken about adipocyte hypertrophy and how problematic that is, as well as visceral adiposity, storing that fat tucked within the organs, which itself undergoes greater hypertrophy, all in an effort to prevent it from getting too big. But suffice to say, if it does start to get too big, that will be reflected in an exaggerated waste circumference. You can measure your waist circumference, the biggest part of your belly, just with a loose tape, measuring tape, and times that by two.
- 5.30 If that number is less than your height, it's a good sign that you're doing OK. Maybe you could be doing a little better, depending on how close it gets. But if this times 2 is less than this, you're doing all right. If this times two is more than my height, than your height, that's a problem. You have too much stored here relative to your body size. Naturally, a bigger body will have a bigger waist circumference. But as the body grows, the height is getting bigger. It shouldn't be matched to a point where the waist circumference times 2 is exceeding the height.
- 5.31 All right. So, these tests, doing it at the clinic or doing them at home, will allow you to know where you are. And once you know where you are, then you can know how aggressive you need to intervene. When we introduce the concept of insulin resistance and its origins, these were the three primary causes. Now, you'll recall that I did not suppose, and still don't, that they are the only ones. These are the primary causes validated across all three commonly used biomedical models. Cell cultures, in a little petri dish, lab rodents and humans, the pinnacle of all creation. We validate them in all of these. These are the three.

- 5.32** And if you will, imagine that these three are levers on a wall, on this panel of metabolism. And we would say, OK, I have insulin resistance, and all three of these are a problem. Unfortunately, not all of them can be gripped and turned down very well. For example, inflammation. I'm not going to spend time talking about it because it's difficult to really know, one, do you actually have a problem with inflammation? And two, what can you do about it? The clinician could say, well, hey, you have insulin resistance and your inflammation is high. The patient would say, well, how do I reduce my inflammation? It's just going to be a shoulder shrug. It's not really clear. That's a slippery lever.
- 5.33** But the other two, stress and particularly hyperinsulinemia, are ones that can be grabbed very firmly and turned down. And so let's focus on them, starting with stress, because it's a little simpler. And then we'll spend the rest of the time on the big one, which is hyperinsulinemia. So with stress, we've spoken previously about two causes of stress that contribute to insulin resistance. Sleep deprivation, through its elevation in cortisol, and then excessive caffeine consumption and its increasing epinephrine.
- 5.34** Remember, what both of those hormones have in common, despite their many differences, practically, the only thing they have in common is that they both want to increase blood glucose. And that, of course, puts them at odds with insulin in ways that we've spoken about previously. Now, this is another paradigm you've seen, but I want to emphasize it again when we are talking about solutions. Commonly, people will indulge in frequent and heavy snacking in the evening. And these are going to be snacks that are spiking their glucose levels because, let's face it, that's what we crave at night.
- 5.35** No one is sitting around in the evening about to turn on a show and relax, and they're thinking, boy, I would murder a plate of bacon and eggs right now. No one says that. If only they would. No one. We want something that is salty and crunchy or sweet and gooey, and that means it must have some starches and sugars to it. So, just the fact of human temptation and indulgence is that we want to spike our glucose then, but that is maybe the worst time to do it.
- 5.36** When we spike the glucose right before going to bed, we have both the glucose spiking our temperature, and we have the high insulin activating the sympathetic nervous system. We go to bed hot and bothered. The heart is beating hard and fast, and we feel like we're anxious, and we're not. We just indulged in the wrong things. We need to reduce that glucose curve before bed. Whatever we've eaten for dinner, try to give our bodies enough time, about three or so hours before going to bed, where the glucose and the insulin has been to come down. And that will resolve the temperature and the racing, pounding heart.
- 5.37** Now, with regards to caffeine, I appreciate that this is a charged topic. We love our caffeine. It is the most widely consumed drug worldwide. And I'm not supposing to tell anyone to not enjoy it. But we know that caffeine will increase epinephrine, which will

not only increase body temperature, but it will also increase heart rate. So my modest suggestion, in all humility, to be diplomatic, is that whenever you know you're going to be going to sleep, give yourself at least a 6-hour window.

- 5.38** That has been shown, if it's at least a 6-hour gap, to allow the person to metabolize much or all of the caffeine and go to bed without some of those consequences of hyperglycemia or racing heart or higher body temperature. Caffeine is a stimulant. We all know that, and we think about it in the context of, I got to calm my brain down. Yeah, I don't know. I won't speak to that. I will speak to the consequences of the metabolic side, the glucose and the insulin issues. Resolving the caffeine consumption before bed, about 6 hours, will help take care of those.
- 5.39** Now, let's go to the big one. Hyperinsulinemia is the one worth focusing on because everything else can be going wrong, and if you're correcting the hyperinsulinemia, you are resolving the most fundamental issue here, because it is the primary cause. And the greatest issue, the greatest efficacy, is to lower insulin. And I want to highlight for you, spending varying amounts of time on each of these, four interventions that can be leveraged all in an effort to improve insulin sensitivity, because that should be the key. Although I'll introduce some context and complications to this.
- 5.40** So let's just start with drugs, because we, unfortunately, have a philosophy of medical care, which is, basically, don't ask the patient to do something uncomfortable. Let's just give them a pill. We all can have issues with that, and I hope I'm making mine quite obvious. But these are the different classes of drugs that are commonly used to resolve insulin resistance and the type 2 diabetes that goes along with it so commonly. One mention, though, is so important, which is, these compounds are foreign substances coming into the body.
- 5.41** And so, what you need to do is evaluate that everything this pill or injection is going to do is a consequence, a side effect. It is really a matter of you asking yourself, "Are the side effects I want balanced or worth the side effects I don't want?" Because that is always there. There will always be side effects that you don't want. And you need to just match that or compare it to the ones you do. Sometimes it's difficult to see the consequences you don't want. You want that immediate gratification. It is human nature, but it is worth asking. It is very much worth determining what are the side effects you don't want because you're going to get them sooner or later.
- 5.42** All right, now this is my own classification. This is the gospel according to Ben. And when it comes to diabetes or insulin sensitizing medications, where some I consider to be a little more worth some of the consequences, some are decidedly split down the middle where I have to admit that they do have a beneficial effect, but the consequences are maybe not worth it. And then there are some that I consider to be, there's no benefit whatsoever to them, that they're purely pathogenic, they purely induce harm, all because they're leveraged to improve the wrong outcome.

- 5.43 OK, now metformin. Metformin is the most widely consumed insulin sensitizing drug in the world, in part because it can work. It can, in fact, directly improve insulin sensitivity, due to its actions on various cells throughout the body, like the muscle and the liver. Helping the muscle use glucose better, helping the liver dump less glucose into the bloodstream. Even still, with this most effective and most widely prescribed anti-diabetic insulin sensitizing drug, even modest lifestyle changes have been shown to be more than two times as effective at resolving insulin resistance.
- 5.44 So, as quick as we may be to prescribe and take this pill, I think we should be at least as quick to note some of the consequences, which I'll highlight in a moment, but then also admit that, yeah, lifestyle change will work twice as good anyway. Are you sure you don't want to do that? Well, that's the hard route, unfortunately. With the regards to the consequence, so me disclosing that metformin does work, it will improve insulin sensitivity.
- 5.45 However, part of its mechanism of action is actually damaging the mitochondria. It's a bizarre set of events, a sequence of events that results in the improved insulin sensitivity. But one of the mediators, at least at muscle, is to slightly inhibit the degree to which the mitochondria can work. I know it seems like a bit of a paradox, and there is more biochemistry involved than I'm elaborating on, but it does bear a particular—it places a particular burden on the muscle. The muscle, if it can't rely on its mitochondria as well, it doesn't work as well. And muscle has a lot of energetic need. It has a high demand, especially as we are up and animated.
- 5.46 Given that consequence, it is particularly interesting to note that there are studies in humans to show that, due to the effect on the mitochondria, metformin, if consumed when someone is exercising or if someone engages in regular exercise, that exercise will improve mitochondria health, and the exercise will improve insulin sensitivity on its own. But if a person who is exercising is also taking metformin, metformin will blunt both of those beneficial effects. So the exercising body wants to improve its mitochondria, metformin doesn't let it. The exercising body wants to improve its insulin sensitivity, metformin blunts it.
- 5.47 So it's almost like you have to do one or the other, which is a terrible way to look at this drug, because any drug that discourages you from exercising to improve your metabolic health, well, that might be one to second-guess. So there's pronounced considerations when it comes to metformin that we need to balance with. If this were a person who's purely sedentary, maybe they are wheelchair-bound or bedridden, well, then that might be worth it then. That the improvement could be worth any consequence because it's not going to be offset by the tendency of the individual or the ability of the individual to exercise. So considerations with metformin.
- 5.48 Now, let's get to a drug that is the most famous drug on the planet right now, or a class of drugs, which are what's called the GLP-1 receptor agonists. These are drugs that are injected, typically weekly, and they act like a hormone called GLP-1, which I'll talk about

more in a moment. I just want to set the stage and help you understand why I would spend time even talking about it. These are drugs that came to market about 10 years ago, and since this time, they have gone up orders of magnitude in popularity and use. And I emphasize the word popularity because it has been, much to my amazement, a point of bragging. People are proud. They brag about being on these drugs. These are drugs like Wegovy and Ozempic. They've become fantastically famous, and partly for good reason, because the results are remarkable.

5.49 But there are considerations. There are consequences that need to be considered that I worry are not. So the GLP-1 agonists are drugs that end with the suffix -lutides, like semaglutide or liraglutide. That'll be the drug name, although it will have another trade name placed on top of it. These drugs seek to mimic the effects, well, mimic and amplify, the effects of a hormone called GLP-1, which is released from the small intestine. It's a natural release. Certain foods are coming through the intestines, and we'll revisit that later, and it will stimulate the release of GLP-1. GLP-1 will have significant metabolic effects.

5.50 This is what happens when GLP-1 agonists are kept at a lower level— so a slightly amplified version of what our own endogenously-produced GLP-1 would do. GLP-1 will improve insulin sensitivity in large part because of its actions at the liver and its actions at the brain. With regards to the liver, GLP-1 inhibits the liver's dumping out of glucose. You'll recall that we discussed a molecule called glycogen in the previous lectures. Glycogen is the stored form of glucose within the liver. And the liver, if blood glucose levels are changing, it's in large part because the liver is breaking down its glycogen into glucose to flood the bloodstream with glucose.

5.51 In someone who's struggling with insulin resistance or type 2 diabetes, they have too much of it. The liver is doing it too much, for reasons that you already know. The liver has become insulin resistant. GLP-1 helps the liver remember that if blood glucose levels are high, it should be storing glycogen. It shouldn't be breaking down the glycogen. So that is one primary mechanism of action that improves insulin sensitivity. It reduces glucose release from the liver, thereby lowering blood glucose, thereby lowering insulin. And if insulin comes down, the body becomes more insulin sensitive.

5.52 Now, there is also a pronounced central effect at the brain, which is to reduce hunger. And so here we have two benefits, where the GLP-1 agonists are both lowering insulin through changes at glucose level at the liver and reducing hunger, having the person eat a little less. Low insulin, low energy, that's a pretty winning combination. However, this is what it's like when it's used at low doses for insulin resistance. Nowadays, people have leveraged these findings to bump it up by multiples. It could be five or even 10 times higher doses to use for obesity. So current strategy nowadays is to take it at very, very high doses.

5.53 And basically what this does is it takes the modest impact on reducing hunger and makes it substantial. I'm saying this in a very polite way, reducing hunger. The more

accurate term would be extreme nausea. The person just feels sick to their stomach very, very often. It's not that they look at something and think, "I'm good." They look at something and think, "It makes me sick to even think about eating." They feel nauseous. They're nauseated constantly because of this high dose. It's a bit of a miserable way to correct your appetite, but there's no arguing that it works. But again, what people call improved satiety or reduced hunger, another person might call, "I feel really, really sick, so I just don't want to eat."

5.54 Beyond this, though, these are all still—someone could still say, "Well, that's all good. I'm fine with any of those options because it helps me." It's this last part that's relevant and totally overlooked, which is the effect of these high levels of these agonists, these activators of GLP-1 receptors, at the fat cell, particularly how they induce adipocyte hyperplasia. Now, you're familiar with that term. This is the ability of the fat cell, which had been limited in how much fat it could store, and the fat tissue couldn't store anymore. These come and say, "Hey, I notice you're having a hard time storing fat. Let's just have you store a little more by inducing the synthesis of new fat cells." But it's a remarkable set of events, and I highlight them here.

5.55 One, it does stimulate new fat cell growth, which you would say, well, how does that allow the person to lose weight? Well, it doesn't. That alone doesn't, although it does make them more insulin sensitive, right? Because if you have new fat cells being produced, you've changed that fat threshold, and the body becomes more insulin sensitive because of it. But at the same time it's stimulating the growth of new fat cells, it's stimulating the breakdown of fat. So that means that, while it is growing fat cell number, the fat cell size is collectively shrinking. And so the fat mass on the body will follow that shrinking.

5.56 But what happens when they stop taking the medication? You lose the activation of the fat breakdown, so the fat breakdown stops. But now you have more fat cells than you did before. And so now your potential to store fat is greater than it was before you ever started the drug. That is a heck of a trade-off that the patient needs to be mindful of—that, one, you start taking this, I'm basically locking you onto this drug for life because you're never going to want to stop. Well, that's a wonderful—that's really good for business, right? It's a patient for life because, boy, we got you hooked. And that is because of the potentially horrific metabolic consequences.

5.57 I don't mean to be dramatic. But again, just to state that all again, the GLP-1 agonists at high doses will stimulate new fat cells, which will allow the fat cells to be insulin sensitive. That's a good thing. But at the same time, it gives you this artificial sense of success because it keeps the fat cells small. Once the drug has stopped, now you still have all the fat cells and you've lost the fat breakdown effect, the stimulation of lipolysis. So my suggestion is a little complicated because I acknowledge that the metabolic benefits of GLP-1 agonists are substantial. It really is a remarkable effect. Might it have a place in improving metabolic health? It very well could.

- 5.58** My suggestion would simply be, use it at the lowest possible dose, knowing that you only want to be on it temporarily, and couple it with robust dietary changes, which we'll get into. That is probably the best way to use it. All right, now let's move into the less green, less potentially beneficial drugs and go into the more problematic. This next class are what's called SGLT inhibitors. And we have, SGLT is a transporter for moving glucose around the body. SGLT-1 are transporters within the intestines that allow us to take the glucose we just ate and move it into the bloodstream.
- 5.59** SGLT-2 are transporters in the kidneys, where the kidneys have pulled out the glucose from the blood, and now SGLT-2 will allow the kidneys to bring the glucose back in, in a normal healthy person. We want that to happen. We don't want a lot of glucose in our urine. And so both of these work by reducing the amount of glucose that's coming into the body, coming in from the diet or coming back in from the kidneys. And as you're pushing down the glucose, you push down the insulin, and the body becomes more insulin sensitive.
- 5.60** However, there is something within the realm of physics and biology called osmotic gradients, which is the ability of molecules like glucose to pull water with them. And so the consequences here become a little obvious. If you allow someone to take this inhibitor, blocking SGLT-1 in the intestines, and they're eating a big starchy, sugary meal, that means all of that glucose is staying in their intestines, pulling in water throughout the whole process. And now they never know whether they can trust any pressure building up in their colon. In other words, they can have terrible diarrhea and horrific consequences, which are not life-threatening, but socially threatening.
- 5.61** And then, next is what happens when you block the kidneys' ability to allow the glucose to come back into the blood. Now all that glucose is staying in the urinary tract, pulling in more and more water. So the person ends up developing a polyuria, a urine production that is like they have diabetes. They feel like they to urinate all the time, and indeed they do. But also, they can feed bacteria, and so UTI risk skyrockets because the urinary tract is potentially a little bit of a dirty place. Urinating helps push the bacteria out, but at any moment, bacteria are trying to climb their way back in. Well, bacteria eat glucose. That is their fuel.
- 5.62** With SGLT-2 inhibitors, flushing the urinary tract with glucose, it basically is giving the bacteria a buffet. It's like the bacteria are running a marathon and you're saying, "Hey, you're getting tired. Here, stop at an aid station and have some glucose on your way up my urinary tract to my bladder." Not a very ideal outcome. And underlying all of this, if we are willing to do this to someone or we're willing to do it to ourselves, where we are willing to have potentially terrible diarrhea or terrible UTIs in urine production, all just to either block the glucose from getting digested or to push the glucose out, there's this rational little voice hovering around the shoulder, this little shoulder angel that just says, "Why not just eat less glucose?"

- 5.63** All right, now let's move on to the thiazolidinediones. These are the glitazones. Thiazolidinediones are fascinating because they act right at the fat cells, directly taking the hypertrophic, insulin resistant fat cells, turning them into smaller, yet more abundant hyperplastic, insulin sensitive fat cells. So basically, it takes a person with this level of fat and that level of personal fat threshold, and the body is saying, "I'm above my personal fat threshold, so I'm insulin resistant." And the TZD medication will push the threshold up. But at the same time, by activating hyperplasia, what do you think happens to the amount of fat this person has on their body? Well, that goes up, too. But it stays below the new threshold that the drug has set.
- 5.64** And so, paradoxically, while we are improving this person's insulin sensitivity, we are making them very fat. And I mean substantial fat gain. They can easily gain 20 pounds of fat in a month. That's the main reason people will stop taking it. They look at the new wardrobe they have to buy every month or so, and they say, "This isn't worth it. I'm getting off this pill." And then at that point, the fat cells go back to being hypertrophic, and the insulin resistance comes back. But again, a fascinating mechanism of action, which academically I appreciate. But as a realist, I appreciate that's not an optimal trade-off.
- 5.65** Now, let's talk about these last ones—sulfonylureas, which is a class of medication. Now, we have a class of medication of synthetic insulins. These basically act by, I mean, one directly in the form of insulin or second-hand in the form of the sulfonylureas. But it takes a body, a beta cell, that is producing some amount of insulin and boosts up the insulin within the body. You can already start to appreciate the consequences of this because of what you've been taught over the past few lectures. But let's revisit this paradigm that you've now seen a few times.
- 5.66** A clinician will feel justified in bumping up insulin because the clinical paradigm isn't insulin-centric. The clinical paradigm is obsessively focused on glucose. Glucose is the marker. Indeed, a clinician needs to account—if a clinician sees patients, all Medicare or the insurance provider cares about is their fasting glucose or other markers of glucose. There is no standardized metric that relies on the insulin. And so the clinician may not even know what the patient's insulin levels are, will only know that the glucose is high. And because we have such a glucose-centric paradigm, to hell with insulin, it doesn't matter. Let's just push insulin up even higher.
- 5.67** Unfortunately, there is no evidence of beneficial clinical outcomes. Nothing gets better when you give an insulin resistant person, who now has moved into type 2 diabetes, more insulin. Indeed, giving a type 2 diabetic insulin is like giving an alcoholic another glass of wine, hoping that the wine will solve their alcoholism. Giving a hyperinsulinemic type 2 diabetic is giving them more of the very thing that's causing the problem. So it's no surprise that they get much fatter. They are much more likely to die from cancer, and they're much more likely to die from heart disease, because the high insulin is driving those pathologies.

- 5.68** And what do you think happens as we flood the body with even more insulin? What happens to insulin resistance? Of course, it gets worse and worse. Now, I want to highlight some studies that emphasize those consequences. What happens to insulin dose? These are patients who were just diagnosed with type 2 diabetes. In other words, the insulin resistance has gotten bad enough that they can't control their glucose anymore. And so they put them on insulin. And this is what happens over six months. The insulin dose goes up and up and up the entire time.
- 5.69** No surprise, right? Because too much insulin causes insulin resistance, just like too much alcohol can cause alcoholism. Then let's look at what happens to body weight over this period of time. They're getting steadily fatter over these months, as insulin is driving this storage of fat within fat cells. Interestingly, as they track the caloric consumption of the patients, it's not going the same direction. They're eating fewer calories, by all accounts, within this published study, than they were at the beginning, and yet they're fatter than they were before.
- 5.70** This reflects the fact that, through mechanisms we've discussed, insulin is so determined to store energy, especially in fat cells, that it will reduce the metabolic rate. And indeed, that happens. As you put diabetics on insulin, their metabolic rate will go down. I highlighted previously in a lecture, looking at what happens in type 1 diabetics—you may recall that, I hope—how insulin will slow their metabolic rate by 20%. It even happens in people with type 2 diabetes, whose insulin levels are already elevated. Put more insulin, and now you slow their metabolic rate. This just enables insulin to have its day in the sun, to have its way, where it wants to promote the energy to be stored in the fat cells.
- 5.71** Now, let's get something a little more consequential than just fat gain, like, for example, cancer mortality. Type 2 diabetics who are treated with insulin or sulfonylureas, which increase insulin, are 90% more likely to develop cancer. So you've almost doubled their risk of cancer, all in an effort to lower their glucose. With heart disease, the diabetics who use the most amount of insulin, even though they will have perfect glucose, the type 2 diabetics, the insulin resistant diabetics who have to use the most insulin, are up to three times more likely to die from heart disease. There are consequences to bumping up insulin.
- 5.72** And indeed, just to bring us back to these plagues of prosperity, these consequences of insulin resistance, these are not driven by the glucose. In some instances, glucose might provide some modest contribution, like type 2 diabetes or cancers, for example. But what they all have in common isn't something wrong with glucose, but it's the high insulin, reflective of insulin resistance. So it's no surprise that when we bump the insulin even higher, things go wrong quite quickly.
- 5.73** All right, and as a final reminder, this last sentiment on drugs, as popular as they are, and we are a drug-obsessed culture, trying to allow drugs to put Band-Aids over the problem, they do just act as Band-Aids. They only treat the symptoms. These drugs can

never resolve the fundamental metabolic defect. It is the food we eat that is the— it is the culprit or the cure when it comes to these problems, which we'll get to in just a moment.

- 5.74** I just want to briefly mention exercise. Briefly, not to put it in a dimmer light than the rest, exercise is tremendously impactful. It's also just so simply obvious that it's not worth spending too much time on. So if the question were, what is the best exercise to improve insulin sensitivity? The very justified answer is, the one you'll do. And I mean that. I live in the beautiful mountains of Utah. And one of our family activities is cross-country skiing. If I were talking to an 80-year-old, and I said to the 80-year-old, the 80-year-old asks me, "Hey, Ben, what should I do?" And I'd say, "Hey, skate skiing is, minute for minute, one of the best things you can do."
- 5.75** Or maybe I'm even more insane and I say, "CrossFit, you should go do CrossFit." They will laugh at me and they should be justified in laughing because that's insane. But if this 80-year-old is willing to take a few laps around the block with her girlfriends, wonderful. Or if the 80-year-old is willing to go golfing with his buddies and walk through the course, incredible. Whatever the person will do is the one they should do.
- 5.76** And that's because, as the muscle starts moving, it becomes so hungry for glucose that it uncouples itself from the demand for insulin. Normally, insulin will be the signal that comes to the door of the muscle cell, knocks, and opens the doors for glucose to come in, escorting the glucose in. But when the muscle is exercising, it gets so selfish that it says, "Insulin, I don't have time for you."
- 5.77** And indeed, insulin levels come down during exercise, but the muscle, by contracting, is able to activate that. It's an insulin-independent pathway. And so the hungry muscle begins consuming the glucose, and as glucose levels come down in the blood, so too does insulin. And that, of course, improves insulin sensitivity. So this is a very beneficial effect. Exercise, it's a winner, no doubt, but it's simple.
- 5.78** All right, now let's talk about surgeries for just a moment. There are two surgeries that I just want to highlight briefly. Gastrointestinal surgeries, nowadays called metabolic surgeries, and then also liposuction. So with GI surgeries, these are things like gastric bypass or the gastric sleeve. There are all kinds of variations on this, where it's either something that has changed the plumbing of the stomach or it has removed almost all of the small intestine, which is where a lot of the digestion and absorption happens of the food.
- 5.79** Basically, regardless of the intervention, they act by forcing the person to starve a little bit, whether it's because it has changed the size of the stomach, meaning the person eats one little handful of nuts and they feel so full that they're going to vomit, or whether it has altered the ability of the food to come through the small intestine and get digested. Regardless, it compromises digestion. Why wouldn't it? These are

perfectly healthy tissues until modern, much more modern times where we do surgeries in other ways on tissues that are perfectly healthy.

- 5.80** I used to say these surgeries are the only surgery I know of where you are removing and mutilating perfectly healthy tissue. There's nothing wrong with this person's stomach and intestines. They're perfectly healthy, but we go in and carve them up anyway, all basically to force the person to starve. And it works. They do starve; they lose a fantastic amount of weight. Now, there can be weight regain, unfortunately.
- 5.81** But interestingly, this was the origin of the discovery of GLP-1, this most obsessed-upon, insulin-sensitizing, anti-diabetic, weight loss-inducing drug that the world is crazy about. It was because we found, as scientists, that when you replumb the intestines, you somehow get higher levels of GLP-1. And that was thought, justifiably, to be part of the improvements in the metabolic outcomes of people who have undergone some gastrointestinal surgery. But suffice to say, regardless of the intervention, you're making the person starve a little bit as energy is reduced. Remember, low energy will lead to low insulin, and so insulin resistance starts to resolve and get better.
- 5.82** Now, liposuction surgeries are an interesting beast entirely. This is taking someone who has, in their mind, too much fat, and they suck the fat out with this clever surgical vacuum. And that works. It sucks out the fat of whatever that fat depot is. In this example, I'm saying, imagine we're sucking out the fat of the person's abdominal fat. Now, as a reminder, something I've mentioned previously, when you undergo liposuction, it is the subcutaneous fat that you're removing. You do not get into that visceral space. That would be catastrophic to try to mess around in there. You are just sucking out the pinchable, jiggable fat, which, remember, isn't always a problem.
- 5.83** Now, what happens to the number of fat cells on this person when you are literally sucking them out? The fat cell number is going down. But is fat cell number the problem? It's not the number of fat cells, you remember. It's the size of the fat cells. And so it's no surprise that you can have an intervention where you remove up to 44% of a person's fat from their belly in particular—that's what this study did. And they find that there is— but it's subcutaneous. There's no improvement in any metabolic outcome. They're just as insulin resistant as they were before it ever started.
- 5.84** Because liposuction is an artificial solution, they've not learned to actually change their eating, there is still that same pressure to store the same amount of fat that they had before. But as you've gone in and decimated the fat cells from, say, the abdominal space, you still have intact fat tissue elsewhere, which is when, as the person continues to eat the way they were eating before, it's no surprise that they don't gain the weight back where they used to have it. They will gain it back in other areas.
- 5.85** So if someone has had an abdominal liposuction, they can fully expect to get fatter on their arms or their buttocks or hips. If they've had the hips liposuctioned, we'll then expect it to come under the belly or the arms. The body is too clever. And insulin is so

determined that if there's adequate insulin or high insulin and adequate energy, the body will find a way to store the fat somewhere.

- 5.86** All right, now, let's get to the big one, which is diet. In order for diet to improve insulin resistance, it must change the size of the fat cell. Remember, not the number, it's the size that matters. Two interventions will work, either lowering energy or lowering insulin. As we've discussed previously, typically they will go together. If you start to lower energy, you'll recall that insulin must also come down to prevent the brain from getting hungry and the person fainting, going unconscious, and eventually dying. So low energy will also eventually lead to low insulin.
- 5.87** But I'm going to challenge the temporality of it all. What should be the order? What should be the sequence of events? Both of these work, but it's better to start with one versus the other. Now, with these in mind, low energy and low insulin approaches, as distinct dietary approaches, there are three ideas that I want to touch on with varying depths. Low-fat diets, which are wholly focused on low-energy, low-calorie—it is purely a low-calorie-centric paradigm—versus carbohydrate-restricted diets, which say calories we don't care about, just lower the insulin, and then the happy union of them, which is intermittent fasting. I'll briefly mention that, too. And then once again, put them all together.
- 5.88** So this is—before we get into some of the specific studies that I want to mention, I just want to remind you of the insulin impact of the three macronutrients of the human diet—fats, proteins, and carbohydrate. Fats on their own will have no effect on insulin. If you've ever heard anything different, it's wrong. It doesn't happen. I've done this in my own lab. There is no insulin response to fat. Pure protein may have a modest impact, and it may be a slightly bigger than modest if a person has type 2 diabetes. Then lastly, pure carbohydrate will generally have a significant effect.
- 5.89** Now, let's just start lightly with low-fat diets. Remember, this is all framed in the context of insulin resistance. So what will improve insulin resistance? Unfortunately, while a person is cutting back on fat, they have to eat something. And so a low-fat diet is, by necessity, a higher carbohydrate diet. Well, that's a problem with, say, someone with type 2 diabetes or even just type 1. In the case of type 2 diabetes, we should scrutinize what the effect will be on nutrients within the blood.
- 5.90** Here I'm showing what is the blood-based version of these macronutrients. When you eat fat, it's going to be with triglycerides and free fatty acids. When you're talking about protein, what is actually moving through the blood is amino acids. When you're talking about carbohydrates, we're actually talking about, primarily, glucose moving through the blood. Well, in a person with type 2 diabetes, which of these is the person struggling to keep in a normal range? Well, it's the glucose, of course, right?
- 5.91** And so it's no surprise that interventions from as far back as the '80s found that when you take people with type 2 diabetes who are insulin resistant, put them on a low-fat

diet, which is by necessity higher carb, they get worse. Or as was noted, there were deleterious metabolic effects. That is the polite way of saying it's miserable; they get worse. This is why the American Diabetes Association, as of their 2020 nutritional guidelines, admitted, perhaps reluctantly, because there sure was a mountain of evidence that has been out there for a long time, that reducing carbohydrates is the most validated intervention for improving glucose levels.

- 5.92** All right, so now let's introduce—we just got done talking about the low-fat diet and the inherent problem, which is going to be that, by necessity, it's higher carb. Now let's talk about low-carb diets, where you are deliberately focusing on the carbohydrates, appreciating that they have the greatest effect on insulin. What I want to do over the next few slides is take us through a series of studies. I'm only showing a small sliver of the dozens of clinical trials that have been done comparing low-fat to low-carbohydrate diets. Let's just start with this one.
- 5.93** They look at the glucose changes in these two different diets where the people are split up randomly into two dietary interventions. The low-fat diet group had an insignificant reduction of about 2% of their fasting glucose. Whereas the carbohydrate-restricted, no surprise, had a significantly greater, multiples greater reduction, in glucose levels. Now, glucose isn't what it's all about. It's insulin. Now, they measured the changes in fasting insulin. Whereas the low-fat diets had a reduction with about 16%, the low-carb diets had a reduction by half, significantly greater. Once again, multiples more.
- 5.94** And the benefits continued as they measured insulin resistance. Insulin resistance actually changed to about the same degree that the insulin levels did. So significantly greater improvements in insulin resistance with a low-carb diet. This next study is one of the most interesting because of so much nuance to it. They took people and split them up into three different groups. Now, I have highlighted the calorie load that came with each diet. The low-fat diet, they were told to limit the amount of calories they were eating. Same with the Mediterranean diet, which was slightly higher fat and lower carb than the low-fat, but still calorie-clamped. Only eat this much calories.
- 5.95** In contrast, you wanted to be part of the low-carb group because they were told, eat as much as you want, but keep your carbs limited to 20 grams. Now, what's interesting is, I have noted, and the study authors actually admit this in the publication, while they started with 20 grams of carbs per day, they considered that the weight loss was so dramatic that it was too much, and it was confounding the study. And so they had them increase their carb consumption by 6 times, up to 120 grams, which isn't particularly low-carb at all.
- 5.96** You can actually see this in the study, the weight loss curve. The other two diets are losing weight. This low-carb diet, which is calorie-unrestricted, the weight loss was boom. Significant weight loss. Then you can see the moment they said, "Hey, you're losing too much weight," and they started gaining some weight back up. But it was still

less than the other groups. Now, I'm not intending to talk about weight. That was just such a dramatic effect. Let's look at these outcomes.

5.97 In every case, the low-carb diet outperformed the others. In every case, the low-fat diet was the worst for improving these metabolic markers that we should all care about. Now, let's keep going. What I'm showing here are the fasting levels across the groups before the intervention. We see that glucose didn't change much, tended to go up a little with the low-fat, tended to go down a little, but it's already at a pretty good range, and both of those numbers are, too.

5.98 Now, this is picomoles, which is more of the international units, whereas in the US, we use micro units per mL. But regardless, the trend is the same, where we see that there's a trend for insulin to go up and a trend for insulin to go down, with the low-fat versus the low-carb diets, respectively, and similar outcomes with measurements of insulin resistance. This is isocaloric. They're eating the same amount of calories, and yet we start to see these differences. Not all calories are created equal when it comes to improving metabolic health. The insulin impact of the calorie matters at least as much as how much calorie you're eating.

5.99 Now, let's talk about another study. I highlight this one because I appreciate that the authors of the study split the participants, the subjects, up into two groups based on their insulin sensitivity. At the beginning of this lecture, I highlighted the fact that fasting insulin levels should be around or below 6 micro units per mL to be considered insulin sensitive. So, these guys were. But interestingly, they were the same general body weight. Not all people carry fat the same way. Remember, fat cell size is different from fat cell number. So it's not all about fat mass, but the size of the cell. So nevertheless, an insulin resistant group and an insulin sensitive group.

5.100 And then they looked at what happens to insulin levels in a low-fat diet, as the subjects were pushed into a low-fat diet. Well, there was no real change in either. The insulin sensitive group didn't have a lot of room to improve anyway, so it's no surprise that they didn't change their insulin levels much. The insulin resistant group, they have a lot of room for improvement, and yet they didn't see any. However, when the patients were switched onto the low-carb diet, once again, you see that the insulin sensitive group didn't really have an improvement. After all, they don't need to. The insulin resistant group now all of a sudden had a significant reduction in their insulin levels, so a significant improvement.

5.101 They went a little further and actually measured body fat. And because fat does influence insulin resistance, I want to highlight this effect. So at both levels, starting levels, there was no significant difference between their body fat levels. I'm highlighting the exact number, but the standard error was such, there was no statistical difference between them. So for all intents and purposes, they're the same. Then they had the subjects move on to the low-fat diet, and there was an insignificant 1% reduction in the insulin resistant group. There was a significant almost 3% reduction in the insulin

sensitive group. Then they had the subjects engage after a washout period, randomized, on a low-carb diet.

- 5.102** Now they found, once again, that darn insulin sensitive group still responded, responded just as well as before. They clearly respond to both interventions. But the insulin resistant group now finally really had a response. They lost a little over 2%, whereas before it was a statistically insignificant 1%. So clearly in the insulin resistant group, one of these strategies is outperforming the other. And then one final study to mention before we wrap up this section in just a moment is what happens when you take, when you further understand the underlying metabolism of the individual.
- 5.103** This study gave people an insulin challenge and noted how much insulin is released in response to that glucose load, kind of like that oral glucose tolerance test we started with at the beginning of the lecture. They found and split the people up into two groups. Some people have a very low insulin response because they're so insulin sensitive. Some people have a very high insulin response because they're insulin resistant. Then they looked at the weight loss that occurred in these groups.
- 5.104** They found, reflective of the other study I just showed you, that insulin sensitive people can lose weight on either diet. Low carb, low fat, they're going to be fine. Although the sustainability might be an issue, but they will respond to both diets with weight loss. In contrast, the people that were insulin sensitive lost more than twice the amount of weight on a low-carb diet compared to a low-fat diet.
- 5.105** And then in summary, there have been, to this point, 58 randomized controlled trials published. Now, a randomized controlled trial is the gold standard that allows you to conclusively state, we did this and this is what happened. So you know whether the intervention works or doesn't. And among these 58 studies, they found that 48 had greater weight loss. Now, you'll notice I have a difference between greater and statistically significant. So this is just a trend for greater weight loss on the low-carb diet, whereas seven showed a trend for greater weight loss on the low-fat, and three found no difference.
- 5.106** However, when we go one step deeper and say, all right, is there statistical significance here? Well, now we have 29 studies that found statistically greater weight loss on the low-carb diet, and none. No study has ever been published that finds a low-fat diet will result in statistically significant greater weight loss than a low-carb diet. Not one.
- 5.107** Now, the final thought here before we move on is the oldest recorded study that was ever published on low-carb versus low-fat diets. We go back to the Old Testament. I'm a religious man, and so I study the Old Testament religiously, but I never thought I would need it as a scientist. And yet here I am, citing the story of Daniel and the other young Israelite boys who were taken by King Nebuchadnezzar from Israel. And the king is so impressed by these Israelite youth that he wants them to become part of his court, part of his kingdom.

- 5.108** But they, of course, live a kosher law. And so when they are brought into the king's household, they say, "We're not going to eat the meat. Give us pulse to eat. Give us grains." And then you will see at the end of the study whether we're healthier than the others you brought in who are eating the meat. And the way they are described, fairer simply means prettier, but fairer and fatter in flesh. So here we have Daniel and his companions who, at the end of this research study, it wasn't randomized, they selected themselves into the study, so it's a little problematic. We can criticize how the study was set up.
- 5.109** But even still, we have ancient proof that the low-carb diet outperformed the high-carb, low fat diet, because after all, again, Daniel and his low-fat-eating companions were the ones who got the fattest. All right, now, one final comment then is intermittent fasting, which really encompasses both. When you intermittent fast well, and we'll talk about this, but I won't show any more than this, it can be remarkably effective.

Lecture 6: Mastering Insulin Control

At a glance — Mastering insulin control.

- 6.1** Having covered a small sliver of the mountain of data that have been available for decades, supporting the efficacy of carbohydrate restriction at improving insulin resistance, let's now transition all of that into application. One of the things that I've taken quite seriously throughout my career as a scientist and a professor is how, sometimes academia is too isolated, where we refer to academia as the ivory tower. That's a beautiful comparison. But what good is the ivory tower if there aren't ideas that are capable of coming out of the ivory tower and actually benefiting the real world? The ivory tower is this almost fake world.
- 6.2** This is where we really put these things into practice. Now, again, the emphasis is to shrink the fat cells. I have highlighted multiple times throughout our time together, the value of both low insulin and low energy with some considerations to both. There should be an order to this. As you look at the chaos of improving metabolic health, the low-insulin approach should be first, and that can be followed, when necessary, by a deliberate low-energy approach.
- 6.3** Unfortunately, the lack of awareness of the value of lowering insulin has led to too many people starting their fat-shrinking journey with the first steps being the low-energy step. They've gotten the order of events wrong. This, unfortunately, is typified. It's reflected in the obsession with low-fat or low-calorie diets. Unfortunately, this is so often resulting in failure because of hunger. This approach sets a person up to fail, and they go back to the beginning of the journey, perhaps worse off than they were before.
- 6.4** Now, to make this point hopefully very clear, let's reason together for a moment. Imagine if we were invited by the world's most famous chefs to come to this most delicious and glamorous buffet we could possibly imagine, and they invite us to come hungry because we are going to want to try everything that they have to offer. Why not? They're the most talented chefs. We want to try everything. We want to indulge and enjoy that food. What would we do to come to this glorious buffet as hungry as possible, so that we can just really indulge? That's the point of the hunger in this case. The point is the indulgence.
- 6.5** I bet we would do two things in the days or some period of time preceding this beautiful buffet. One, we would eat less. At the same time, we would start exercising more. These would be this two-pronged approach to make sure, and it would work, that we came to this meal as hungry as possible and we could just really enjoy ourselves. Do you see the problem with this? After all, I was just asking you the recipe for coming as hungry as possible. And yet this two-pronged approach is the underpinning of all of the metabolic strategy that has existed for the better part of 60 years.

- 6.6** We've been telling people for decades now that your way to resolve your insulin resistance, to shrink your fat cells, is to eat less and exercise more. How can we expect that clinical advice to have any different effect than the logical conclusions we could just come to, with regards to perfectly setting ourselves up to be as hungry as possible? There is no difference in outcome. This approach, which again has been dogmatically embraced and codified in clinical practice, is the perfect recipe to make sure someone is as hungry as possible.
- 6.7** And hunger always wins. In this environment in which we find ourselves, where we have ready access to food all the time, hunger always wins. If our paradigm, if our journey is set up in such a way that it starts with low energy, and low insulin is either just a surrogate, a corollary consequence, or not a focus whatsoever, then the journey will fail. We will stop, and hunger will have us turn back. So, simply, we need to put the right order to these events.
- 6.8** The first steps of this fat cell-shrinking journey to improve insulin resistance should be to deliberately lower insulin, which is not a matter of calories consumed, quantity-wise, but rather the type, the caliber of the calorie, not the quantity. Then second, we worry about energy consumption if needed. This progression, this series of events, in my mind, I strongly submit, is best served by first embracing a diet that controls carbohydrates and others that we'll talk about, other interventions, to really have a good all-encompassing view of low-carbohydrate diets.
- 6.9** Then it moves into intermittent fasting as a mechanism to control energy if needed. There are four very simple ideas that allow us to implement these concepts. Now, I say simple. We're all mature enough and rational enough to know that simple is not the same as easy. While I am describing these hopefully elegant and simple principles, I don't want to leave you with an impression that I am lording over you and assuming that, well, I can do it so anyone can. Not at all. I do not want to convey that impression.
- 6.10** They're hard to implement only because, when we talk about changing habits, we're changing addictions. We're changing behavior, we're changing the self and the way we perceive how we live our lives, what we eat, when we eat it, how much we eat, and so on. Nevertheless, four simple pillars of an approach that will enable long-term success at improving insulin resistance, the point of which should be to be leaner and to reduce the risk of chronic disease, all of those plagues of prosperity that are derivative of the insulin resistance. Just always keep that in mind.
- 6.11** It's not resolving insulin resistance just for the sake of insulin resistance. It's resolving insulin resistance with the hope of reducing greatly our risk of Alzheimer's disease or improving breast cancer outcomes or improving fertility. First step, in order, control carbohydrates. Second, prioritize protein. Third, don't fear fat. Then lastly, my alliteration couldn't hold. Watch the clock. Make sure you're paying attention to when you're eating as a fourth and final step if needed.

- 6.12** Now, firstly, let's dive into a little more detail with each of them. With regards to control carbohydrates, I do not intend to declare war on all dietary carbohydrates. At its simplest, control carbohydrates, the sentiment is perfectly expressed by just watch out for the carbohydrates that have the greatest effect on insulin. A general next level of interpreting this is focus on whole fruits and vegetables much more than grains, breads, cereal, rice, etc. Those will have a much greater insulin impact than whole fruits and vegetables.
- 6.13** Now, there are other layers to this, but we don't need to go that far. But when it comes to fruits and vegetables, you must avoid the temptation to blend them or juice them. If God wanted us to drink an apple, then the apple would have come in a little slurpy form from the tree. That would change the whole Bible story. But these are meant to be consumed by chewing, with a lot of fiber. Fruits and vegetables are intended to have fiber in them to generally offset the digestible starches and sugars. So whole fruits and vegetables will be a prudent strategy.
- 6.14** I also think there's another reason to appreciate our ancestors' approach to food, which is fermentation. Now, it's funny for a guy like me, who's never touched a drop of alcohol in his life, to be such an advocate of fermentation, but I am because I appreciate the benefits of the little bacteria and how the bacteria do some of the work for us. For example, an apple nowadays is essentially just a big sugar bomb, and it might be a bit of a silly example. But if we enabled the bacteria to ferment the apple to a degree, the bacteria eat the starches and sugars. That's the only thing they eat. They only eat the glucose.
- 6.15** Now, this is not the greatest example. Let's take dairy, for example—dairy, which consists of both fats and proteins and carbohydrate. Well, if you ferment dairy, the bacteria eat the carbohydrate. They do not touch the proteins, they do not touch the fats. That's perfect for us because those are the two things we actually really want in order to control our insulin. Now, regardless, back to the sillier example of the apple, the bacteria will lower the amount of glucose, the starches and sugars that we're actually getting from that.
- 6.16** Not only do they remove some of we don't want, they also give us some things that we benefit from, namely the short-chain fats. We touched on this in the previous lecture when I introduced short-chain fats and their total inability to be stored by the body. They cannot be stored. They must be burned, which is what makes them ketogenic. The shortest of all of the short-chain fats is a fat called acetic acid, also known as vinegar. The amount of evidence that has shown a benefit to vinegar consumption is substantial.
- 6.17** I consider it one of the most overlooked little hacks, if you will, that if you are going to eat something starchy, then just take a little bit of apple cider vinegar in water, and it will significantly blunt the glycemic response, not only because it changes the digestion of the starches and sugars, but it also accelerates the rate at which the muscle is pulling

in the glucose, helping you clear the glucose faster and helping prevent what might otherwise be a big insulin effect. It brings the insulin and the glucose down. So apple cider vinegar is a great way to implement some short-chain fats if you're not fermenting all of your carbohydrates.

- 6.18** Another idea when it comes to controlling carbohydrates is to control the timing. This study actually analyzed the insulin and glycemic response to the exact same meal, but changed the order of consumption. On one hand, carbohydrates were consumed first and then the proteins and fats. On the other, the proteins and fats were consumed and then the carbs came last. Well, that's where they belong. In this strategy, when carbs were consumed after the protein and fat, the glycemic response was 50% of what it was otherwise. A significant reduction by just ordering it and again putting carbs where they belong, put them at the back.
- 6.19** Now, one little tangent here when I talk about controlling carbohydrates. Thankfully, we have more and more ways to indulge in the sweet sensation of something that we eat or drink, without having to rely on sugar. Sugar, no surprise, has a substantial insulin effect. There are a lot of low-calorie sweeteners that are artificial or natural. There's an enormous variety of them, but they all have a lower glycemic insulin load than sugar itself does. Sugar alcohols, these are things like xylitol or malitol, mannitol, erythritol. To varying degrees, they can have an insulin effect. It will be less than sugar, but it will be there. And there's so many, there's such a variety of these that I can't list them all, but some are better than others.
- 6.20** Then next, we go to this other group like allulose or aspartame, stevia, monk fruit extract, which have no effect on insulin whatsoever. Regardless, I would say they're all better than sugar when it comes to the insulin impact. But you have to mind the extras. Many of these, depending on how you consume them, some of them may come in a little sugar packet. It looks like a sugar packet. For example, stevia supplements will commonly have this.
- 6.21** Not to cast any shade on stevia, it's a perfectly good sweetener, but many of the packaged stevia supplements will have as the filler a molecule called dextrose, which is just pure glucose. The actual amount of stevia in it is so modest because it's so sweet that they put something extra in there so that, when you open it, it all comes out more easily. Well, that's unfortunate. So just mind the extras. Make sure that you're only getting what you want. When you want stevia, make sure that it's just stevia and there's no dextrose filler, or the same with allulose or aspartame or whatever else it may be. But they are good substitutes.
- 6.22** The next rule, the dietary rule, is prioritize protein. As I showed you at the very beginning of all of our lecturing, the consumption of protein in the United States over the past several decades has been going down steadily. We don't want that trend to continue. Protein shouldn't even be considered a calorie that we should ever worry

about. Protein is a building block. It's fats and carbohydrates that are fuels. Protein builds, and those fuel the body.

- 6.23** Now, when it comes to protein, any animal source protein is superior to any plant source protein. Soy is the best plant source protein, but it is still inferior to any animal source protein, be it from dairy or meat. We should appreciate that. And as we prioritize protein, try to honor or to meet that reality. The problem with some of the plant proteins, due to how we consume them, is the presence of anti-nutrients, which are molecules that get inadvertently enriched in these plant protein powders, which block the digestion of the protein.
- 6.24** Also, there's a third party group, a nonprofit organization that quantified the amount of lead and arsenic that came in popular protein supplements. Wouldn't you know it, the plant source proteins were the greatest offenders. There may be a lot of these things you don't want. Now, also with regards to protein, a target that we should be focusing on that I'll mention again at the end is trying to get about 1.5 grams of protein per kilogram body weight. Now, I could add ideal body weight. If you're starting this intervention and you know that you are 50 pounds overweight, then go with that ideal weight. But it is hard to get too much protein. Your body will begin to limit how much you want to eat.
- 6.25** Importantly, our trends set us up to fail. One of the greatest predictors of mortality is muscle mass. We need to keep our muscle mass to have a longer health span, living functionally and then just dying, which is much better than slow decay, slow decay, and then dying miserably, having been an invalid for 20 years. We need to make sure we eat adequate protein.
- 6.26** Interestingly, as we age, our ability to convert consumed protein into muscle and bone protein is diminished. We actually need more protein at 65 than we did when we were 25 because of our growing inability to metabolize it and handle it properly. So the need only grows. Thus, I further emphasize, prioritize protein in your meals. When you're eating, ask yourself, am I getting adequate protein in this meal?
- 6.27** One other comment about the plant proteins. I don't intend to throw them under the bus entirely. I think there can be a place for them, particularly if they are fermented. It is not common, but once upon a time, it was. For example, soy was the primary source, or in Latin American cultures where some of the proteins were coming from legumes, it was fermented in big pits over time, over days, and the fermentation enhances the digestibility of the proteins. If you can find plant proteins that have undergone some process of fermentation, which removes both the anti-nutrients and the heavy metals, then you are dealing with a much better protein. I, as a scientist, have a much easier time supporting that.
- 6.28** Now, the third rule I keep together with prioritize protein, because in nature, the best proteins come with fat. That's how we should eat them. I'm not saying there's never a

place for things like coconut oil or butter, which is pure fat. Absolutely, we shouldn't fear fat. But we should also remember that while we are prioritizing protein, those proteins will come with fat naturally. That's how God intended them to be. In our hubris, we've pulled them apart. We should allow them to come together. When fat and protein come together, we digest the protein better, and it is more anabolic.

- 6.29 Studies in humans have found that when you give a human a pure load protein, there will be some degree of muscle growth following a workout. If you give them a load of protein and fat, at about a 1:1 ratio, like what you get with an egg, the white and the yolk, it increases muscle protein synthesis by about double. We need fat to use protein. Some of the ancient or the early American explorers experienced this terrible diarrhea as they were forced to rely on very, very lean meats. The situation is sometimes referred to as rabbit starvation. They could trap rabbits, but rabbit meat was so lean that they were eating a diet of almost pure protein.
- 6.30 Without the fat, they would have terrible diarrhea, that you just can't digest the protein well. It is one of the features of human physiology that when we have fat and protein coming down together, the changes in bile acid content within the intestines will facilitate the digestion of the protein. When you're tempted to just take the scoop of whey, ask yourself, maybe I should also put a little fat in there to help my body digest it better.
- 6.31 Now, as much as I want to remove the fear of fat from your minds, as a reminder, there is one type of fat to fear, and that is the refined seed oils for reasons that we've already elaborated on. Also, previously, we've introduced the concept of the saturation of the fat. Saturated fats, monounsaturated fats, and polyunsaturated fats. While we are embracing dietary fat in our day-to-day living, be mindful of the fact that saturated and monounsaturated fats are abundant. They will be abundant in natural fats—natural fats being any animal fat we are well adapted to consuming and any fruit fat.
- 6.32 Fruit fats are the fats that come from the flesh of the fruit. That is things like the coconut or avocado or olive. That is the flesh of the fruit. Our ancestors, anciently, merely needed to get the flesh of the fruit and step on it or somehow crush it. Now we had an oil that we could consume and use. The seed oils, as I alluded to lightly, previously, require significant technology, a significant industrialization, to take a seed. Because if you just step on seeds, you just get a mash. You have to get the oil from the mash. It's a very complicated process. I would say it almost reflects the fact that we're not supposed to get oil from seeds, the sheer difficulty that we have getting it. Be careful.
- 6.33 Now, PUFA, in general, the polyunsaturated fats—I don't mean to say they're purely evil. We're just getting super physiological levels of them. These natural fats, the animal fats, actually contain all three of these. It's just that the actual amount of PUFA in them is very low. That's how we should get it.

- 6.34** All right, now, we have mentioned previously GLP-1. This is that miracle drug now that activates GLP-1, that acts like GLP-1, I should say. But it naturally is produced from the intestines, and it clearly has beneficial metabolic effects. It's interesting to note what these different fats do to GLP-1 secretion. When we eat saturated fats and monounsaturated fats, GLP-1 will go up 15 to 20 times above normal, beyond what it was just a moment before. When we eat a pure polyunsaturated fat, it is at best half of that response or even a third. Another reason to balance the focus when it comes to not fearing fats.
- 6.35** All right, so we've gone through the three that deal with the three macronutrients. Control carbs, prioritize protein, and don't fear fat. Now, let's get to the fourth, which finally starts to introduce this concept of some degree of energy restriction. Because when you're not eating, your energy coming in, your calories, are diminished. This is the familiar dietary paradigm that we've viewed now a few times. This is the average individual's insulin levels throughout the day.
- 6.36** Why I'm such an advocate of these dietary rules is because we have a culture that is obsessed with eating the exact opposite of the rules that I've mentioned. We're spiking our insulin all the time, and we're making it even worse by snacking in between all of our meals. At a minimum, the best idea of embracing this watch the clock is to at least cut out the snacking. If we like to call them, we can have mini fasts throughout the day. You're eating three square meals.
- 6.37** But I want to highlight some studies that employed an 18:6 fasting protocol. But actually, let me back up. On this slide, I've highlighted some of the most popular mechanisms of fasting. An 18:6, which takes a 24-hour window and has the individual eat typically two meals within a 6-hour window, fasting 18 hours, so you're fasting through one meal. Other people do one meal a day. Another alternative method is a 1:1, where you fast 24 hours and you eat normally 24 hours. Or in other instances, what has been published in the literature, eating normal 5 days and fasting for 48 hours, a 2-day fast. Of course, drinking water and electrolytes, possibly throughout.
- 6.38** Now, back to this, at a minimum, mini fast. Then I want to show you some of the studies that have compared 18:6 fasting patterns, intermittent fasting. Within 24-hour periods, you either cut out the breakfast, so you fast through breakfast, which is easy to do, and then eat a lunch and a dinner as normal, or alternatively, you fast through supper and have a normal breakfast and lunch. Unfortunately, the easier one of these has the less metabolic benefit. If we have a situation where we're fasting through breakfast, it increases the likelihood that our hunger is mounting and mounting and mounting, and we might not eat the right way at the end of the day.
- 6.39** Regardless, fasting through breakfast and stacking the calories into the evening and afternoon does increase hunger the next day, and it increases fat synthesis. The fat cells even become apparently a little hungrier, compared to the alternative, which is when you fast through dinner, fast through supper, and then you have greater satiety

the next day. It is reflected in this ideology, this mantra of breakfast like a king, lunch like a prince, and then dinner like a pauper. This idea of, you stack your calories earlier in the day and then you're slowly tapering off.

- 6.40** Again, I said, I noted, that's not a convenient way to eat. I know for me personally, it is very easy to fast through breakfast, just as part of my family dynamic, but also as a part of my family dynamic, I want to have dinner with my family. I don't want it to be weird where dad's just sitting there looking at everyone while everyone else is eating. I've not wanted to create that weird paradigm or process and have a tradition within my family. But there's still a way to do it, which is at a minimum, control your dinner. Let dinner be a small meal, perhaps because your lunch has been bigger.
- 6.41** Now, with regards to fasting through dinner, this has been time-tested as the better way metabolically, albeit the worst way socially to eat. With this pattern, other studies have looked at the consequences of having early eating windows where it is based on low-carb meals versus low-fat meals. Even though they are isocaloric, when you do this in human studies, they're eating the exact same number of calories. But reflective of the fact that it's not all about calorie number, the low-fat version of the breakfast and early eating window causes significantly greater hunger throughout the day.
- 6.42** That's because, if you spike insulin, the brain will sense the insulin spike and the lower nutrients in the blood, and it will drive you to want to eat sooner. If you do engage in any form of intermittent fasting, I still strongly suggest that you not forget the previous three rules. If you can couple the "watch the clock" paradigm with controlling carbs, prioritizing protein, and not fearing fat, now you're cooking with gas. That's the best way to implement these ideas.
- 6.43** Now, I've introduced the four concepts, and indeed, I just repeated them. But let's look at how you might bring them all together. First, with control carbs, just as a reminder, focus on whole fruits and vegetables. Put the carbs at the end of the meal if they're in the meal at all, and appreciate the value of fermentation, not only of consuming carbs that have undergone fermentation or just avoiding the food entirely that's been fermented or because you can't get it, and then using things like apple cider vinegar to force the metabolic response, the beneficial response, to consuming the short-chain fats that come from fermentation.
- 6.44** Second was prioritized protein. Focus on getting that target of around one and a half grams of protein per kilogram body weight. The source of the protein matters. Animal source proteins will trump plant source proteins. But if the plant source protein has been fermented, it brings with it a benefit. Don't eat the protein alone. Embrace the fact that in nature, proteins come with fats. Then with regards to fats, don't fear fat. Focus on the natural sources of fats being animal and fruit fats, and just be wary of seed oils. That's the simple guideline there.

- 6.45 Then lastly, the watch the clock. Eating the majority of your calories earlier in the day is easier or better, rather, metabolically speaking, although not as easy, socially speaking. Then perhaps at a minimum, just have a small dinner and then no snacking throughout the day, particularly in the evenings. Do your very, very best that, once dinner is done, you're done. Some strategy, some people find when they're feeling the cravings in the evenings, is to actually sip on a little bit of apple cider vinegar in water. Just that tart, sour taste helps control the cravings and brings them under control. But if you feel like you just have to snack, don't forget the first three rules.
- 6.46 All right, now, with regards to these rules, some may say that I am being unnecessarily harsh on carbohydrates. Well, I hope I've defended my view and indeed been diplomatic enough to say that carbohydrates are not the devil here when it comes to dietary view. I think dogmatically, we've embraced carbohydrates too much, in part for the reasons I've mentioned, and in part because we've clearly forgotten what humans actually need. Here, once again, are the three macronutrients of the human diet. Let's just explore what is essential within each of them.
- 6.47 Within the family of dietary fat, there are two fats that appear to be essential. With regards to proteins, there are many more amino acids that are essential. In other words, these are fats and amino acids that we must eat in order to survive. When we move into the dietary carbohydrates, we hear crickets chirping. There is no such thing as an essential carbohydrate. I'm not being—it sounds like I'm being provocative, but even the National Academy of Sciences within the United States admitted in a statement years ago that the lower limit of dietary carbohydrate is zero.
- 6.48 Assuming the person's eating fats and proteins, that's a nice way of couching it. But the fact is, there is no such thing as an essential carbohydrate. I'm not saying we should never eat them. I'm not saying they're all bad, but I am saying it does, in fact, have the greatest effect on insulin, which bears significant metabolic consequence, and it happens to be the one we don't need. Why then would we put that as the bulk of what's sitting on our plate, which is what we've been told to do?
- 6.49 I think the evidence suggests, and human reality further supports this, that that is irrational. We should focus on the things we need and just the things that happen to also have the least impact on insulin. Now, a little more. I'm not done talking about the carbohydrates yet, because while there is no such thing as an essential carbohydrate, that does not mean that blood glucose is not essential. This is not the same. So many ignorant people have conflated these ideas. The fact that there are cells within the body, like, for example, red blood cells, who are totally reliant on glucose as their fuel.
- 6.50 The red blood cell can use no other fuel, and the brain will readily use glucose. Although we don't know to what degree it's essential for the brain. As I showed you, the brain is very readily able to use ketones as its primary fuel. But regardless, we know that there are some cell types that must have glucose. So don't conflate the two. We do need blood glucose. We do not need dietary carbohydrate. That is simply because

the liver, remember, that's the ultimate soccer mom of nutrient metabolism, is giving, giving, giving. It is capable of producing all of the glucose that the body needs. It is one of the wonders, metabolically speaking, of what the liver can do.

6.51 The liver knows some parts of the body need the glucose, so like the soccer mom, it has little snack packs ready to go. The moment needed, the moment the family starts crying out for some glucose, the liver has it ready to go. How does the liver do this? It is this beautiful wonder of human metabolism where it mostly comes from the muscle. This is a process called the Cori cycle, C-O-R-I. I only mention that because it's actually a beautiful instance where the Cori family, the Coris, were a husband-and-wife pair. They won the Nobel Prize in part because of this discovery of this so-called Coricycle.

6.52 What they found was that the muscle releases a lot of lactate. Now, just as an aside within an aside, because we are, after all, in a tangent, there is no such thing as lactic acid within the human body. Despite all the myths that we've heard about lactic acid, the muscles make a molecule called lactate that has nothing to do with muscle soreness. It's not making the muscles feel sore or anything like it. The whole idea of lactic acid is born from bacteria that are fermenting sugars. Well, that doesn't happen within the body like it does in bacteria.

6.53 When the muscle is working, it produces lactate, and it dumps this lactate into the bloodstream. The liver is so clever that it can take the carbon skeleton of the lactate, which itself is a product of burning glucose for fuel, take those carbons, send them to the liver, and the liver repays that by dumping it back as glucose into the bloodstream. Most of the glucose that the liver is making when someone is restricting carbohydrates or fasting, which is also a form of restricting carbohydrates, it's coming from lactate, which itself is coming from the muscle. It's a beautiful instance of this recycling within the body.

6.54 Now, one last idea, and we'll leave diet behind, although I desperately hope you will not forget those dietary rules and the ability to implement them, your ability to implement them. I want to also touch on the benefits of changing the temperature around the body. This is a lever that we can grab and does have significant benefits. When it comes to temperature changes, we can go either way and expect significant metabolic improvements. Whether we're making the body too hot or we're making the body cold, either strategy, for reasons we'll get into, will lower the glucose level, which in turn will lower the insulin level, ultimately resolving insulin resistance to a substantial degree.

6.55 Well, let's start with the hot side because that's how most people prefer it. Some like it hot. Now, in this case, the benefit comes twofold, either through changes in blood flow or through sweating. Those are the two interventions or the two mechanisms whereby this heat intervention will elicit metabolic benefits. This study actually found that the response to sauna time and the changes in blood flow were so substantial that it actually mimicked the blood flow that happens during exercise. It was so substantial

that they suggested that, for people who are unable to engage in exercise, heating up the body may be a suitable replacement.

- 6.56** That is so impactful. Once again, if we're talking about the person who's wheelchair-bound or generally bedridden, even if it is putting them in a warm bath or the person is getting into a warm bath, it doesn't have to be a beautiful finished sauna or steam room. That will improve the blood flow. And as more blood is going to the muscle, the muscle just starts consuming more glucose. As the glucose consumption continues, insulin comes down, and the body is more insulin sensitive as a result.
- 6.57** Now, in the midst of this thermal challenge of heating up the body, we also generally will start to sweat. Sweat itself is a metabolic challenge to the body, where we know that every few drops of sweat, just the very act of the body changing blood flow, opening some blood vessels, narrowing some others, is a challenge that it requires energy to do it. So every few drops of sweat requires about a half a calorie of energy in order to produce.
- 6.58** This means that during a normal sauna session, where a person can lose up to—it's not too much to lose about 500 grams of sweat—it's like they will have just spent about 300 calories, which is comparable to running a few miles. I'm not saying that it's a perfect replacement. Running has other benefits, sure. But if that becomes limiting, or you want to add something else, well, then heating up the body is a great way to get some of the benefits. It will certainly improve insulin sensitivity.
- 6.59** Then one final mechanism, whereby warming up the body benefits, is through changes in cortisol. I want to highlight the effects of one study, which, briefly, it took people, subjects, in over four different days, just had them go into a sauna for 12 minutes. That's all. It was just four 12-minute sauna sessions, and their cortisol levels came down by 30% at the end of the study period. That's relevant, remember, because cortisol is one of the primary stress hormones that induces insulin resistance very readily. And so, all together, we have multiple layers, multiple facets, to this benefit of heating the body up, of making the body get a little warm and sweating a bit.
- 6.60** Now, let's go to the opposite end of the spectrum, or at least the opposite end of the thermometer, which is looking at cooling the body. When we cool the body, we have two benefits to note. One is shivering of the muscle, and the other one is the brown fat. Of course, what both of those have in common is that it's trying to heat the body up. With shivering, this is obvious. Where the muscles begin to work, just like when we get warm while we're exercising, when we enhance the rate at which the body is undergoing any chemistry, any biochemistry, you just produce heat. That's a byproduct of any biochemical reaction.
- 6.61** When the muscle is working, we have a lot of those reactions, and we start to heat things up. When the muscle is shivering when we're cold, it's doing so to heat the body up, it will start pulling in energy, just as if it were exercising, perhaps to a lower degree.

But the benefits become quite substantial. When you're running, you're really only working your legs. When you're shivering, you're doing everything. Every muscle, every skeletal muscle, could be spasming uncontrollably, all in an effort to try to heat the body up.

- 6.62** Well, it is like you are exercising those muscles a little. As they shiver in an effort to create heat, they become hungrier and hungrier. They start pulling in fats and glucose, and of course, that will help lower the insulin level, thereby improving insulin sensitivity. Now, also, as we've discussed before, the human body has various depots of fat. One of them is the brown fat that we have up in this thoracic cavity.
- 6.63** When the body is thermally challenged, when it is sensing this pressure to get cold, brown fat comes to life. It will fire up in order to start burning nutrients, like fats and glucose, all in an effort to create heat. It's basically this little campfire, and now we're throwing logs onto the fire in the form of feeding the brown fat cells, fats and glucose, thereby lowering the insulin levels.
- 6.64** Now, I emphasize this fellow's belly. Something we alluded to earlier in a Q&A is that we have some potential, when we are thermally challenged and cold and shivering, to have white subcutaneous fat, where we pinch and jiggle, start to act more like brown fat. You'll recall that was the process that is referred to as beiging. It's not quite brown. It doesn't have the mitochondrial density to actually change its color to be that dark, but it is definitely darker in color than it was before. It has gotten a little more tan or or beiged, if you will.
- 6.65** It starts to act a little more like the brown fat cells. Whereas white fat has a metabolic rate here, and brown fat's metabolic rate is 10 times higher, we can increase that to about three or four times the white fat. It almost gets halfway, but not quite. All because the body is thinking, we're really cold, and I'm going to now recruit this dormant, inert fat to help me heat things up. Now, one final tangent on this, and indeed, it is the final tangent of all of the lectures, is work from my own lab, which is us, we were exploring the differences and to what degree you can leverage brown fat versus white fat.
- 6.66** We found two contrasting findings. One is that insulin levels will make brown fat act more like white fat. Insulin will slow down the metabolic rate of that very metabolically active fat depot. That, of course, is not a good outcome, is it? But it further reflects the fact that insulin is so determined to store energy that it wants to try to block any wasting possible.
- 6.67** Then on the other hand, ketones, which are only produced when insulin is low, has the opposite effect. Ketones will take white fat and have it start acting more like brown fat. Everything I just showed you, if you looked at the cooling of the body and the shivering, and you are to any degree like my wife, which would rather laugh and slap me in the

face than actually immerse herself into some cold water, this gives you some of the benefit.

- 6.68** Where we found in humans that if a human was in ketosis and we pulled fat biopsies to measure metabolic rate, if you're in ketosis, the metabolic rate of that white fat tissue from the belly was about three times higher, almost right around the same range as full cold immersion. If you see the sentiment about cold immersion and you say, "Ben, I'm never going to do that, come hell or high water," as cold as the water may be, but ketones maybe. Well, then that could be some strategy, or you do both, then you're really amplifying the benefits.
- 6.69** But nevertheless, right back down to the nutrient biochemistry and the signaling of insulin blocking fat-burning and producing ketones, we have these contrasting effects. Insulin wants to slow the brown fat down to act more like white fat. Ketones want to accelerate the metabolic rate of the white fat to make it act more like the brown fat. That's a little easier to leverage, perhaps.
- 6.70** Now, I am a professor, and I love giving homework. So as we wrap up the lecture, as I have some assignments for you, this is my homework. Firstly, I want you to know where you are with regards to the risk factors regarding insulin resistance and infertilities in a obvious way, although these apply to everything we've talked about. One, get your fasting insulin measured, and ideally, it is below about 6 micro units per mL. Second, get your HOMA measured. Once you've measured your fasting insulin, you can measure your HOMA. You want that score to be about 1.5, although remember, there were some ethnic differences, where it can be around—you want it to be closer to 1 or closer to 2, depending on your ethnicity.
- 6.71** Then lastly, we have the adipose insulin resistance index. You'll recall that was my favorite one because not only is it such a beautiful reflection of nutrient metabolism and how insulin controls the fuel we burn, but also because it highlights, it goes right to the very first domino that's falling. We might not have manifested full, whole-body insulin resistance yet, but we can detect it at the fat cell. Remember, there are differences across the sexes. Among the many differences between males and females is the rate at which we inherently burn fat. Because women inherently burn fat faster, their adipose insulin resistance score will naturally be higher.
- 6.72** Next, with regards to the cardiovascular markers, the heart disease concerns. One I didn't talk about, but you will on your own learn a little more about is the coronary artery calcium score. We did speak about plaque formation within atherosclerosis. A plaque is formed by those big fatty foam cells, but eventually they become calcified. The plaque goes from being a soft plaque to what is called a hard plaque. You can image this with a CT scan. It is, in my mind, one of the best investments. If you worry about your blood lipids or you have been told that you should go on a lipid lowering medication, my strong and polite submission, despite not being your clinician, is to

measure your coronary artery calcium score first. That will directly tell you whether you have any plaque forming. Ideally, it'll be around zero.

- 6.73** At the same time, from any blood test you ever get, you can determine your triglyceride to HDL ratio because you will always measure your triglycerides, you will always measure your HDL. Those are standards on every blood test. On their own, they have limited value. Together, there's some synergy where you get a little more value than the sum of the parts. Look at that ratio, and again, you want it to be at around 1.5. A slight difference is according to ethnicity.
- 6.74** Then lastly, we come to the neurological disorders where we particularly focused on Alzheimer's disease as the primary form of dementia. But then I also mentioned migraines at the end, just teasing you with some centuries-old body of evidence. In both of these instances, once again, I remind you to pay attention to your insulin, but I also encourage you to control your dinner, your evening meal, and snacking, to such a degree that at least several times a week, when you wake up, you could prick your finger and measure your ketones with a ketone meter and detect your ketones at around 3 millimolar or higher.
- 6.75** If you aren't getting into states of ketosis within a week, particularly in the morning, you should be concerned that your insulin levels are too high and you're insulin resistant, and insulin is inhibiting the ketogenesis, the production of the ketones. That's a reflection of insulin resistance, but you should also begin to worry about your brain, because the brain is among the most hungry tissues when it comes to ketones. It loves ketones, and by all accounts, cognition will be improved if the brain is given some of those ketones from time to time.
- 6.76** Now, my final sentiment is, don't wait. I appreciate the fact that among all the things we've been talking about, the concept is simple, but the implementation is not. You need to take an honest assessment of what you can do, what you want to do, and what help you can recruit. My final homework is to start changing things, even tonight. Don't snack when you go to bed tonight. After you have dinner, stop eating. Do everything you can to control your evening habits when it comes to snacking on foods. Or if you have to snack, remember those dietary rules. Control carbs, prioritize protein, and don't fear fat.
- 6.77** Next, when you wake up, change your breakfast. I showed you one of many studies that has found that, even if you eat the exact same amount of calories, a breakfast that spikes your insulin will lead to much greater hunger throughout the day. Remember, so often, hunger always wins. So don't set yourself up to fail. Change breakfast tomorrow.
- 6.78** Then lastly, to whatever degree you feel comfortable, get a little uncomfortable, and shiver or sweat. Whether it's through cold immersion or cold exposure, or whether it is through some heat challenge, there are substantial well-documented benefits to

improving insulin resistance beyond what you can do with controlling what you're putting in your mouth.

- 6.79** Then in conclusion, this is what we've been talking about today, the interventions to improve insulin resistance. Determining where you're at currently with those blood tests and those at-home eyeball tests, what you can just see. Now we've ended with some homework, all while we were describing the myriad interventions and highlighting their varying degrees of efficacy with regards to improving insulin sensitivity.
- 6.80** Then never forget, these plagues of prosperity, as much as they are ruining the health of people around the world, we can control them, and we are tempted to put them in silos. Imagine someone, if you will, opening up their medicine cabinet every morning. They take one pill for their migraine headaches. They take one pill for their infertility. They take one pill for their hypertension and one pill for their diabetes, little knowing that, while they all may have distinct noxious stimuli that are causing them, we know that among these, they all have one deep core in common. I would submit that there is tremendous value to addressing that one singular cause.
- 6.81** Then in short order, the person may find that they don't need that medication anymore, and then that one, and so on. In the time we've had together across all of the lectures, I've broken this up deliberately into three key parts. We described on the first set of lectures, what is insulin resistance, emphasizing where it comes from, some of the causes, the origins of it, and the role of fat cells in particular in other tissues. We then moved on to why does it matter? Why is it such a relevant issue beyond its prevalence, how it is impacting so many of these chronic diseases?
- 6.82** Then lastly, the happy ending to what otherwise seems like a horror story, albeit, hopefully, an entertaining one, we have the solution, which is that these problems are very real, but they are very rapidly resolved. Thank you so much for listening and participating. I'd be delighted to take any questions.
- 6.83** In the last couple of years, especially, we've seen an increase in alternative diets such as the carnivore, paleo, and given one of the points of the non-essential intake of carbohydrates, I'm wondering if you could shed some more light on that.
- 6.84** Unfortunately, there are not a lot of clinically validated and published studies on the carnivore diet. I want to be careful because I always want to make sure that I'm speaking as an authority based on the published literature. Unfortunately, I cannot cite specific studies. However, anecdotally, and I try to view anecdotes in a deeply rational way, the results, the response metabolically that someone will have to cutting carbohydrates out of their diet completely is fantastic. There's no other word for it. It is remarkable how rapidly someone can start dropping medications for their psoriasis on their skin or their hypertension or their infertility.

- 6.85** My view on the carnivore diet in particular is that it can be a remarkably effective way of eating. It's limited. I would encourage someone to be honest with their family dynamic and their social and living dynamic and to what degree they could do that. But there's no question, it is a perfectly healthy way of thriving.
- 6.86** One of the greatest myths is that you need fiber, for example. The dogmatic dietitian's view is that fiber is this manna from heaven. Well, whatever manna was to the children of Israel, it wasn't fiber because there's no nutritional value to fiber. Basically, you consume fiber if you want big bowel movements, at the risk of being scatological. A carnivore diet is a perfectly healthy way of eating because, especially if it is ruminant meat, you are getting literally every nutrient you need.
- 6.87** I have two questions on low-carb diet. So, the first one, say you're trying to lose fat or weight on a low-carb diet. Does your body get used to it at some point? Then if it does get used to it, to even all the markers that get better, what do you do at that stage? Do you want me to ask the second question?
- 6.88** When you say get used to it, do you mean plateau?
- 6.89** Plateau, that's right.
- 6.90** Yeah. So that is—I love that you're asking that question because it lets me revisit a concept that I worry I didn't impress upon you well enough. The first intervention should be low insulin, which enables the body to begin using its own fat for fuel. It allows us to finally tap into these energy bars that are strapped to our bodies waiting to be used. Now, at that stage, a person may lose all the weight they want, just on that alone.
- 6.91** If they find that they get to a new plateau and they think, "I've still got more to lose," that's why I put the order of events the way I did. If you have lowered your insulin, you've become accustomed to using your own fat for fuel. Then if you want to start implementing some calorie restriction, ideally in the form of intermittent fasting, then it's not such a shock. The brain looks at the reduction in calories coming in and says, "I'm fine, I'm good. I'm burning so much of our own fat, and that's making ketones. I don't even feel the difference. I can do this all day. Let's keep fasting." But that is a nice method of implementing calorie restriction.
- 6.92** OK, thank you. My second question is, could there be certain ethnicities that, say, don't do well on low-carb? Say, you grew up in Asia and you ate a lot of rice or they've been eating rice for decades. Do they do less well on low-carb diets?
- 6.93** Yeah, that's a great question. Unfortunately, there aren't a lot of studies that have implemented the same low-carb diet across ethnicities within one group. But I would answer, with some degree of speculation, a resounding yes. We do know, there are studies showing benefits across virtually every ethnicity at this point. But I would emphasize the fact that if we were to revisit, at the very beginning of our lectures, that

map of the world, you'll notice that it is, in fact, Asia and the Middle East and cultures that rely heavily on carbohydrates that are the most insulin resistant of all of the populations around the world.

- 6.94** That is where, per capita, the problem is the worst by far. The United States isn't even in the top 30, I don't think. Maybe not even top 50. The problem is so much worse in those parts of the world. So this idea that, well, in Asia, the consumption of these grains or rice, whatever, that's a physiology that's well adapted to it, I would say the evidence doesn't support that.
- 6.95** In a context of insulin sensitivity, no resistance, is there some situation in which carb deprivation can induce some stress? Maybe in physiologic hyperinsulinemia, like poverty or pregnancy, for example?
- 6.96** Yeah, it's a really good question. I actually had about 10 slides in this lecture that I removed, that would have answered that exquisitely well. But it was this idea of a so-called physiological insulin resistance that accompanies carbohydrate restriction, which is not true. Some people say that, but what happens is that the beta cells adapt to the environment. Basically, at the very beginning of our time together, I highlighted the fact that when a person eats glucose, carbohydrates, there's two insulin bumps. What's called the first phase of insulin, and then a second phase of insulin.
- 6.97** The first phase of insulin secretion is the beta cells releasing the insulin that they already have on hand. The beta cells have prepackaged insulin ready to go. When glucose comes in, the beta cells will dump that insulin into the body. And then later it will make a little more insulin from scratch. But if someone has been controlling carbohydrates, restricting carbohydrates, even as short as a period, of a period of time as eight hours. So just eight hours of not eating carbs is enough to lose your first phase of insulin secretion. The beta cells are basically looking at all this insulin, cluttering up the space, and saying, we don't need all this here. Let's just get rid of it. And it will degrade its own insulin.
- 6.98** Now, all of a sudden, if the person eats a load of carbohydrate, it takes them longer to clear the glucose, not because they've become insulin resistant, but because they've lost that first phase of insulin secretion very, very temporarily. If someone has adopted a low-carb diet and they have to go to the clinic and perform an oral glucose tolerance test, they will likely fail the test, not because they're insulin resistant, because the beta cells have adapted so quickly to the fact that no glucose is coming in.
- 6.99** So the way you pass the test with flying colors is within about a 6-hour period before you go to take the test, eat some carbs. That reminds the beta cell, OK, glucose is coming. I'm going to make sure I have a lot of insulin on hand and ready to go. Then you go do the full oral glucose tolerance test, and it's no problem.
- 6.100** Yeah, I was just wondering, can food allergies be related to insulin resistance?

- 6.101** Yes, certainly. Insofar as the allergy to the food elicits what's called a hypersensitivity. Strictly speaking, it's a type 1 hypersensitivity. You're having this immune response to an insult, a food which your body shouldn't perceive as a problem, which is why it's called a hypersensitivity. You're reacting to something that should be perfectly benign, but that does involve an immune reaction. And anytime we start invoking immunity, we begin producing lots of cytokines, which activates an inflammatory response. And then, yes, anytime the inflammation is up, insulin resistance is up as well. So if a person were continuing to insult their body by consuming a food that they were adversely responding to, they would absolutely have a demonstrable insulin resistance every time the insult were occurring.
- 6.102** You mentioned dementia as one of the conditions that, a plague of prosperity. I wanted to ask you, in regards to brain insulin resistance, what's the best way to diagnose it and how early can we get it? Because I think that prevention in these cases is crucial and health promotion essential.
- 6.103** Yeah, great question. I appreciate you asking this because your authority is the brain more than mine. With regards to diagnosis, unfortunately, the manifestation of cognitive decline will occur later than the metabolic manifestation. Let's focus on that, which would be the earliest. I highlighted in previous lectures the fact that in women, as early as their 20s, and women do experience Alzheimer's disease more than men. It's one of the rare instances. Usually, it's men who die from everything faster than women do. But this is one instance where the woman is the one uniquely suffering, you can detect the reduction in brain glucose uptake in her 20s.
- 6.104** This has been followed through some prospective studies. In people as early as their 30s, before there's any evidence of cognitive decline, you can already monitor and detect reductions in brain glucose uptake. I would say the earliest manifestation of future cognitive decline is reductions in brain glucose uptake. Unfortunately, most of us will never have the opportunity to do that test because it's not a clinically offered test. It's typically just within the realm of academia. It's never left the ivory tower in a way, if you will.
- 6.105** But all the more reason to look at these surrogate markers of insulin resistance. If you can't measure the degree to which the brain is actually pulling in the glucose, at least get a measurement of insulin resistance. That to me, of course, to the man with the hammer, everything's a nail. To this metabolic scientist, insulin resistance is everything, even with dementia. Thank you, guys. Thank you so much.